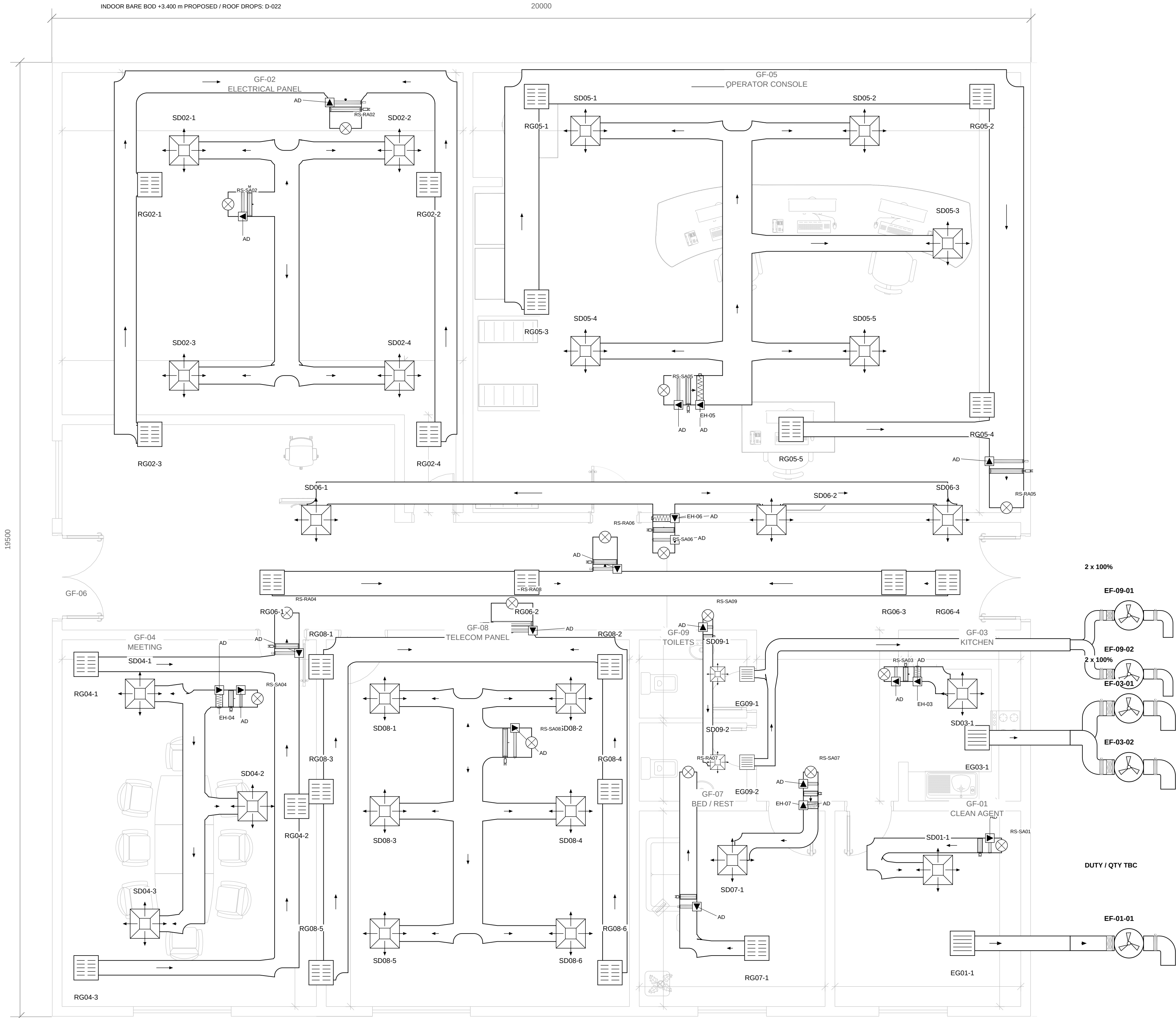


MAIN DUCT AND RETURN - COORDINATED DUCT PLAN

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039



ROOM AIRFLOW SCHEDULE

GF	Room	SA L/s	RA L/s*	EA L/s*
02	ELECTRICAL PANEL	664.8	605.5	-
05	OPERATOR CONSOLE	777.7	718.4	-
04	MEETING	449.2	389.9	-
08	TELECOM PANEL	884.6	825.3	-
06	CORRIDOR	418.2	358.9	-
07	BED / REST	168.2	108.9	-
09	TOILETS	46.2	-	64.9
03	KITCHEN	147.6	-	147.6
01	CLEAN AGENT	141.3	-	TBC

DUCT / SYMBOL LEGEND

	SA - supply duct
	RA - return duct
	EA - dedicated extract
	SD - four-way supply diffuser
	RG - ducted return grille
	EG - extract grille
	VCD - balancing damper
	MFD* - conditional fire damper
	RCD - return control damper
	EH - inline electric heater
	AD - access door
	RS - proposed roof supply / return drop

Terminal aliases: SD02-1 = SD-02-01; RG / EG follow the same format. Full tags and quantities are in the registers.

Roof-main OPTION: 15 independent room drops RS-SA / RS-RA. No indoor service crossings. Finished ceiling +3.300 m; proposed bare BOD +3.400 m. Envelope: D-020; roof/drop holds: D-022.

* Return quantities are assumed balances. Extract references and +50 / +25 Pa targets require final airflow and pressure verification.

Units: mm / L/s / m/s. Terminal faces and clear duct widths are shown. Duct levels, selected fitting development and manufacturer clearances remain pending. No shared return from kitchen, toilet or clean-agent room.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

COORDINATED DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-001

SCALE: 1:50 AT A1

REVISION

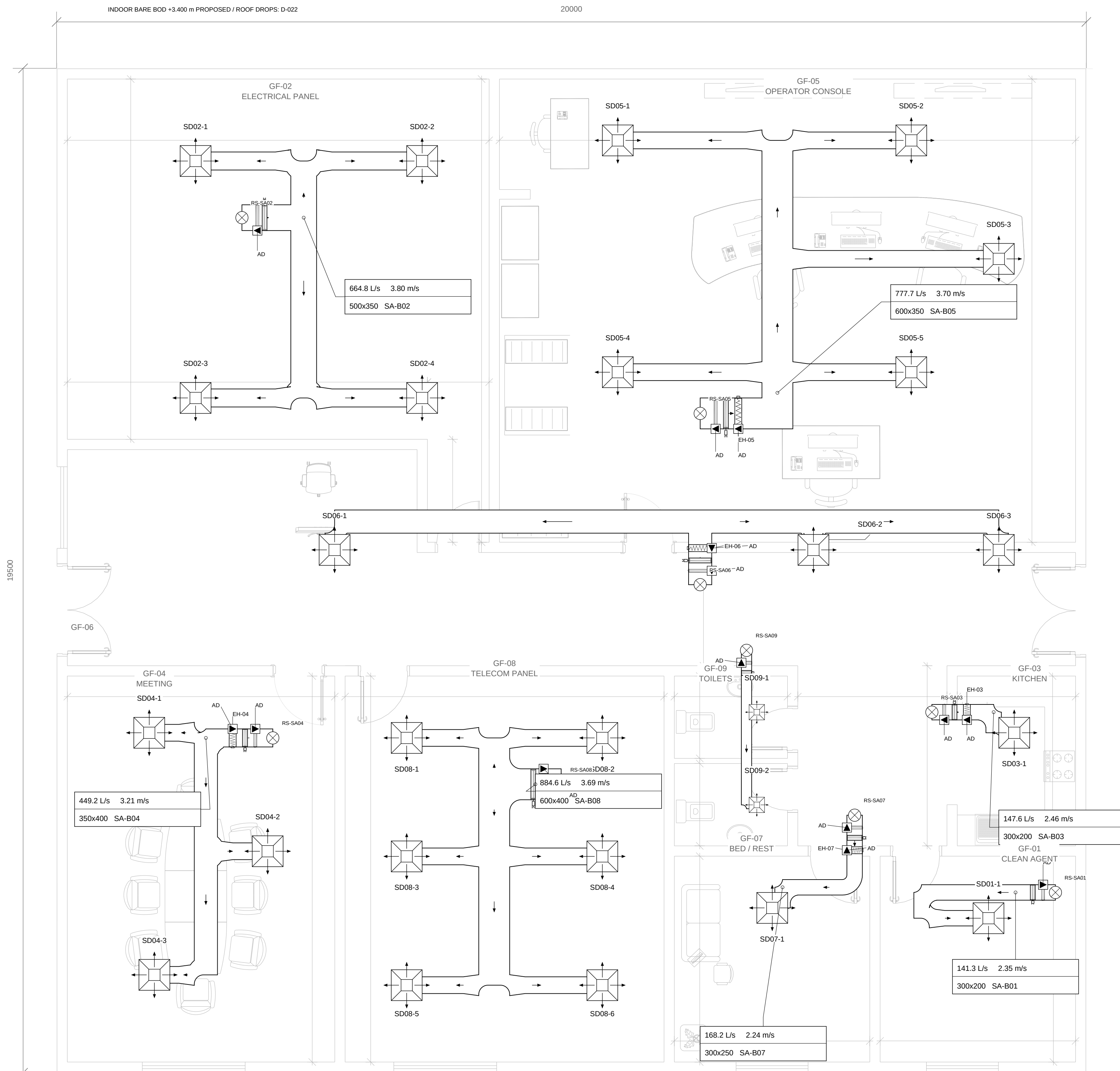
02

08 OCT 2026

SHEET 1 OF 50

MAIN DUCT AND RETURN - SUPPLY AIR DUCT PLAN

REISSUED DX MAIN / RETURN OPTION || ROUTES / DUTIES RETAINED FROM REV12 || D-xxx REFERENCES = MDR-xxx || DX DETAILS MDR-033..035



PROPOSED ROOF SUPPLY MAIN BASIS

Section	W x H mm	L/s	m/s
SA-M01	1200x650	3697.8	4.74
SA-M02	1100x575	3033.0	4.80
SA-M03	650x750	2255.3	4.63
SA-M04	650x575	1806.1	4.83
SA-M05	650x300	921.5	4.73
SA-M06	500x250	503.3	4.03
SA-M07	450x250	457.1	4.06
SA-M08	400x200	288.9	3.61
SA-M09	300x200	141.3	2.35

ROOM SUPPLY BRANCHES

GF	W x H mm	L/s	m/s
02	500x350	664.8	3.80
05	600x350	777.7	3.70
04	350x400	449.2	3.21
08	600x400	884.6	3.69
06	450x300	418.2	3.10
07	300x250	168.2	2.24
09	200x150	46.2	1.54
03	300x200	147.6	2.46
01	300x200	141.3	2.35

Room headers retain their branch size. Diffuser tags identify individual outlets. Round runout / neck diameters and each terminal airflow are listed on D-004 to D-006 and D-010.

MFD* is a conditional rated-penetration provision. VCD-Snn and FSD-Snn correspond to the GF-nn supply branch. Inline heaters are upstream of that room's terminal take-offs.

Continuous main tapers, flared branch boots and terminal adaptors: D-011. AD tags identify maintenance access doors. Proposed taper lengths and device coordinates are in the registers.

Total HAP-reference supply: 3697.80 L/s. Three 50% PAUs: two duty, rotating available alternate. Boxed callouts show branch flow, velocity and clear size. No room sensors are overlaid on this duct plan

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SUPPLY AIR DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-002

SCALE: 1:50 AT A1

REVISION

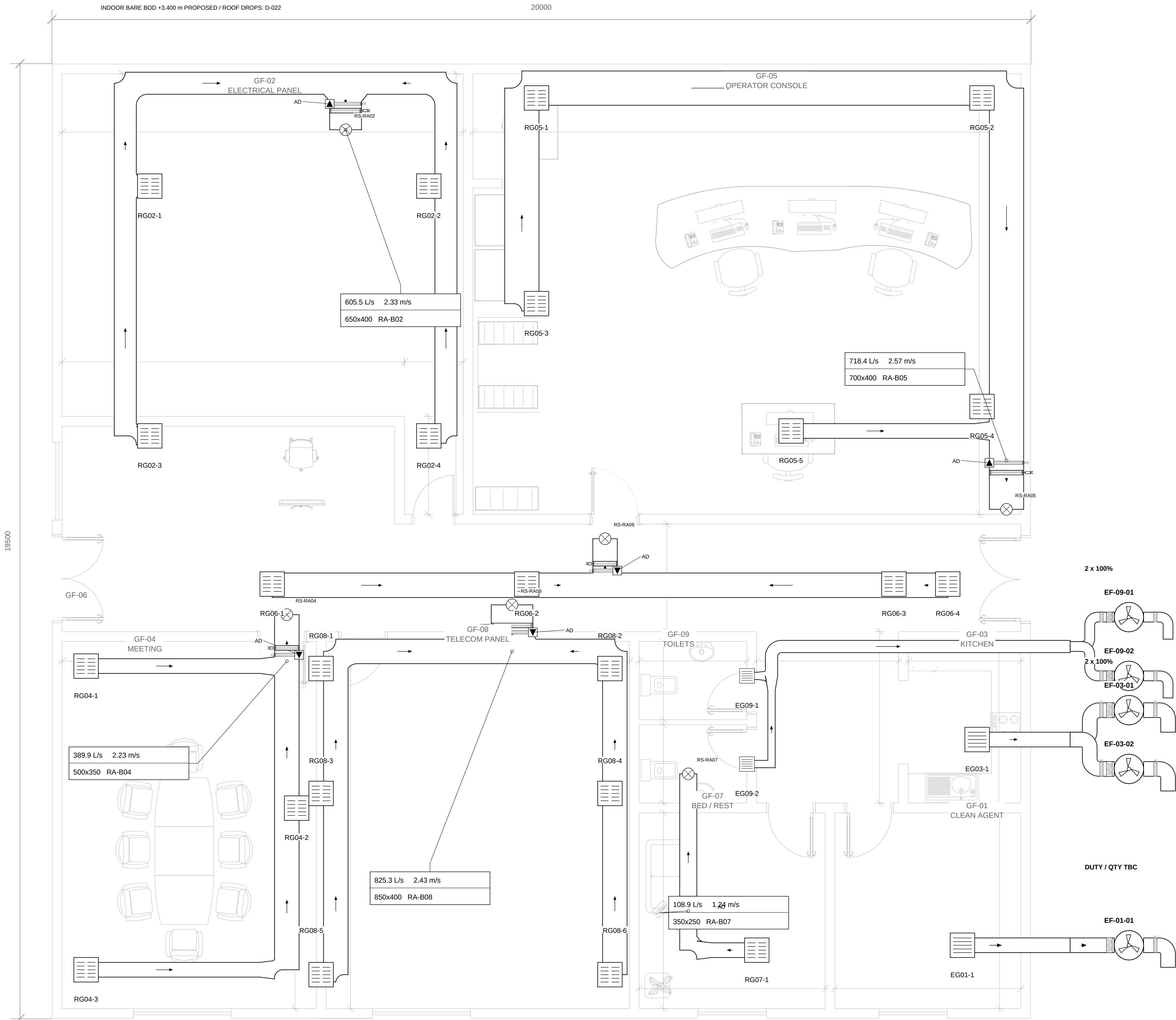
02

08 OCT 2026

SHEET 2 OF 50

MAIN DUCT AND RETURN - RETURN AIR AND EXTRACT DUCT PLAN

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039



PROPOSED ROOF RETURN MAIN BASIS

Section	W x H mm	L/s	m/s
RA-M01	1400x650	3006.7	3.30
RA-M02	1200x600	2401.2	3.34
RA-M03	600x800	1682.9	3.51
RA-M04	600x600	1293.0	3.59
RA-M05	350x575	467.7	2.32
RA-M06	350x300	108.9	1.04

ROOM RETURN / EXTRACT

GF	Air	W x H mm	L/s*
02	RA	650x400	605.5
05	RA	700x400	718.4
04	RA	500x350	389.9
08	RA	850x400	825.3
06	RA	500x400	358.9
07	RA	350x250	108.9
09	EA	250x200	64.9
03	EA	300x200	147.6
01	EA	TBC	TBC

No central return: GF-09 toilets, GF-03 kitchen and GF-01 clean-agent. Kitchen / toilet fans: 2 x 100% with duty rotation. CAG duty and quantity remain unverified.

Extract fan symbols are schematic equipment markers; actual fan and louver dimensions depend on manufacturer selection. Wall outlets face downwards.

* Return total 3006.68 L/s is an assumed starting balance. Existing extract references do not verify positive pressure or outdoor-air compliance. Confirmed fire shuts down all HVAC; automatic fire extraction is not enabled.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

RETURN AIR AND EXTRACT DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-003

SCALE: 1:50 AT A1

REVISION

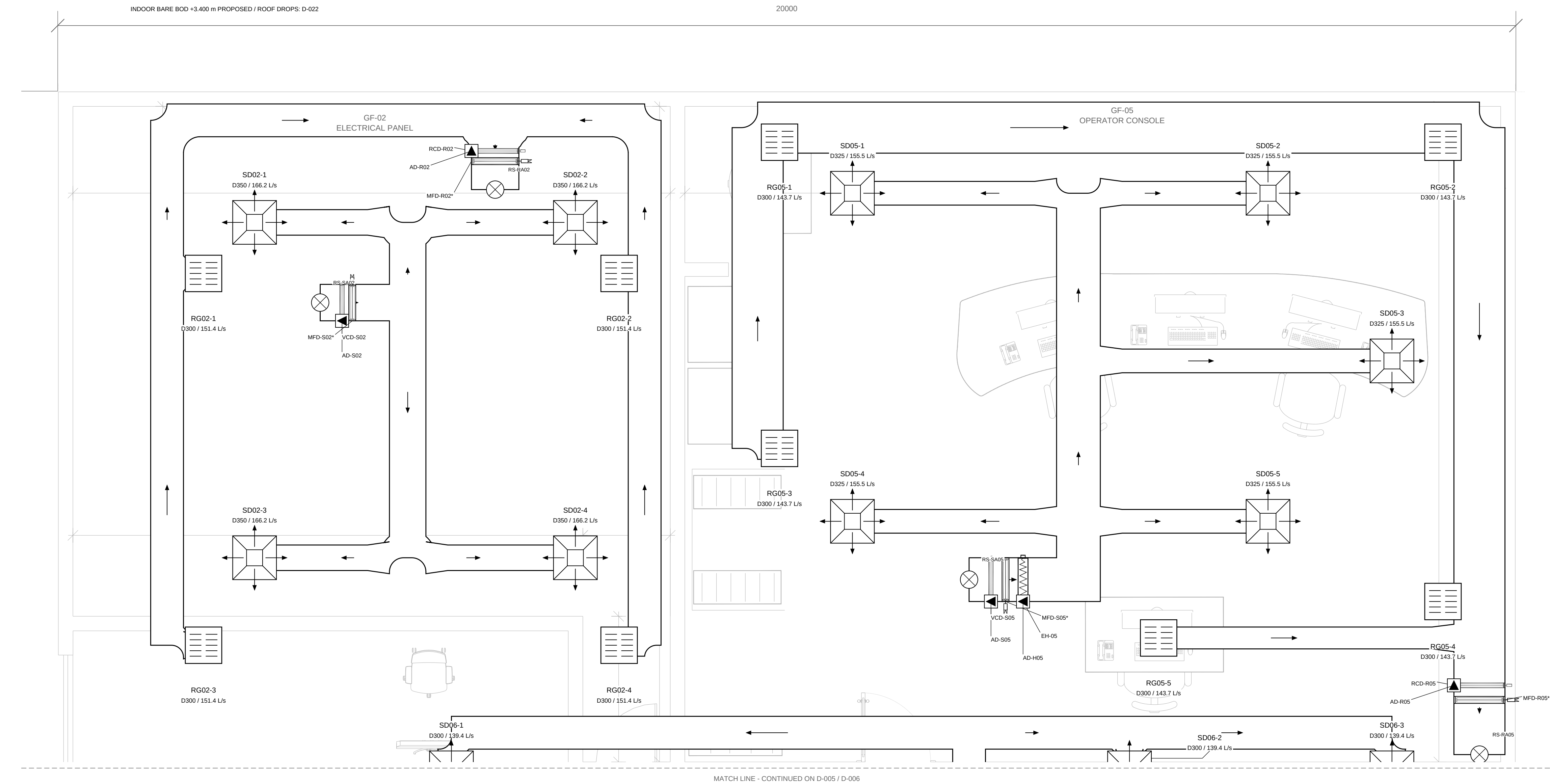
02

08 OCT 2026

SHEET 3 OF 50

MAIN DUCT AND RETURN - NORTH ROOMS - DUCT ENLARGEMENT

REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039



NORTH-ROOM TERMINALS / BRANCHES

GF-02 ELECTRICAL PANEL

SA-B02 500x350 mm | 664.80 L/s | 3.80 m/s

RA-B02 650x400 mm | 605.46 L/s | 2.33 m/s

SD: 4 x 166.20 L/s; neck D350; face 600x600

RG: 4 x 151.37 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-05 OPERATOR CONSOLE

SA-B05 600x350 mm | 777.70 L/s | 3.70 m/s

RA-B05 700x400 mm | 718.36 L/s | 2.57 m/s

SD: 5 x 155.54 L/s; neck D325; face 600x600

RG: 5 x 143.67 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

Plan symbols identify VCD-S02 / S05, conditional MFD-S02 / S05*, RCD-R02 / R05 and conditional return MFDs. The component register retains the FSD tag prefix for the conditional motorized fire/smoke damper symbols.

Deep common mains are proposed on the roof. Separate supply / return drops feed each room tree. Bare BOD +3.400 m is a proposed indoor level. Actual roof routes and structural penetrations require coordination; D-020 / D-022.

Access doors: AD-Snn / AD-Rnn beside branch dampers; AD-Hnn beside heaters. Taper / access arrangements: D-011. Instrumentation: D-007 / D-012. Catalogue selection remains pending.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

NORTH ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-004

SCALE: 1:35 AT A1

REVISION

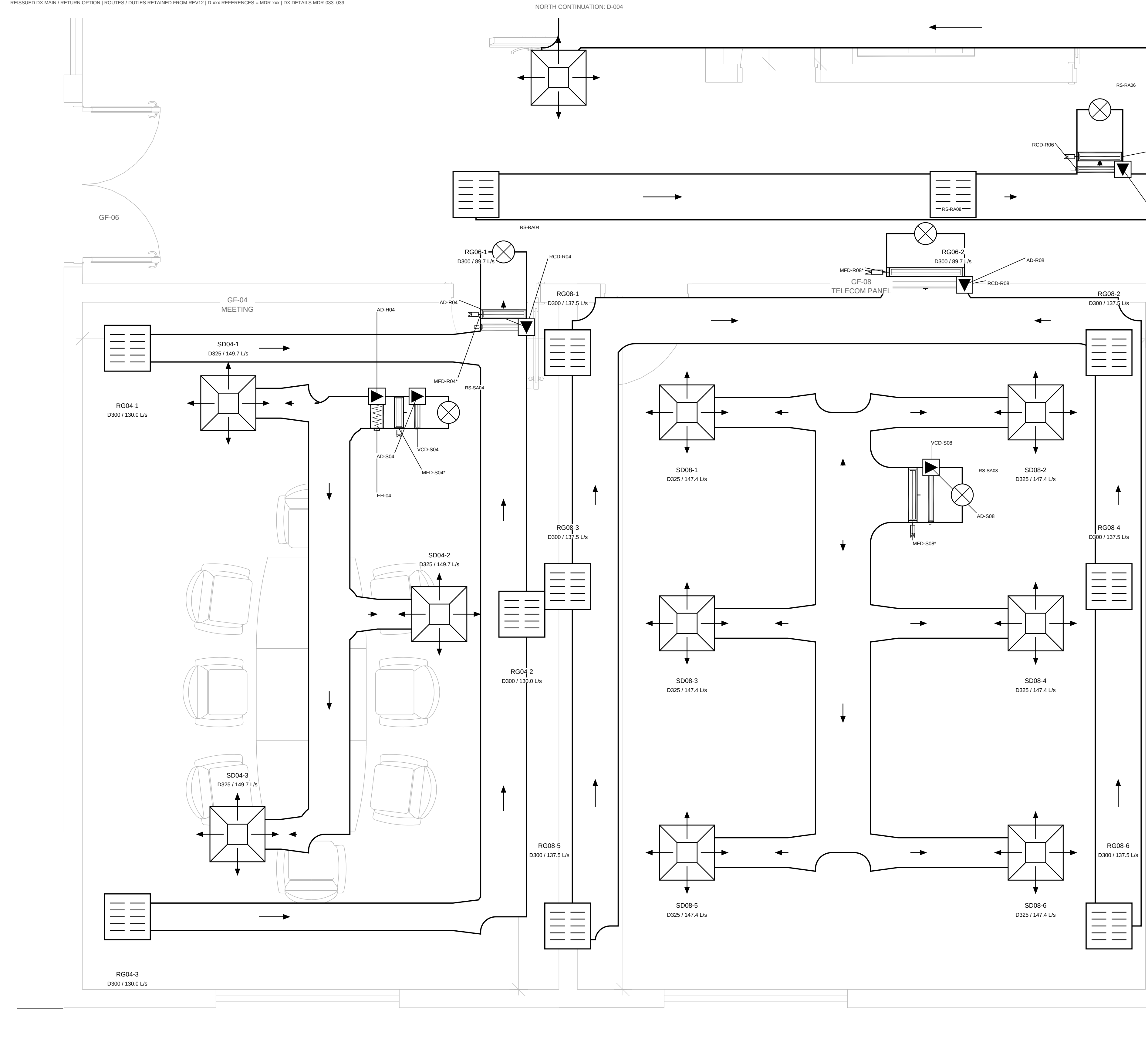
02

08 OCT 2026

SHEET 4 OF 50

MAIN DUCT AND RETURN - SOUTH-WEST ROOMS - DUCT ENLARGEMENT

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039



ROOM BRANCH / TERMINAL DATA

GF-04 MEETING

SA-B04 350x400 mm | 449.20 L/s | 3.21 m/s

RA-B04 500x350 mm | 389.86 L/s | 2.23 m/s

SD: 3 x 149.73 L/s; neck D325; face 600x600

RG: 3 x 129.95 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-08 TELECOM PANEL

SA-B08 600x400 mm | 884.60 L/s | 3.69 m/s

RA-B08 850x400 mm | 825.26 L/s | 2.43 m/s

SD: 6 x 147.43 L/s; neck D325; face 600x600

RG: 6 x 137.54 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-06 CORRIDOR

SA-B06 450x300 mm | 418.20 L/s | 3.10 m/s

RA-B06 500x400 mm | 358.86 L/s | 1.79 m/s

SD: 3 x 139.40 L/s; neck D300; face 600x600

RG: 4 x 89.72 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

DUCT / SYMBOL LEGEND

- SA - supply duct
- RA - return duct
- EA - dedicated extract
- SD - four-way supply diffuser
- RG - ducted return grille
- EG - extract grille
- VCD - balancing damper
- MFD* - conditional fire damper
- RCD - return control damper
- EH - inline electric heater
- AD - access door
- RS - proposed roof supply / return drop

Red arrows show air direction; green outlines are supply and blue outlines are return. Room drops refer to D-022. No indoor service crossovers. Same-service junctions are continuous open tees; headroom envelope: D-020.

GF-04 meeting; GF-08 telecom; GF-06 corridor. Room targets +50 Pa relative to outdoors; actual pressure verification remains pending. Full flow / velocity / horizontal route data is listed on D-010.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SOUTH-WEST ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-005

SCALE: 1:25 AT A1

REVISION

02

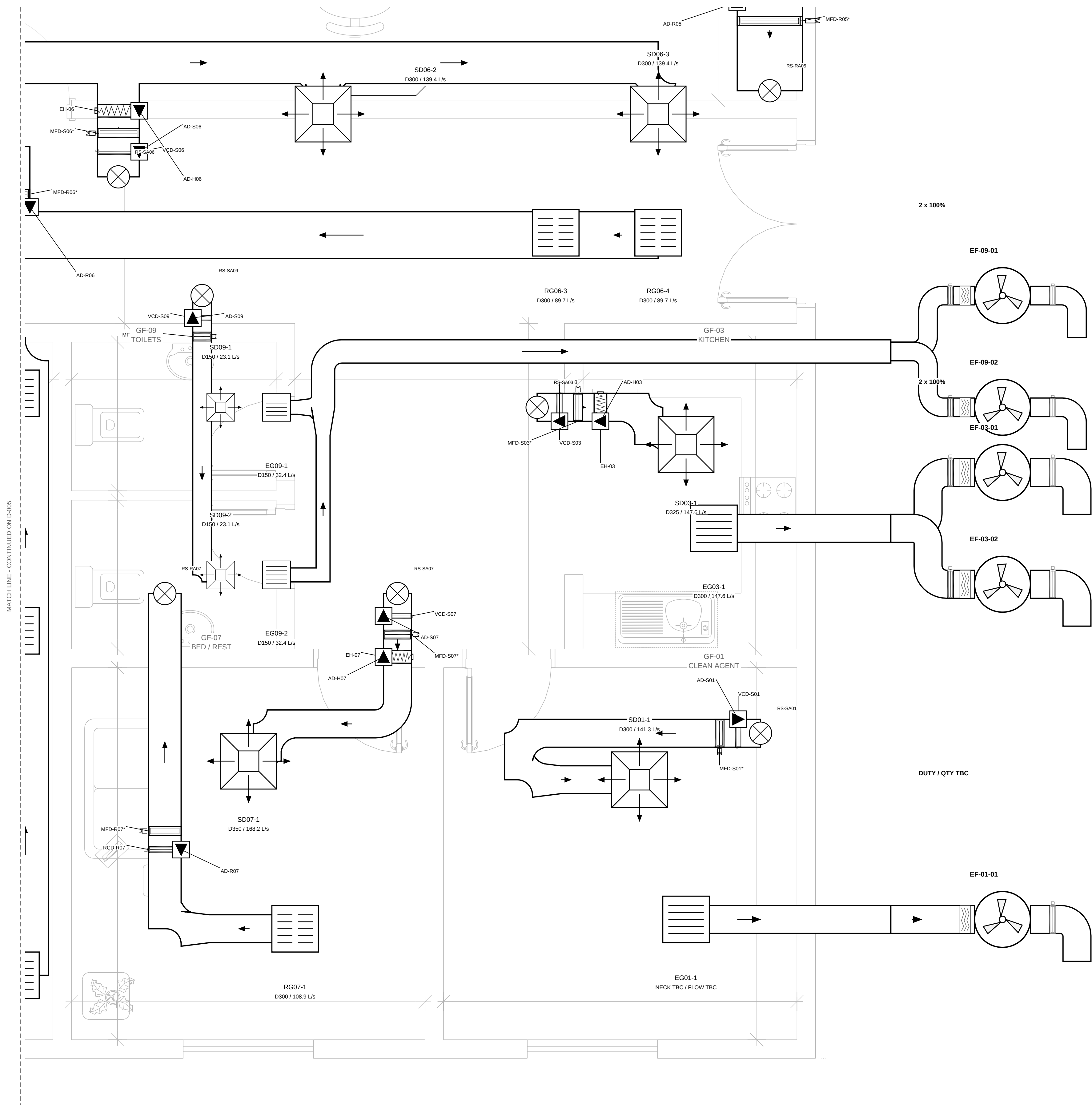
08 OCT 2026

SHEET 5 OF 50

MAIN DUCT AND RETURN - SOUTH-EAST ROOMS - DUCT ENLARGEMENT

REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

NORTH CONTINUATION: D-004



ROOM BRANCH / TERMINAL DATA

GF-09 TOILETS

SA-B09 200x150 mm | 46.20 L/s | 1.54 m/s

EA-B09 250x200 mm | 64.89 L/s | 1.30 m/s

SD: 2 x 23.10 L/s; neck D150; face 300x300

Dedicated extract; no central return

Room header: branch size. Terminal runout: listed neck.

GF-07 BED / REST

SA-B07 300x250 mm | 168.20 L/s | 2.24 m/s

RA-B07 350x250 mm | 108.86 L/s | 1.24 m/s

SD: 1 x 168.20 L/s; neck D350; face 600x600

RG: 1 x 108.86 L/s; neck D300; face 500x500

Room header: branch size. Terminal runout: listed neck.

GF-03 KITCHEN

SA-B03 300x200 mm | 147.60 L/s | 2.46 m/s

EA-B03 300x200 mm | 147.60 L/s | 2.46 m/s

SD: 1 x 147.60 L/s; neck D325; face 600x600

Dedicated extract; no central return

Room header: branch size. Terminal runout: listed neck.

GF-01 CLEAN AGENT

SA-B01 300x200 mm | 141.30 L/s | 2.35 m/s

SD: 1 x 141.30 L/s; neck D300; face 600x600

EA / grille duty and pickup level: TBC

Room header: branch size. Terminal runout: listed neck.

EXTRACT ARRANGEMENTS

EF-09-01/02: toilets, 2 x 100%. EF-03-01/02: kitchen, 2 x 100%. EF-01-01: one CAG fan marker shown; final quantity, airflow and pickup elevation require confirmation.

Kitchen, toilet and CAG targets: +25 Pa to outdoors. Resting room: +50 Pa, with return air. Minimum ventilation: 10 ACH kitchen / toilet; 5 ACH CAG.

Toilet HAP supply is 46.2 L/s; minimum outdoor-air duty is not established by that mixed-air supply. Resolve the fresh-air duty before final HAP and pressure balance.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

SOUTH-EAST ROOMS - DUCT ENLARGEMENT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-006

SCALE: 1:25 AT A1

REVISION

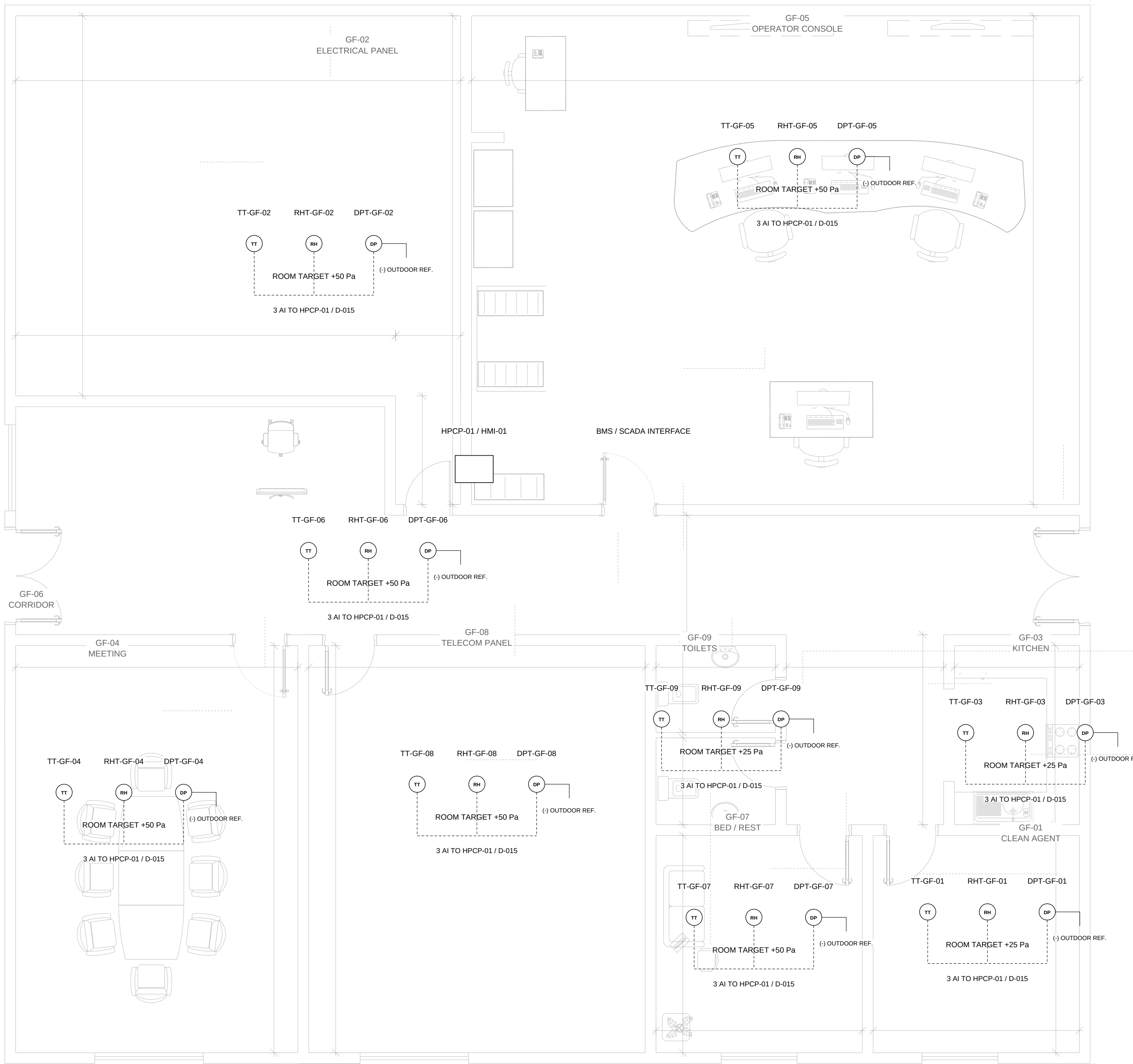
02

08 OCT 2026

SHEET 6 OF 50

MAIN DUCT AND RETURN - ROOM INSTRUMENTATION AND CONTROL LOCATIONS

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039



Room instrument locations are proposed. Plant/filter/fan instruments and heater AFS / TSHH protection: D-012. Final ranges, reference probes, mounting and routes require I&C coordination; operating sequence: S-004.

ROOM FIELD INSTRUMENTS

GF	Temperature	Humidity	Pressure	Pa
02	TT-GF-02	RHT-GF-02	DPT-GF-02	50
05	TT-GF-05	RHT-GF-05	DPT-GF-05	50
04	TT-GF-04	RHT-GF-04	DPT-GF-04	50
08	TT-GF-08	RHT-GF-08	DPT-GF-08	50
06	TT-GF-06	RHT-GF-06	DPT-GF-06	50
07	TT-GF-07	RHT-GF-07	DPT-GF-07	50
09	TT-GF-09	RHT-GF-09	DPT-GF-09	25
03	TT-GF-03	RHT-GF-03	DPT-GF-03	25
01	TT-GF-01	RHT-GF-01	DPT-GF-01	25

MAIN DUCT / PLANT MONITORING

TT-SA-01 / TT-RA-01: main duct temperature

RHT-RA-01: main return humidity

FIT-SA-01 / FIT-RA-01: common airflow

Each PAU filter: PDT remote + PDI local

DPT-SF-A/B/C: PAU fan pressure proof

VS: PAU and ventilation-fan vibration

EM: feeders >7.5 kW, quantity to confirm

Signals report to HPCP-01 dedicated PLC and HMI-01. BMS / SCADA provides the specified monitoring, initial start and emergency-stop interfaces. Protocol, alarm limits, I/O module allocation and actuator fail positions remain pending.

DPT room references are outdoors, not the adjacent room. DPT-GF-06 is the corridor / building reference. Additional room DPTs and CAG humidity sensing are marked as added provisions in the register.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOM INSTRUMENTATION AND CONTROL LOCATIONS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-007

SCALE: 1:50 AT A1

REVISION

02

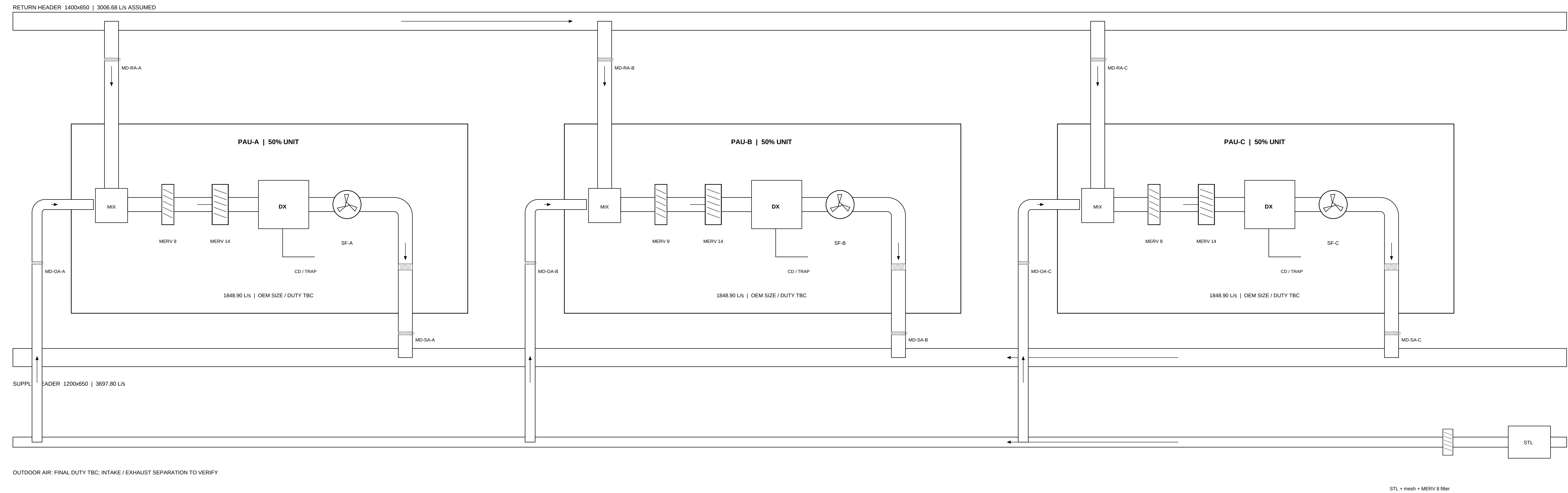
08 OCT 2026

SHEET 7 OF 50

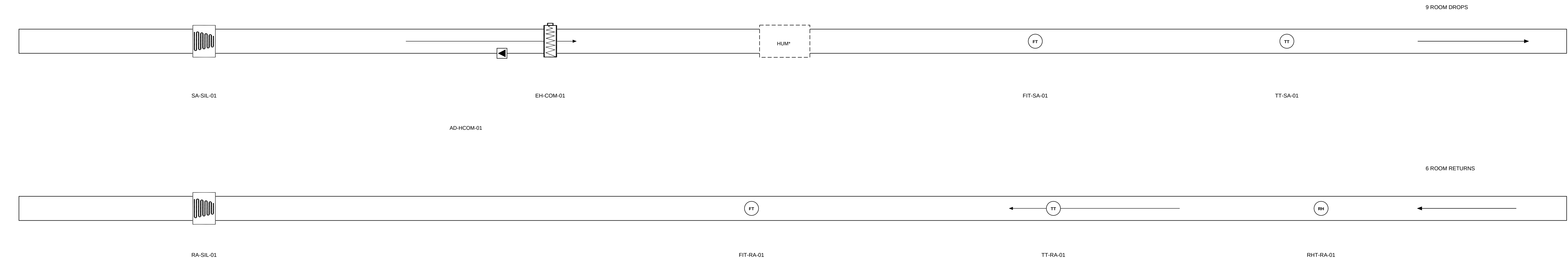
MAIN DUCT AND RETURN - PAU BANK AND PROPOSED ROOF COMMON DISTRIBUTION

REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

A ROOF CONNECTION ARRANGEMENT - SCHEMATIC FOOTPRINTS



B ROOF COMMON PLANT / DISTRIBUTION OPTION - FUNCTIONAL NTS



Common heater / conditional humidifier move to a weather-protected roof plant enclosure in this option. Equipment suitability, safety, intake separation, maintenance space and outdoor insulation require OEM / project specification. Actual roof geometry and vertical development: TBC.

ROOF OPTION, NOT CONSTRUCTION ISSUE. Functional connections shown NTS; do not scale as roof routing. Each room drop and structural penetration is proposed, D-022. Common instrument signals remain on D-012 / S-001.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

PAU BANK AND PROPOSED ROOF COMMON DISTRIBUTION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-008

SCALE: NTS AT A1

REVISION

02

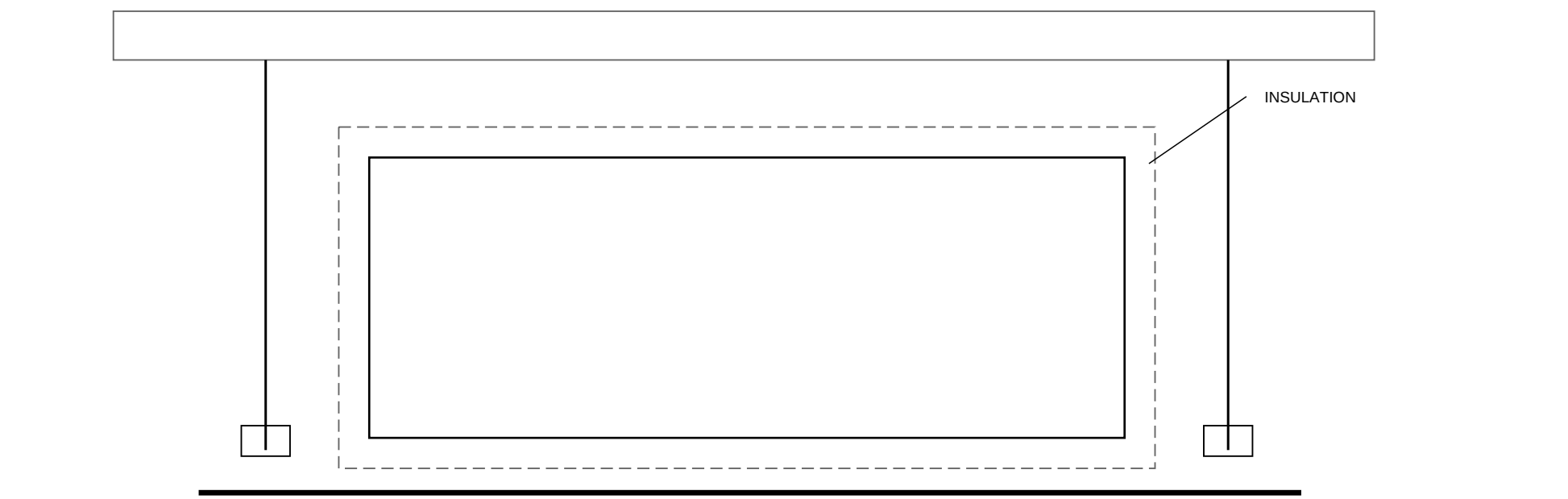
08 OCT 2026

SHEET 8 OF 50

MAIN DUCT AND RETURN - DUCT SYMBOLS AND INSTALLATION DETAILS

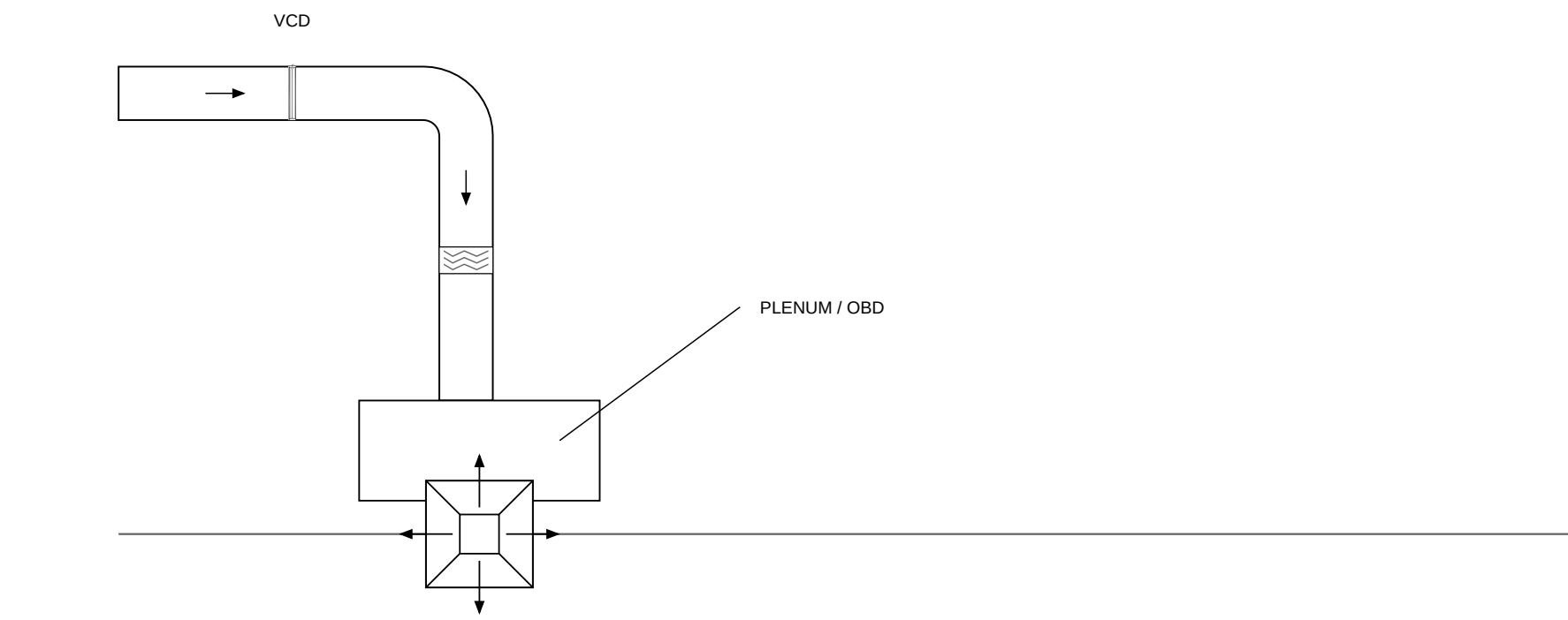
REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

01 DUCT SUPPORT / INSULATION



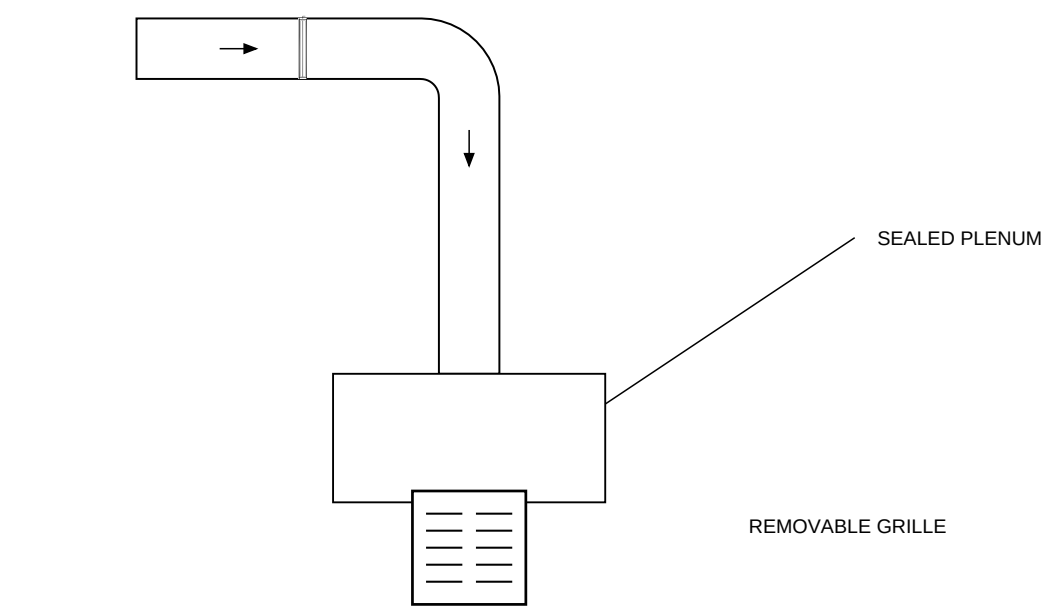
Anchor loads, support spacing, insulation and vapor barrier: approved fabrication / structural selection.

02 SUPPLY DIFFUSER CONNECTION



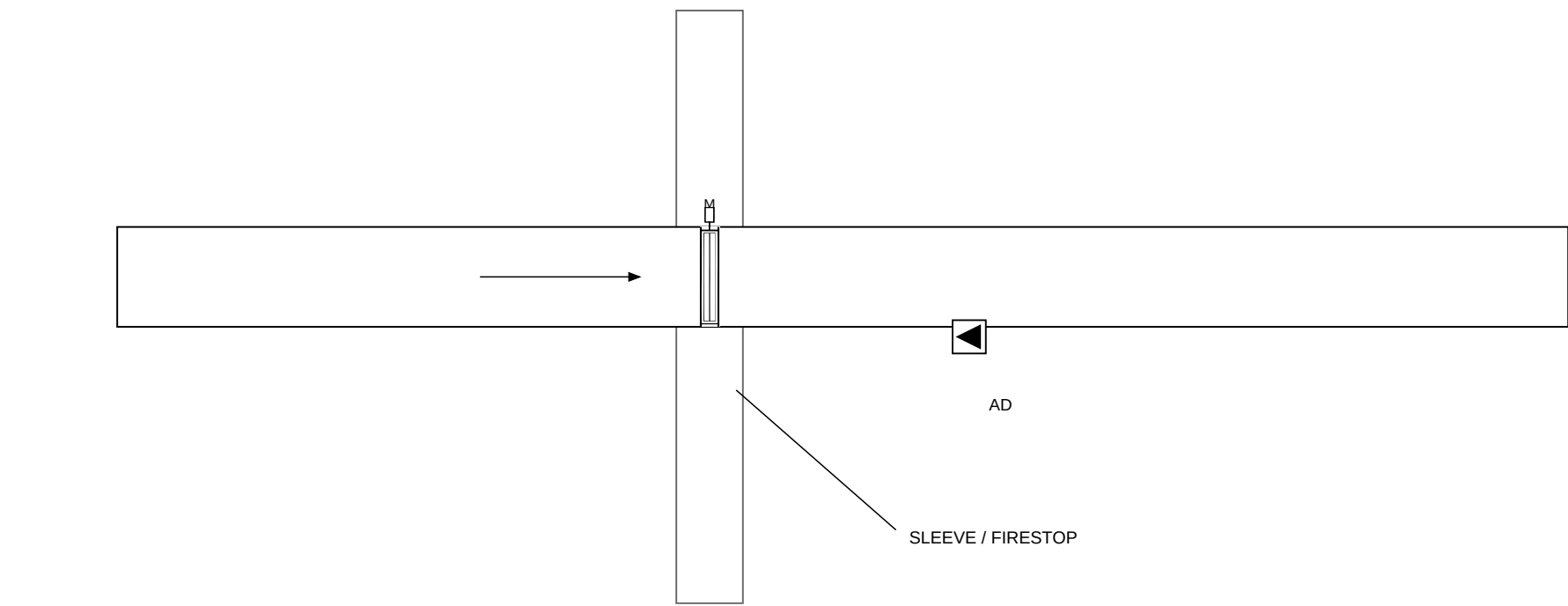
Neck <=2 m/s. Face, throw, noise and loss from selected diffuser catalogue; accessible balancing required.

03 RETURN / EXTRACT GRILLE



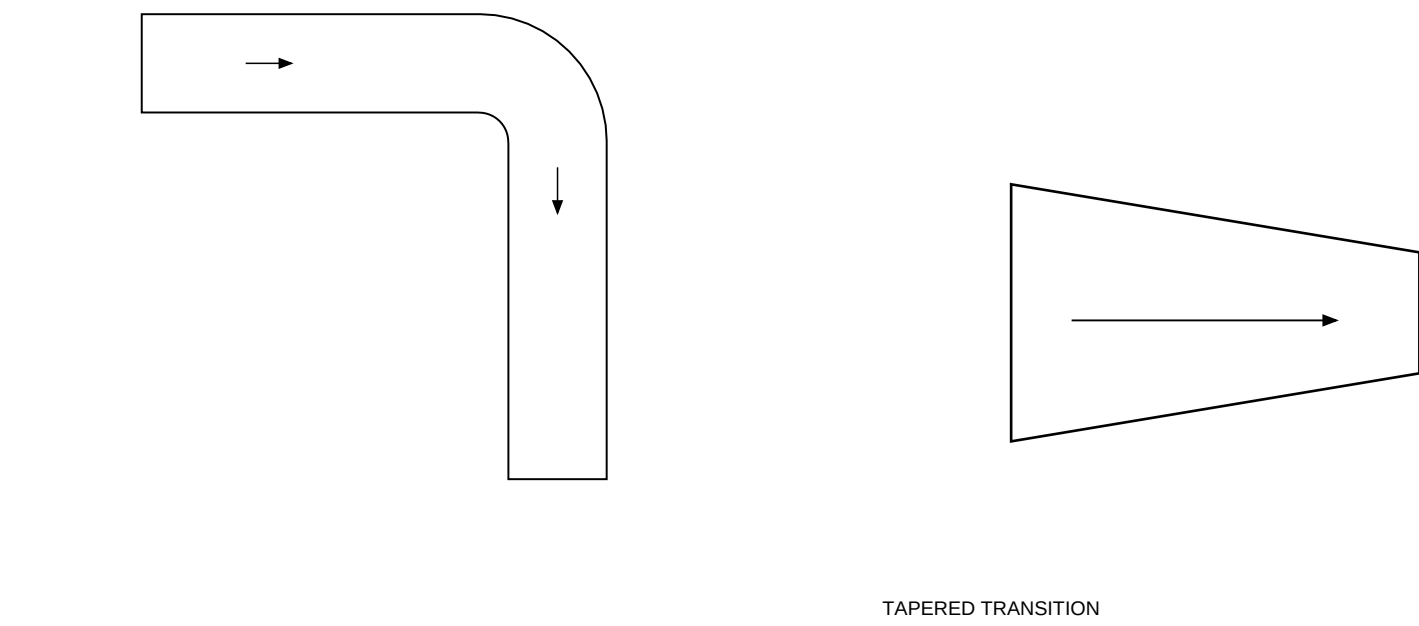
Return grilles connect to RA duct. Extract grilles connect to their dedicated outdoor duct; pickup height TBC.

04 RATED PENETRATION / MFD*



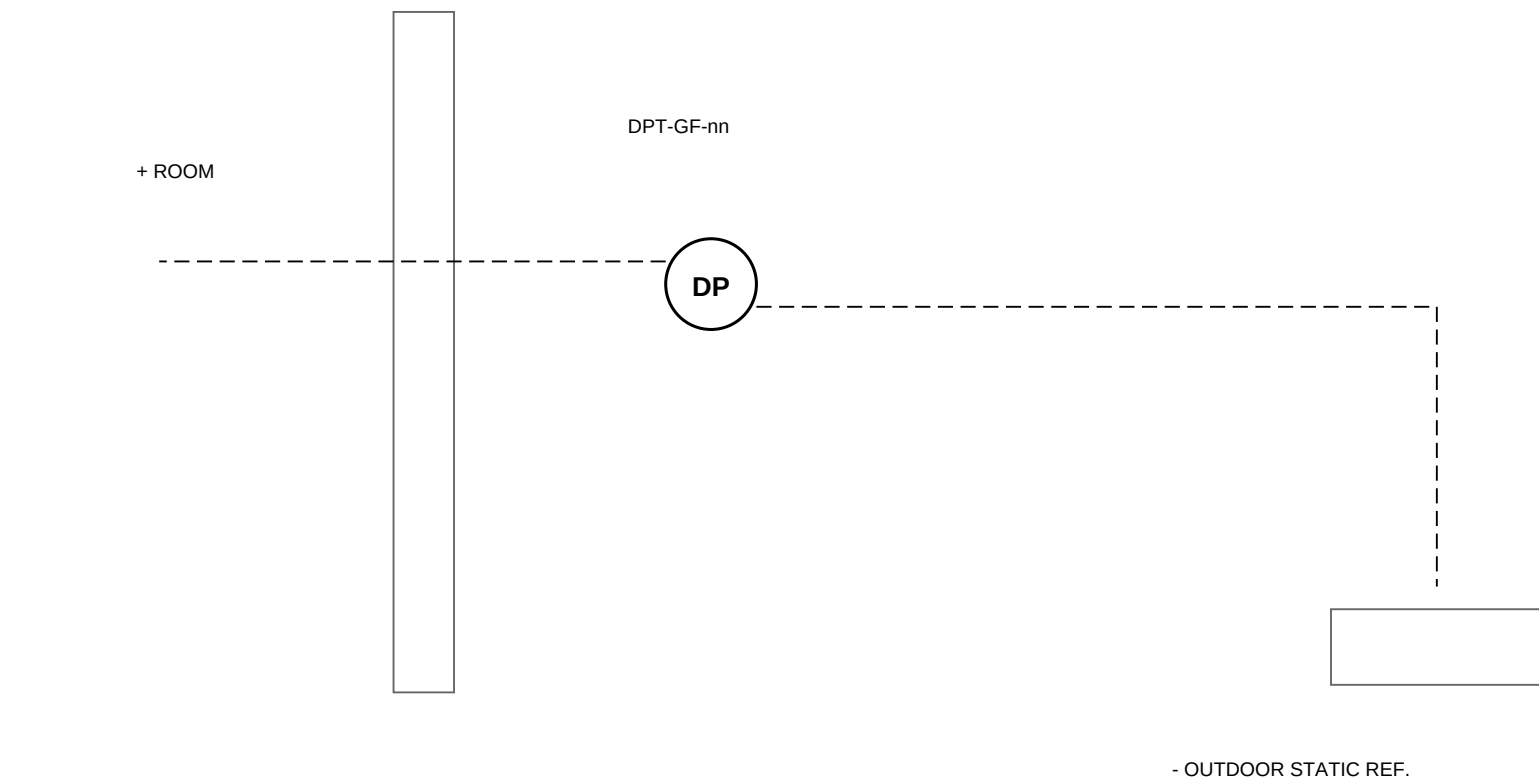
MFD* only at approved rated boundaries. Select tested sleeve / firestop assembly and allow actuator / inspection access.

05 BEND / TAPER TRANSITION



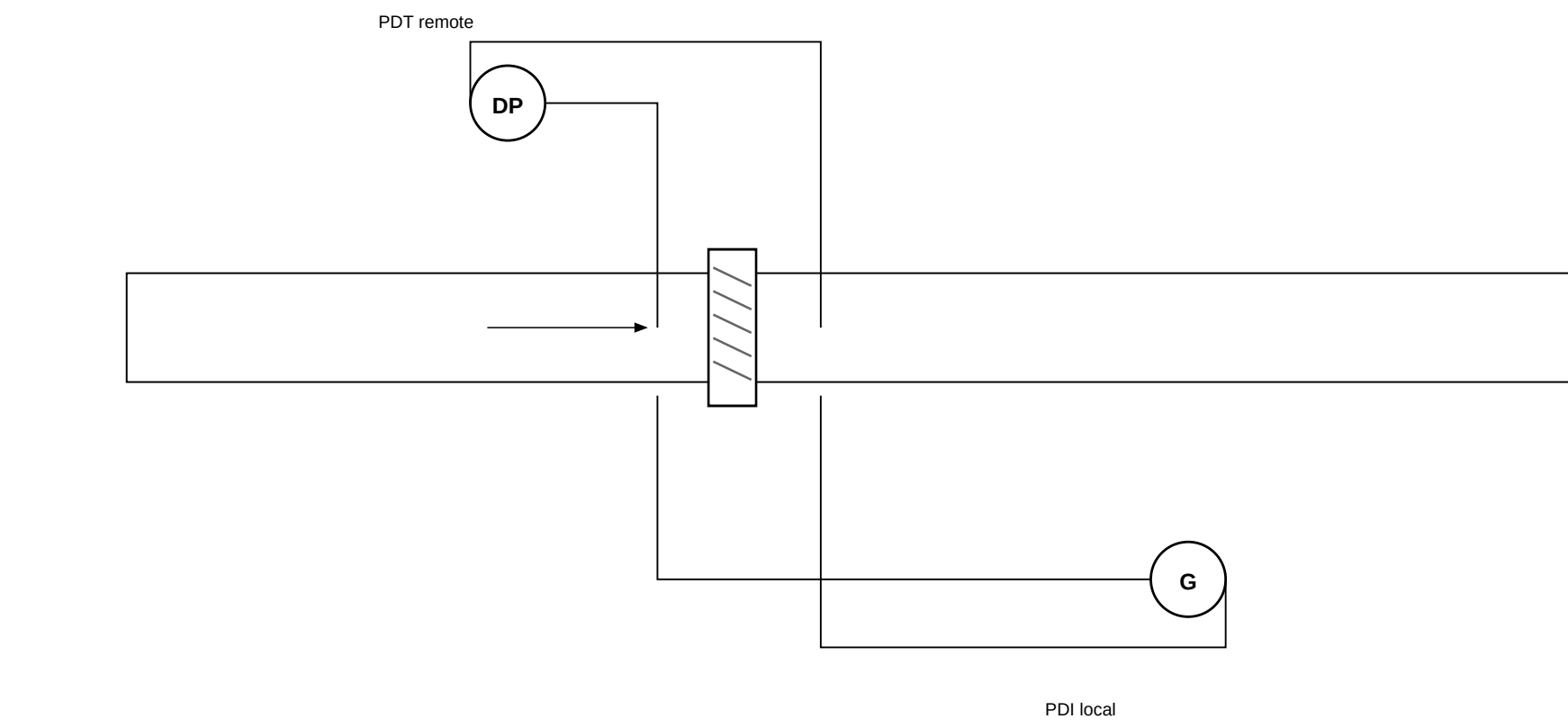
Radius bends or miter bends / turning vanes to selected fitting design. Final loss coefficients and development TBC.

06 ROOM PRESSURE SENSING



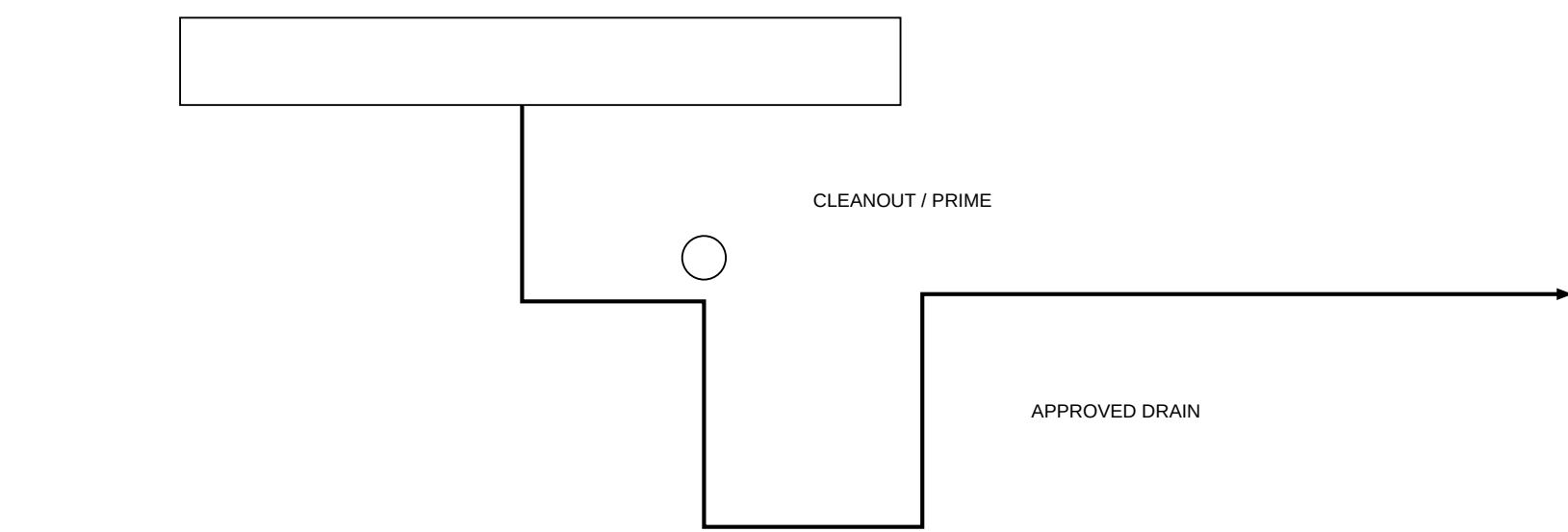
Weather-shielded outdoor static reference; route tubes to minimize wind / water effects. Range, alarm limits and mounting TBC.

07 FILTER DP MONITORING



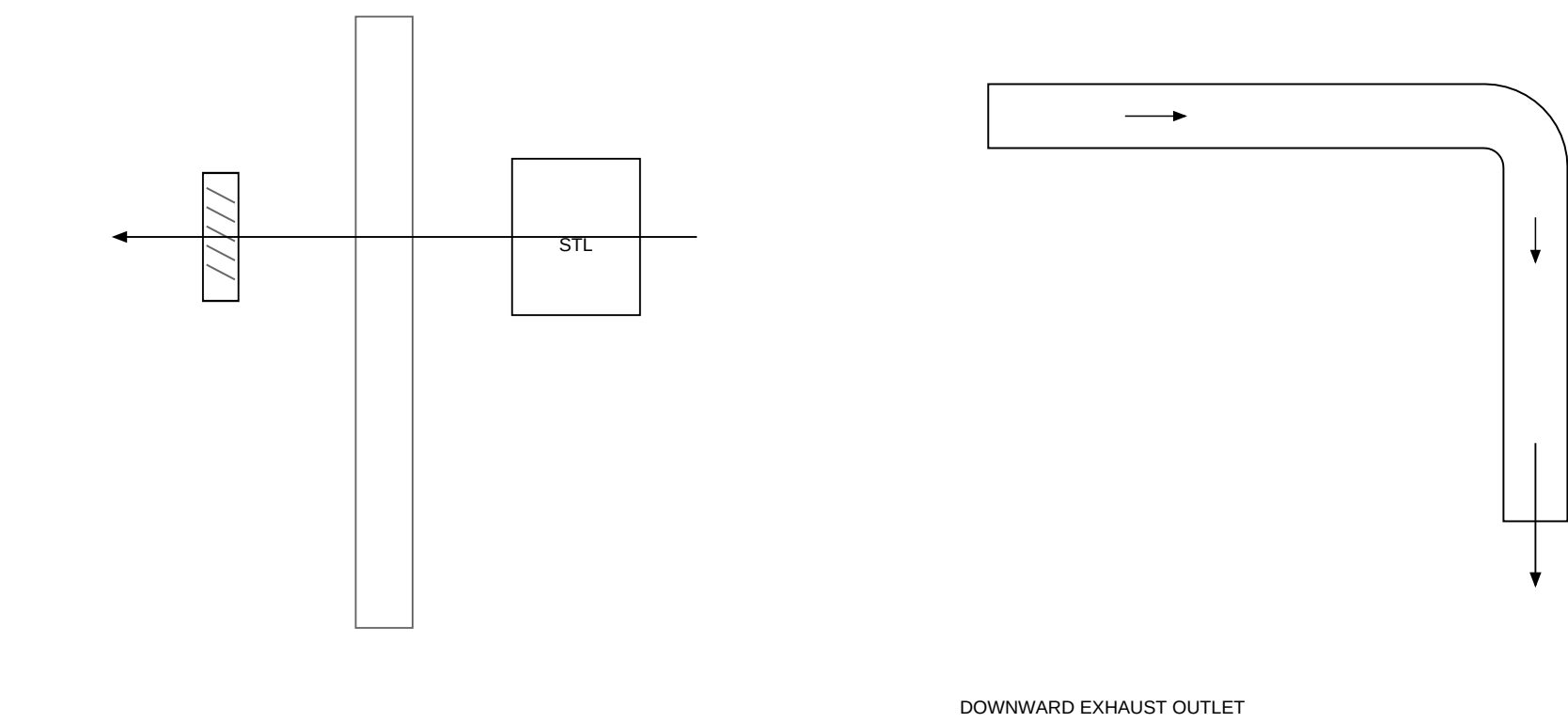
Each PAU filter bank has remote DP and local gauge. Upstream / downstream taps; dirty-filter threshold by OEM.

08 COIL DRAIN / TRAP



Trap seal / invert from actual unit casing pressure. Drain slope, disposal route, cleanout and water seal by OEM / plumbing.

09 INTAKE / EXHAUST OUTLET



STL face <=1 m/s; exhaust louver <=2.5 m/s. Final face areas, loss, materials and intake separation require selection.

Details are NTS engineering concepts. Duct colours / symbols match the plans; body dimensions are not manufacturer selections. No room pressure, fire rating or equipment clearance is certified by this drawing.

MAIN DUCT AND RETURN - DUCT AND TERMINAL SCHEDULES

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

MAIN AND ROOM BRANCH SECTIONS

Section	Size mm	L/s	m/s	Lh m*	Pa/m
SA-B02	500x350	664.80	3.80	1.20	0.399
RA-B02	650x400	605.46	2.33	0.95	0.130
SA-B04	350x400	449.20	3.21	1.30	0.330
RA-B04	500x350	389.86	2.23	0.95	0.151
SA-B08	600x400	884.60	3.69	1.30	0.313
RA-B08	850x400	825.26	2.43	0.95	0.125
SA-B05	600x350	777.70	3.70	1.90	0.349
RA-B05	700x400	718.36	2.57	1.00	0.150
SA-B06	450x300	418.20	3.10	1.23	0.324
RA-B06	500x400	358.86	1.79	0.95	0.093
SA-B09	200x150	46.20	1.54	0.60	0.231
SA-B07	300x250	168.20	2.24	2.80	0.254
RA-B07	350x250	108.86	1.24	2.80	0.081
SA-B03	300x200	147.60	2.46	1.20	0.351
SA-B01	300x200	141.30	2.35	2.60	0.325
EA-B09	250x200	64.89	1.30	6.70	0.123
EA-B03	300x200	147.60	2.46	1.90	0.351
EA-B01	TBC	TBC	TBC	TBC	TBC

* Lh is the source orthogonal horizontal route allowance. Final straight lengths, selected elbow development, risers / drops and fitting losses require coordination. Vertical lengths and BOD are unverified.

Sizes are clear internal dimensions. Catalogue free area, diffuser throw / noise, coil / filter / louver / silencer losses, fitting coefficients and OEM fan curves are pending. Confirmed fire stops the entire HVAC.

TERMINAL GROUPS

GF	Type	Qty	Each L/s	Neck mm	Face mm
02	SD	4	166.20	D350	600x600
02	RG	4	151.37	D300	500x500
05	SD	5	155.54	D325	600x600
05	RG	5	143.67	D300	500x500
04	SD	3	149.73	D325	600x600
04	RG	3	129.95	D300	500x500
08	SD	6	147.43	D325	600x600
08	RG	6	137.54	D300	500x500
06	SD	3	139.40	D300	600x600
06	RG	4	89.72	D300	500x500
07	SD	1	168.20	D350	600x600
07	RG	1	108.86	D300	500x500
09	SD	2	23.10	D150	300x300
09	EG	2	32.44	D150	300x300
03	SD	1	147.60	D325	600x600
03	EG	1	147.60	D300	500x500
01	SD	1	141.30	D300	600x600
01	EG	1	TBC	TBC	500x500

SYSTEM / DESIGN BASIS

Total supply: 3697.80 L/s. Assumed return: 3006.68 L/s.

Six return rooms: 59.34 L/s net outward allowance each, based on the accepted 0.01 m2 illustrative equivalent area.

Final room pressure, IAQ outdoor-air duty and revised HAP loads remain unverified. Current 420.70 L/s OA is below the 691.12 L/s assumed-balance makeup requirement.

All 116 duct sections, 37 fitting entries, 53 terminals, 176 components and 85 instruments are in the editable registers.

Rev11: routing / fitting topology revised, changed sizes and proposed terminal relocations are registered on D-021. Same-service overlaps and duplicated centerlines are rejected.

MAIN DUCT AND RETURN - TAPERED FITTINGS AND MAINTENANCE ACCESS

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

01 CONTINUOUS MAIN REDUCER

Taper replaces the straight duct profile. Both width and height change; actual development / K from selected fitting.

02 FLARED ROOM-BRANCH BOOT

Side connection has a tapered throat and open main wall. Place balancing / fire devices on the branch straight.

03 RECTANGULAR TO ROUND ADAPTOR

PLAN
HEIGHT / ROUND CONVERSION: OEM

Rectangular header to round neck / plenum conversion. Short direct drops use an integral adaptor: SD07-1 / RG05-5; section D-009.

04 TYPICAL BRANCH STATIONS - FLOW DIRECTION / ACCESS

SUPPLY: boot -> VCD -> conditional MFD* -> room heater where scheduled -> terminals

RETURN: room grilles -> RCD / inspection access -> conditional MFD* -> return main

05 ACCESS / PROTECTION

AD is a duct maintenance opening, not an inline damper. Access side and ceiling hatch must reach the damper blade / actuator, heater and inspection points.

AFS proves heater airflow; TSHH independently trips on high temperature. OEM hardwired safety inhibits heating; status reports to HPCP. Wiring: D-012 / S-004.

MAIN TAPER SCHEDULE - PROPOSED GEOMETRY

Fitting	Inlet mm	Outlet mm	L mm*
---------	----------	-----------	-------

ACCESS-DOOR TAGS / BRANCH MAP

GF	Supply access	Return access	Heater access
02	AD-S02	AD-R02	-
05	AD-S05	AD-R05	AD-H05
04	AD-S04	AD-R04	AD-H04
08	AD-S08	AD-R08	-
06	AD-S06	AD-R06	AD-H06
07	AD-S07	AD-R07	AD-H07
09	AD-S09	-	-
03	AD-S03	-	AD-H03
01	AD-S01	-	-
COM	AD-HCOM-01	-	Common heater

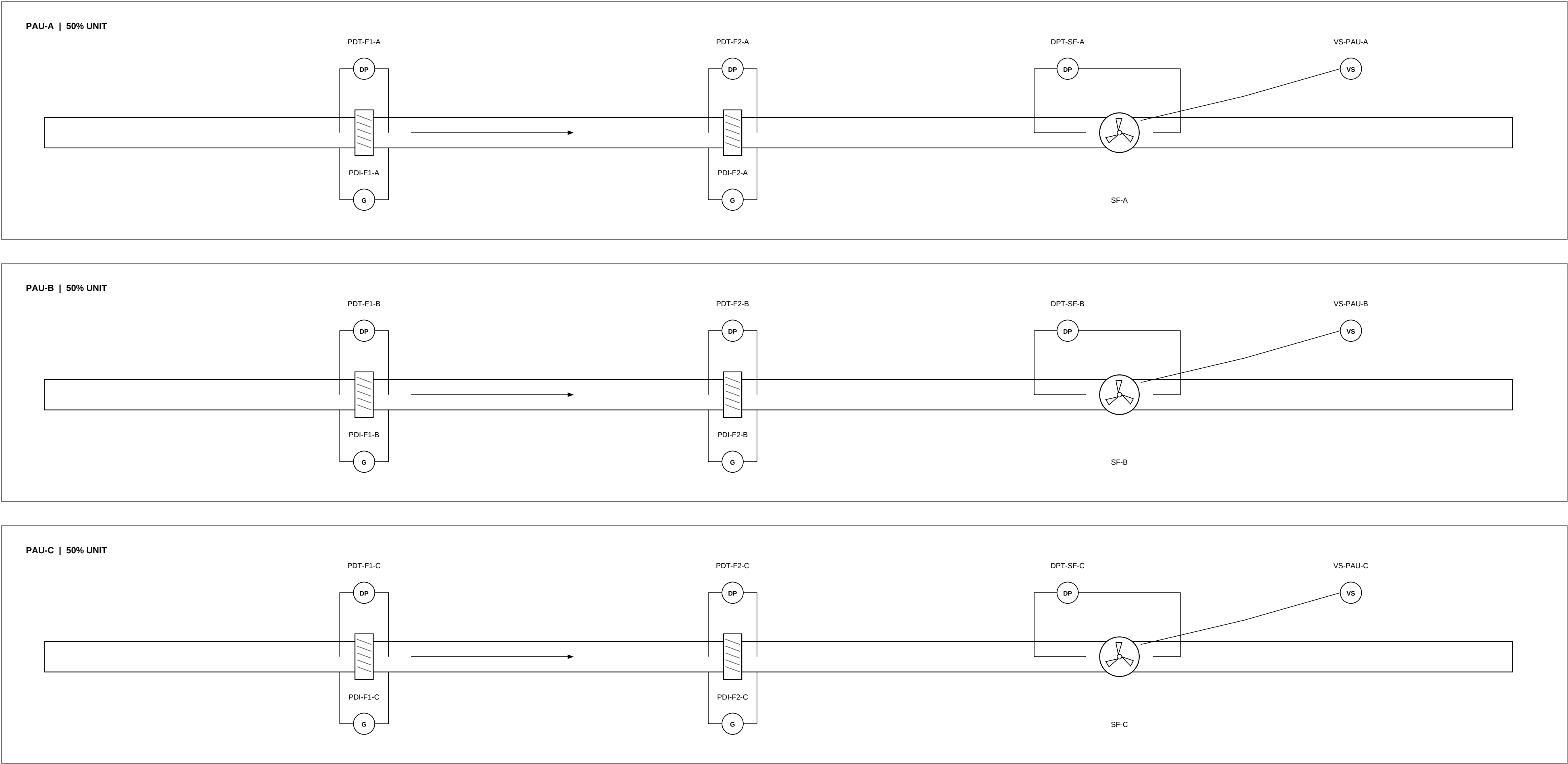
20 indoor AD locations plus AD-HCOM-01 roof plant access are registered. Door clear opening, sealing / insulation, access hatch and OEM working envelope remain to be selected.

* Taper lengths are explicit proposed coordination geometry, not approved fabrication lengths. The indoor register includes terminal adaptors. Main reducers and branch boots above are roof functional typicals; actual roof fittings and vertical drop details remain pending.

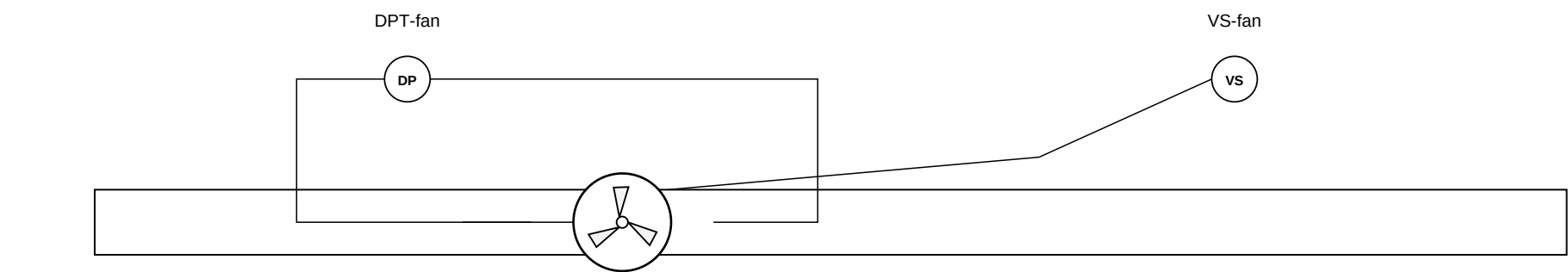
MAIN DUCT AND RETURN - PLANT INSTRUMENTATION AND HEATER PROTECTION

REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

A EACH PAU: FILTER DIFFERENTIAL PRESSURE / FAN PROOF / VIBRATION



C EXTRACT / BASEMENT FAN PROOF

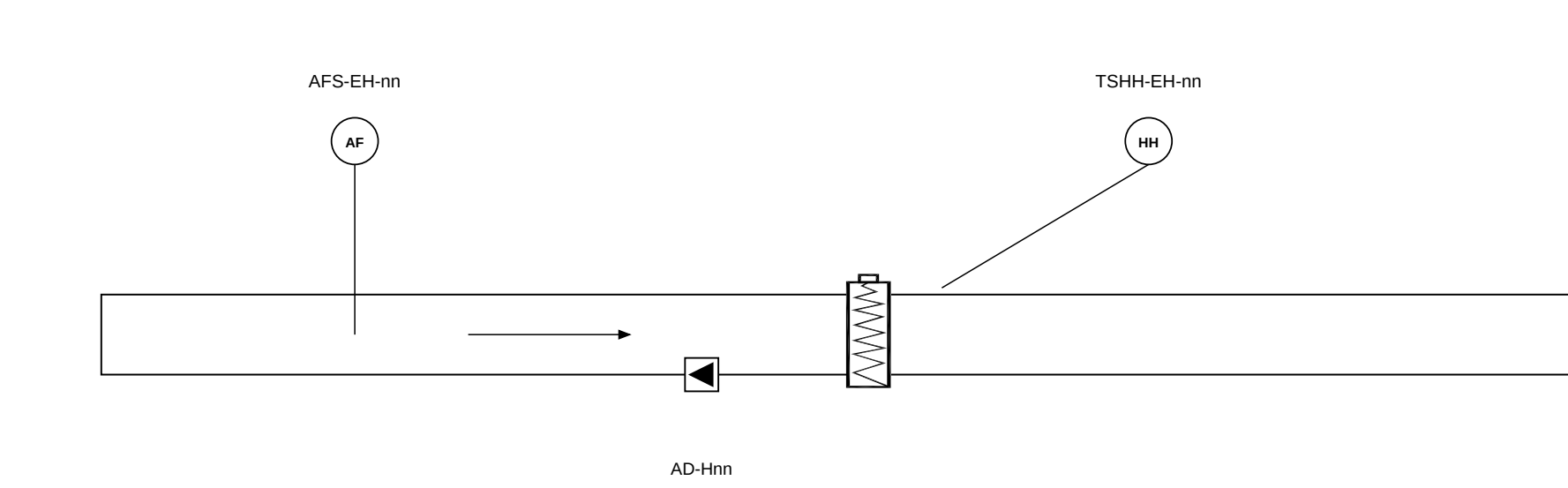


Tag suffixes: EF-03-01/02; EF-09-01/02; EF-01-01. Each has DPT- and VS- tags. Basement: SF-B1-01/02 and EF-B1-01/02 use the same convention.

Fan start / run / fault / ready / local-auto / E-stop and damper proofs are listed in I/O. CAG duty / quantity and basement duties remain pending.

Instrument glyphs are functional symbols, not equipment dimensions. AFS additions are explicit proposed OEM protection / monitoring provisions. Full tags: Excel / CSV registers. Room TT / RHT / DPT locations: D-007.

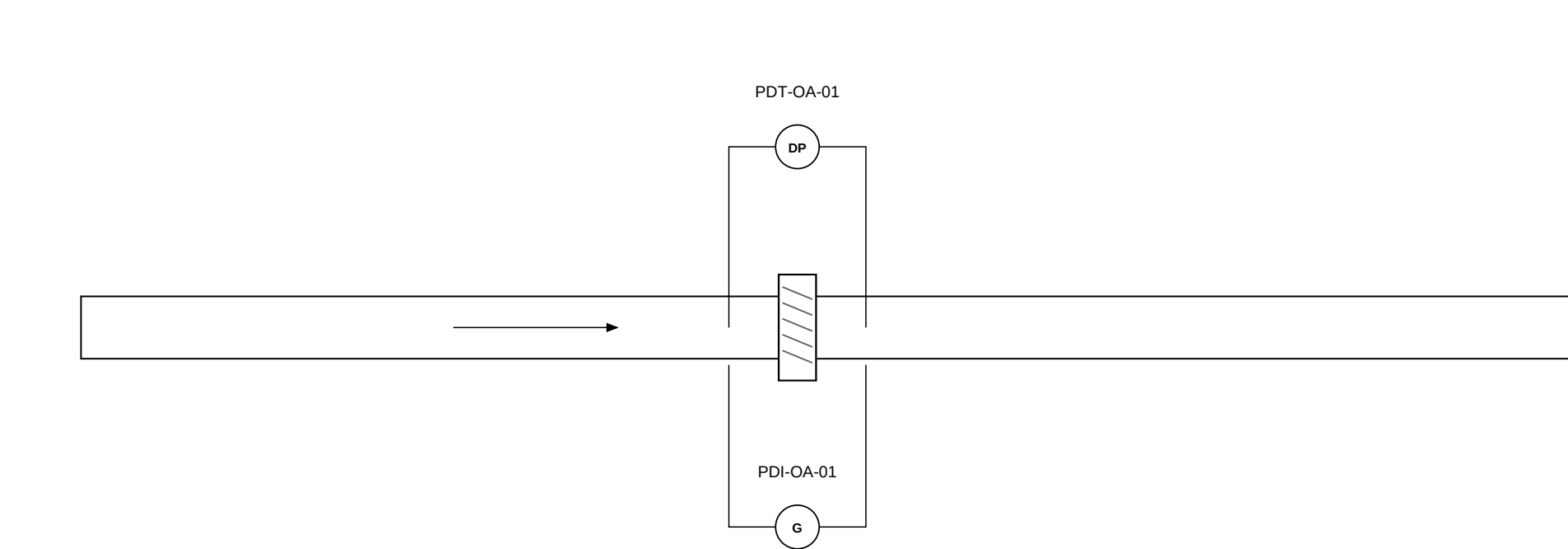
D HEATER AIRFLOW / HIGH-LIMIT SAFETY



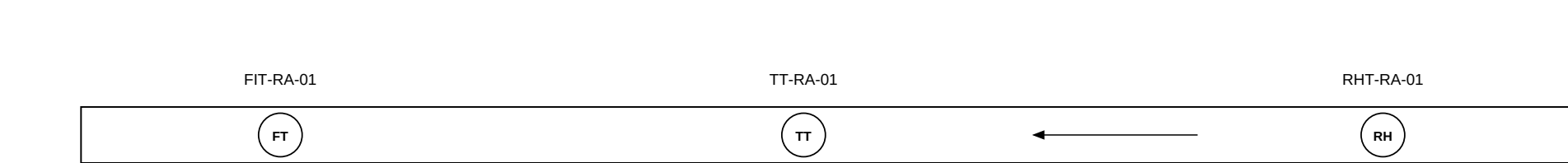
nn = 04, 05, 06, 07, 03, COM-01. Six AFS DI points are retained from Rev07. Each heater retains independent TSHH; loss of proof / high limit switches heating off.

AFS proof technology, setpoint and OEM hardwired wiring to be selected. PLC demand cannot override local safety.

B OUTDOOR-AIR FILTER / COMMON DUCTS



PDT = AI TO HPCP; PDI = LOCAL GAUGE



Filter taps measure pressure across each filter, not between rooms. Fan DP proves the operating fan path; VS monitors fan / unit vibration. Alarm thresholds and OEM signal type TBC.

E PRELIMINARY HPCP I/O

Type	Points	Spare >=20%	Channels
AI	65	13	78
AO	22	5	27
DI	141	29	170
DO	44	9	53

COMM: 3 interfaces; separate from wired channel totals. Signals report to dedicated HPCP-01 PLC / HMI-01 and BMS / SCADA. Final protocol and allocation TBC.

CONFIRMED FIRE: stop every HVAC fan and heater. No automatic fire extraction / post-fire purge. Reset and damper fire positions follow the approved cause-and-effect.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

PLANT INSTRUMENTATION AND HEATER PROTECTION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-012

SCALE: NTS AT A1

REVISION

02

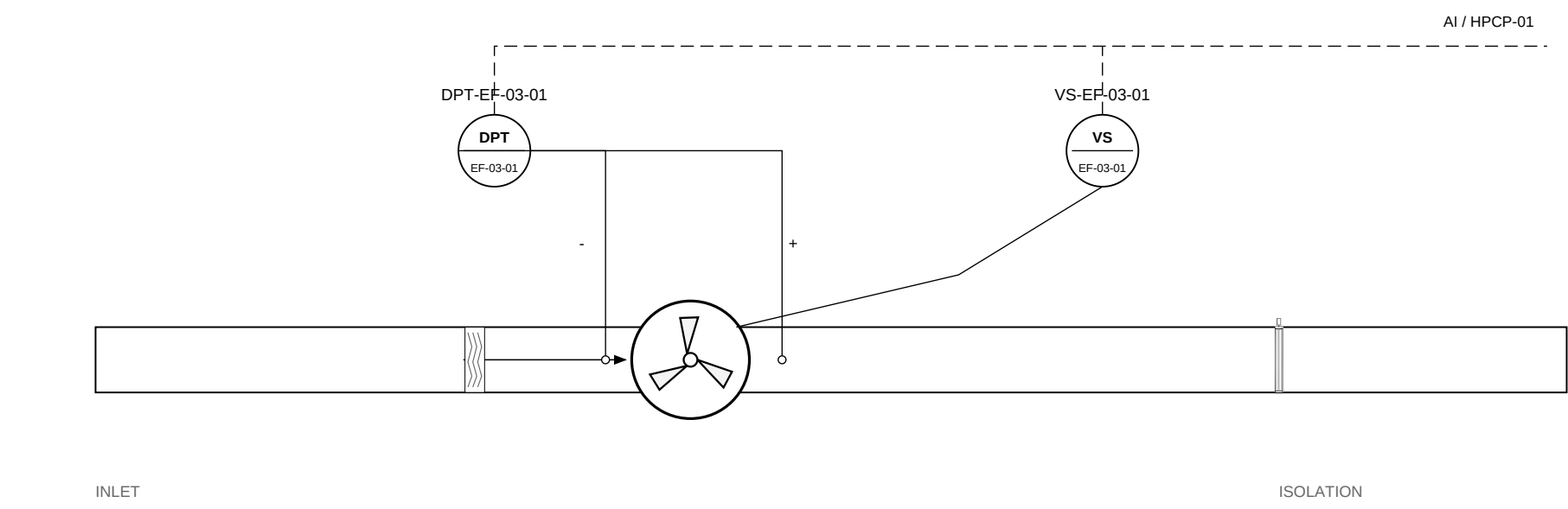
08 OCT 2026

SHEET 12 OF 50

MAIN DUCT AND RETURN - EXTRACT AND BASEMENT FAN INSTRUMENTATION

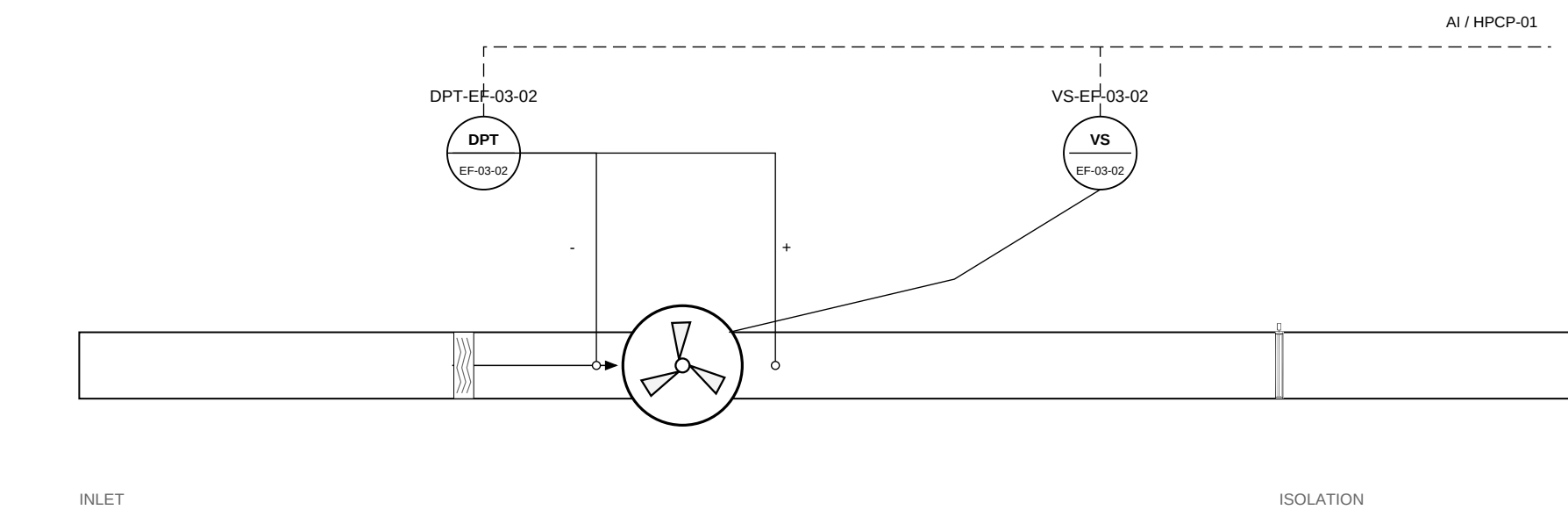
REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

01 EF-03-01



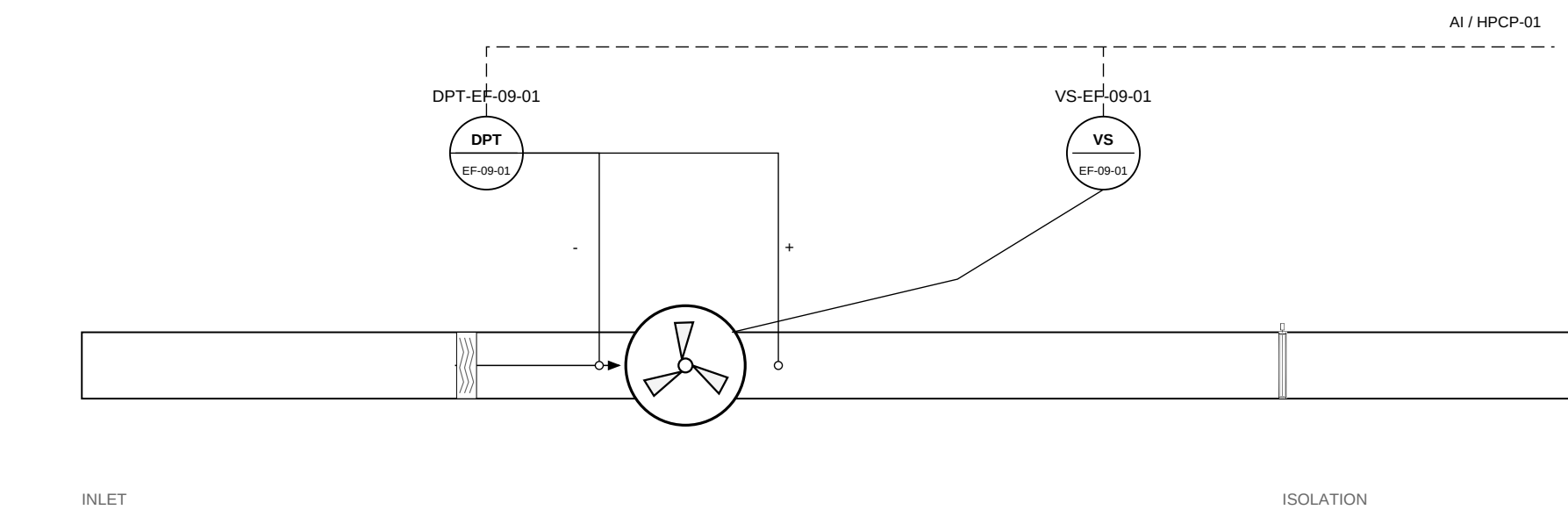
DP taps across fan. VS attachment at OEM bearing / frame. Isolation tag and fail position: approved control register.

02 EF-03-02



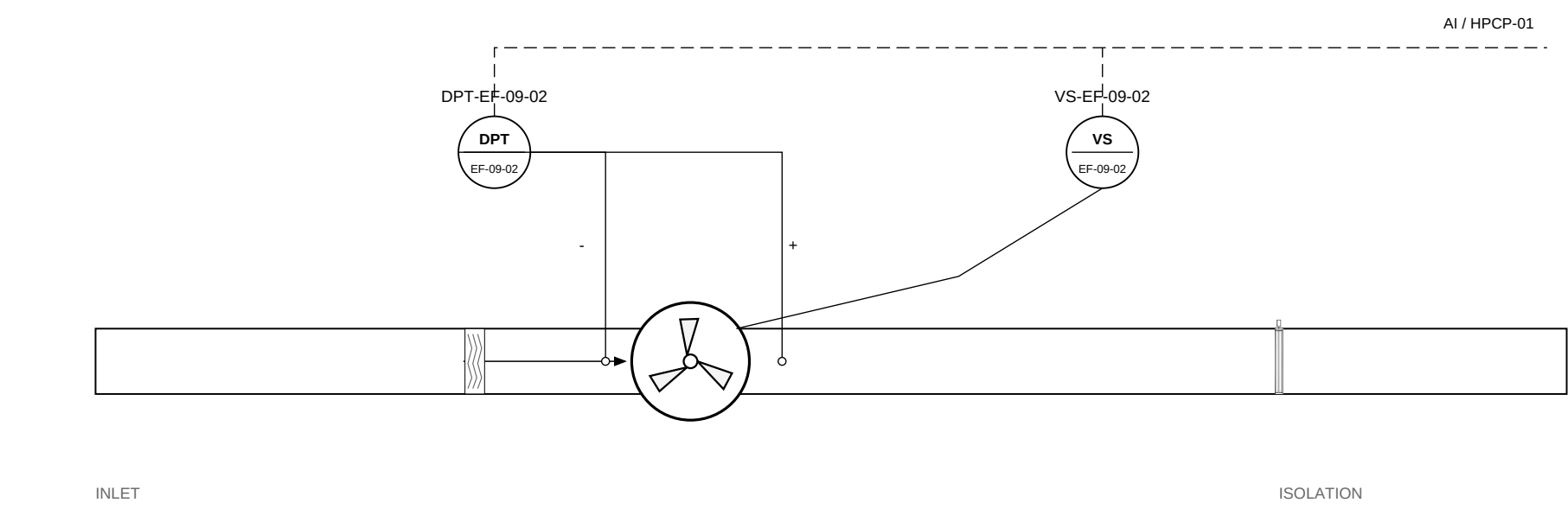
DP taps across fan. VS attachment at OEM bearing / frame. Isolation tag and fail position: approved control register.

03 EF-09-01



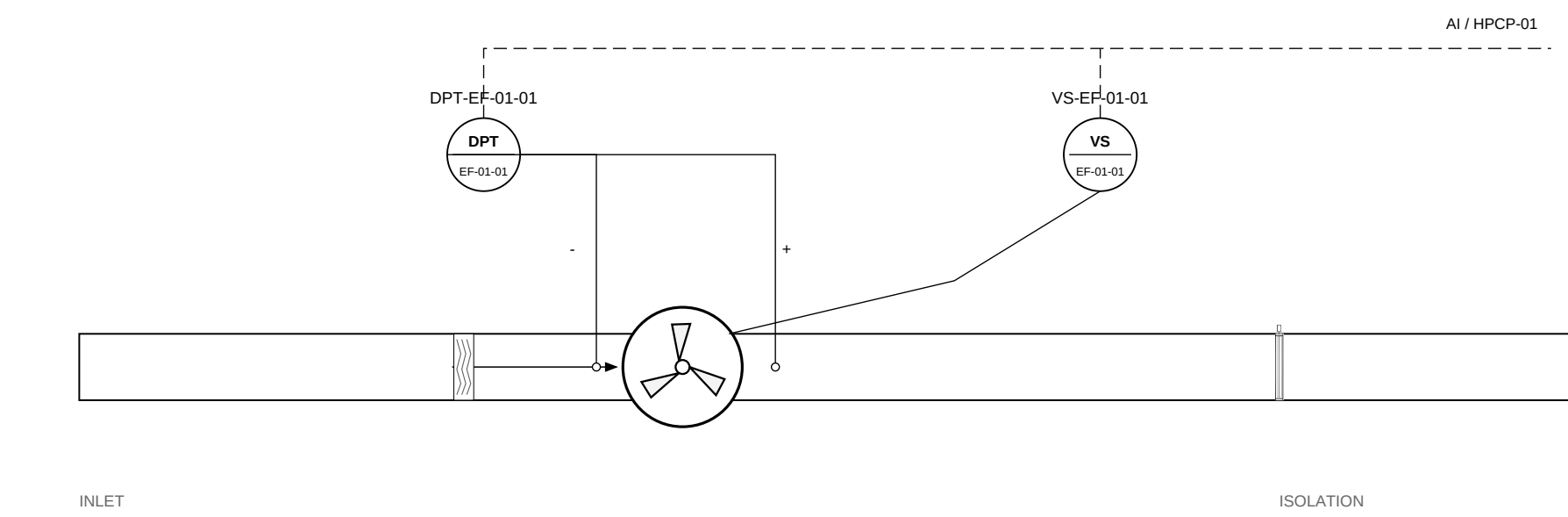
DP taps across fan. VS attachment at OEM bearing / frame. Isolation tag and fail position: approved control register.

04 EF-09-02



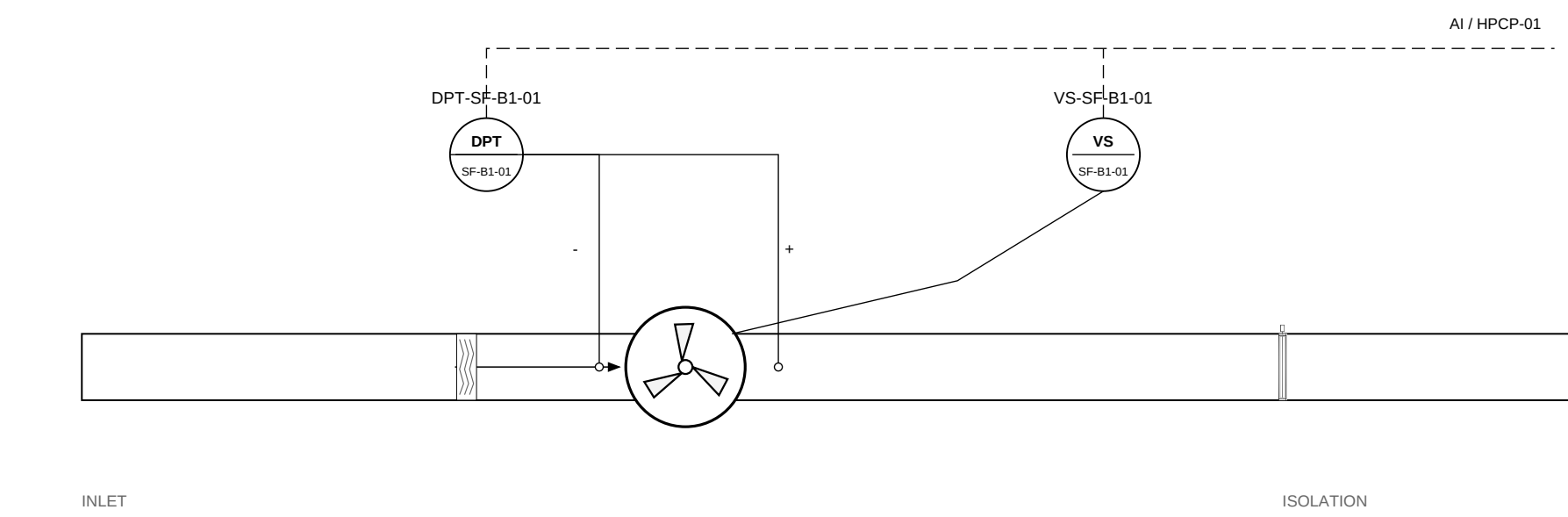
DP taps across fan. VS attachment at OEM bearing / frame. Isolation tag and fail position: approved control register.

05 EF-01-01



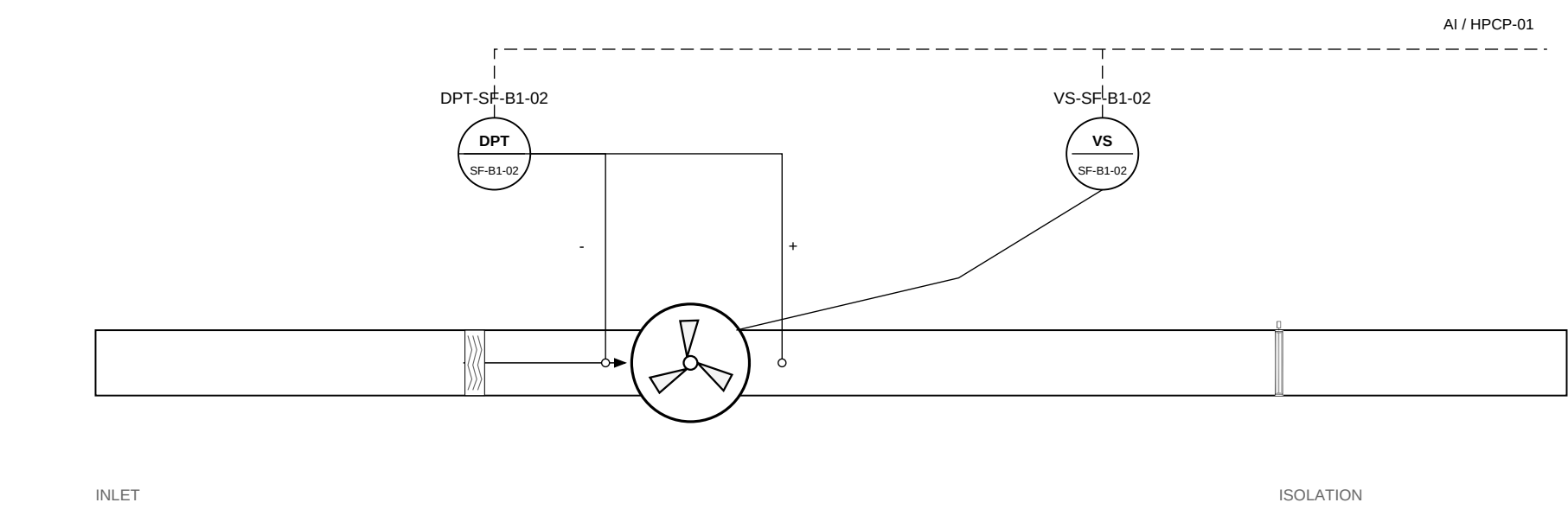
DP taps across fan. VS attachment at OEM bearing / frame. Isolation tag and fail position: approved control register.

06 SF-B1-01



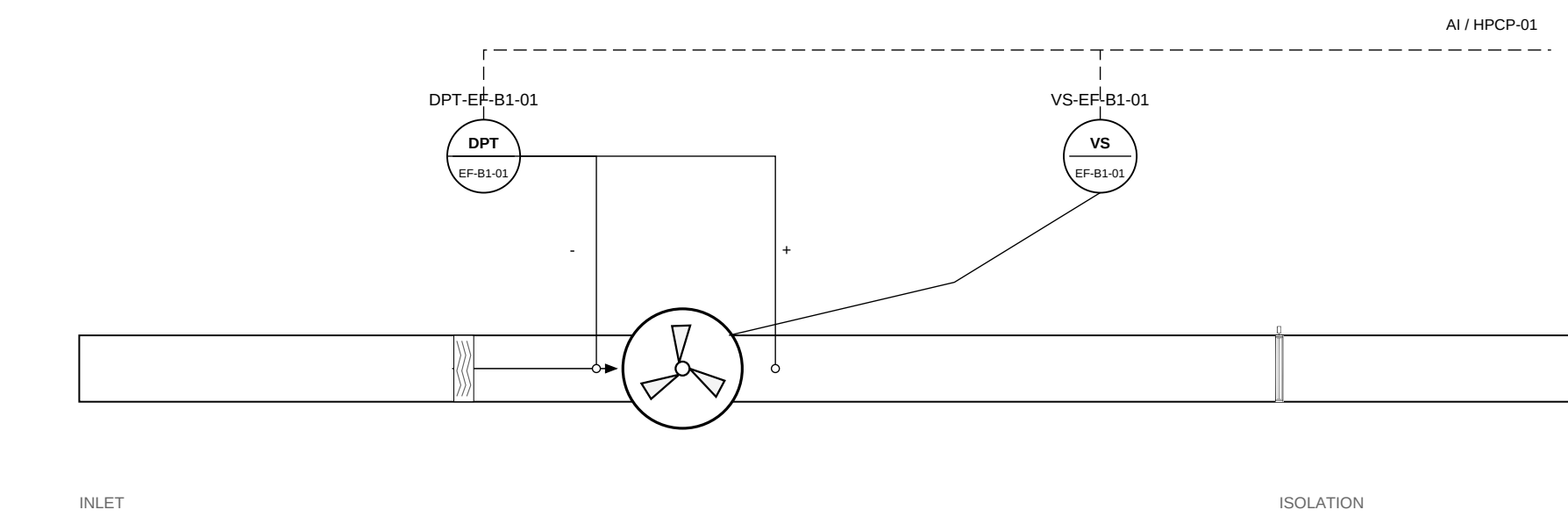
DP taps across fan. VS attachment at OEM bearing / frame. Isolation tag and fail position: approved control register.

07 SF-B1-02



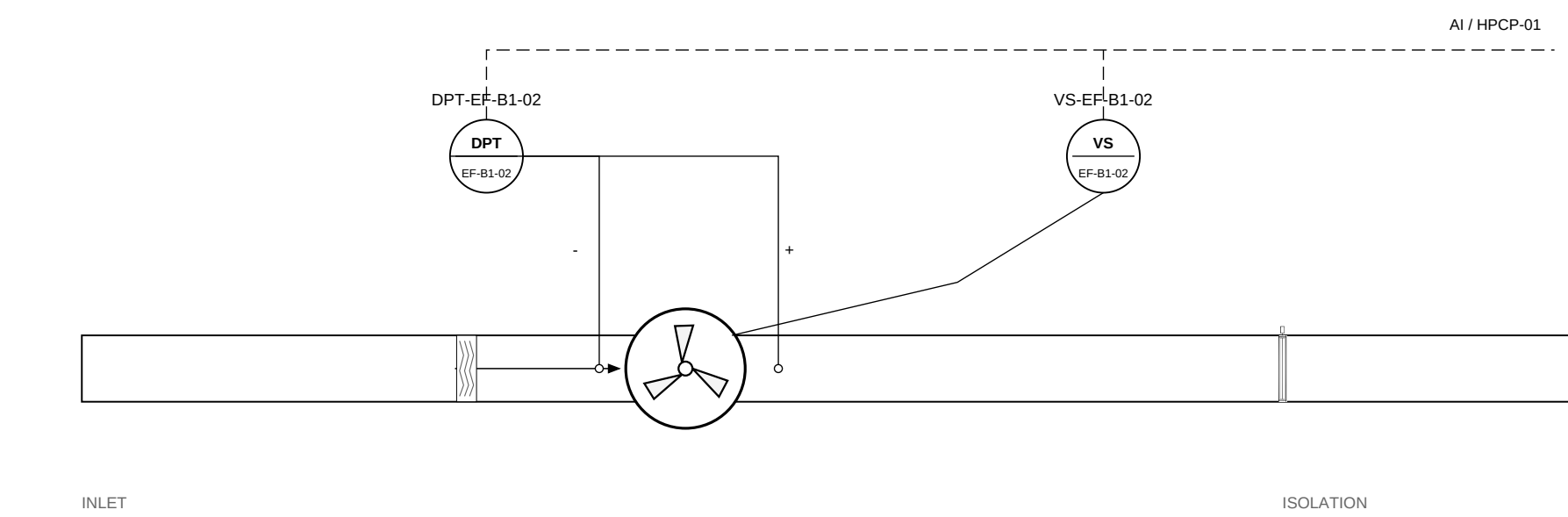
DP taps across fan. VS attachment at OEM bearing / frame. Isolation tag and fail position: approved control register.

08 EF-B1-01



DP taps across fan. VS attachment at OEM bearing / frame. Isolation tag and fail position: approved control register.

09 EF-B1-02



DP taps across fan. VS attachment at OEM bearing / frame. Isolation tag and fail position: approved control register.

NTS functional arrangements. Basement equipment positions are not shown on the Administration plan. CAG fan duty / quantity, tap locations, alarm limits and vibration mountings require OEM / I&C confirmation.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

EXTRACT AND BASEMENT FAN INSTRUMENTATION

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-013

SCALE: NTS AT A1

REVISION

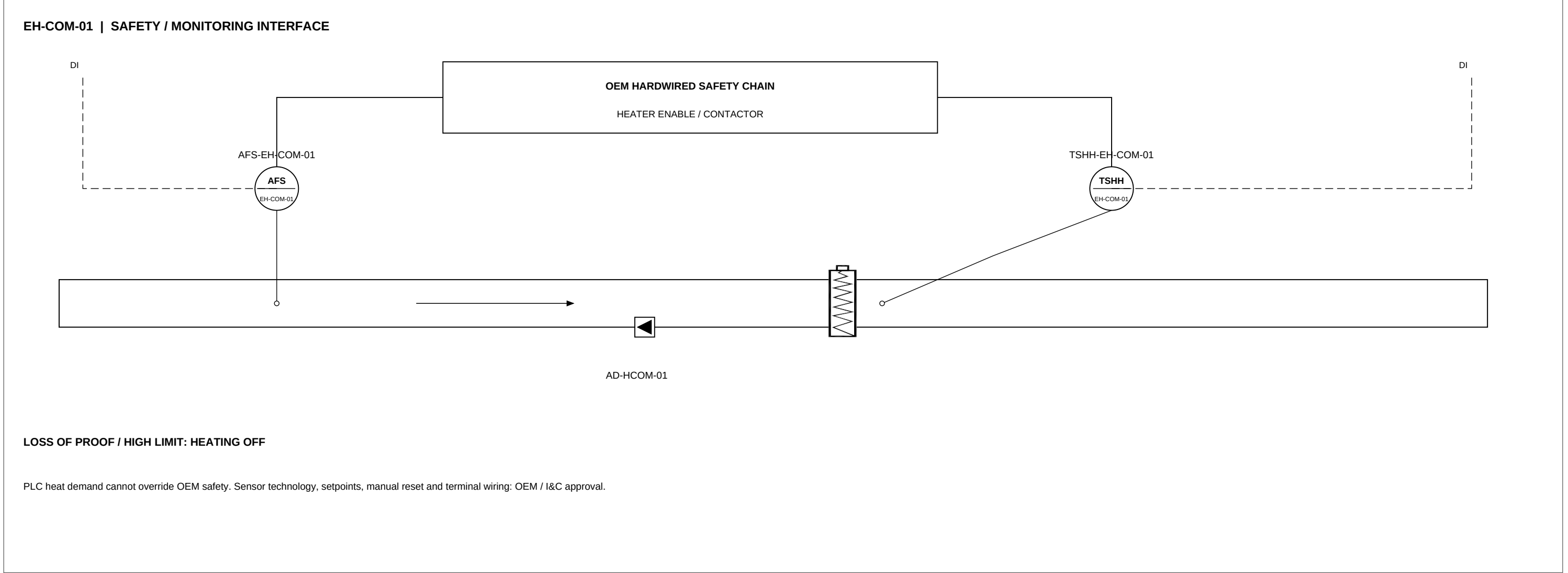
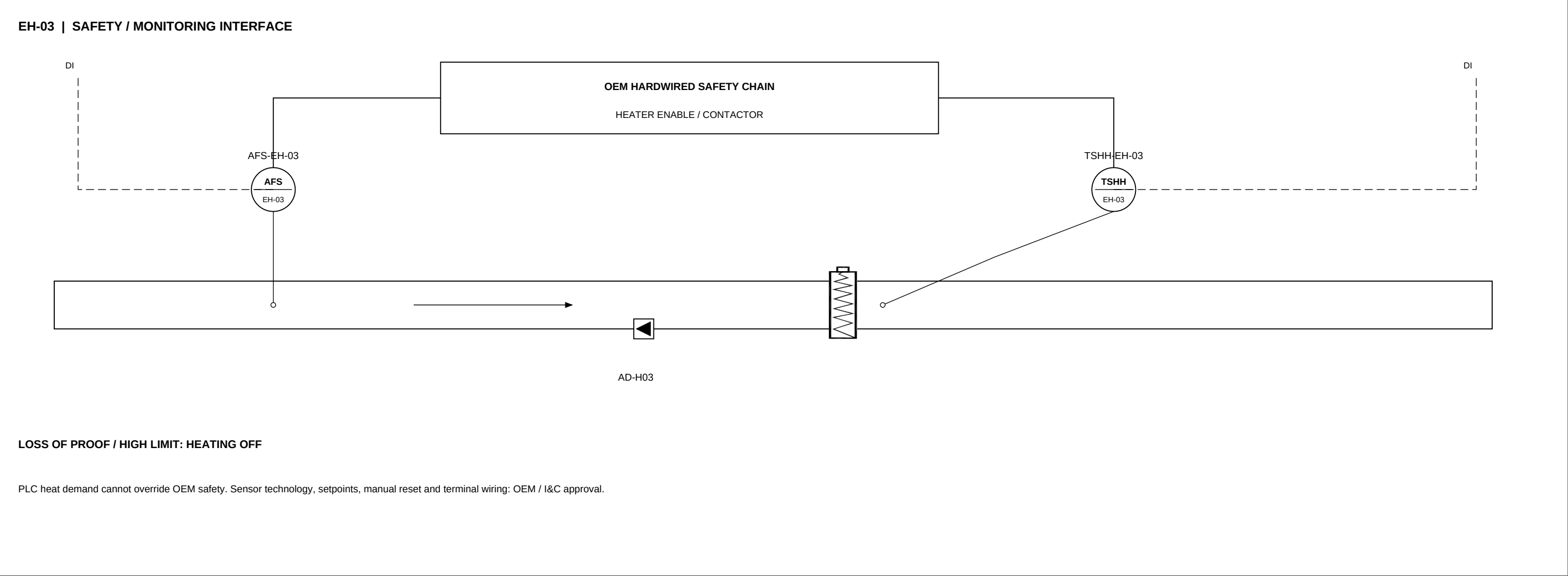
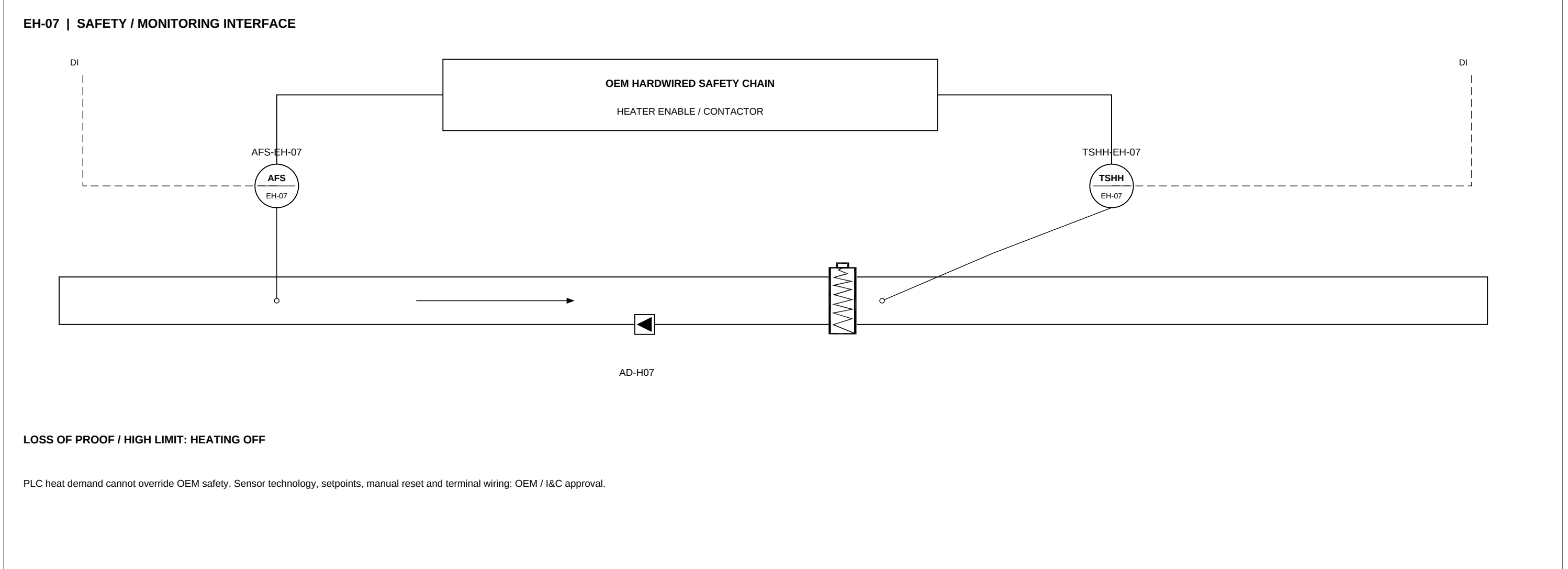
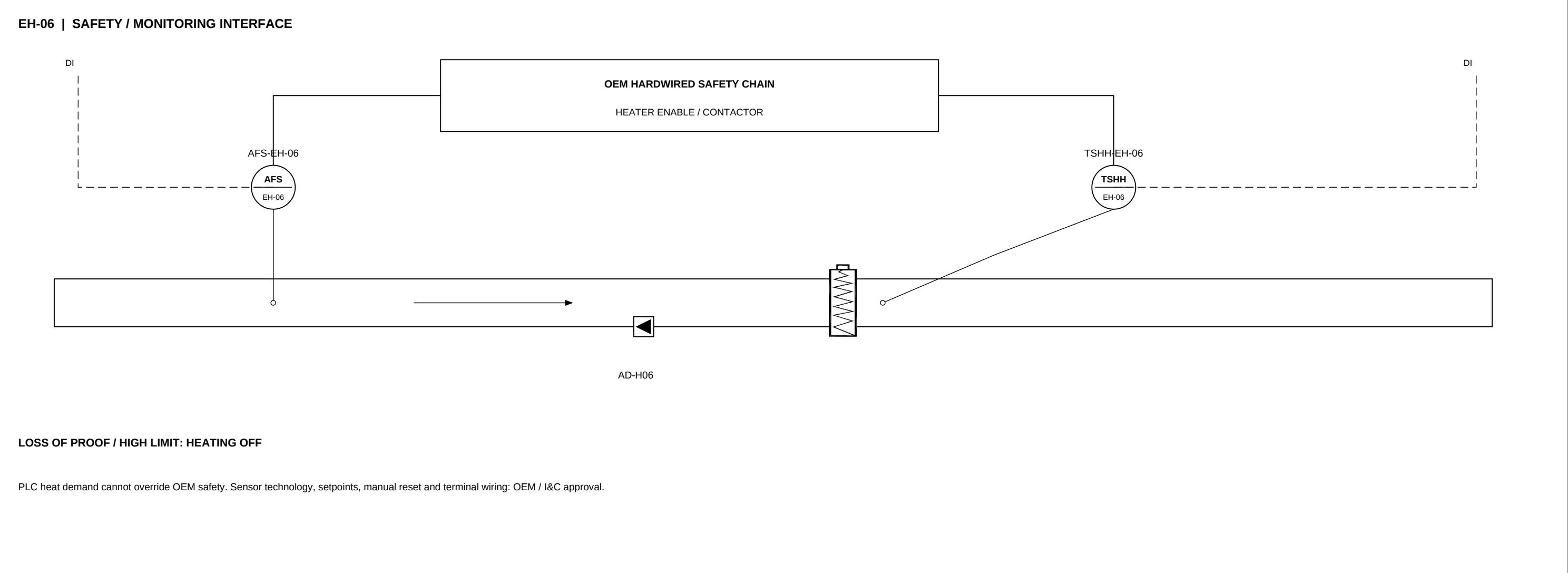
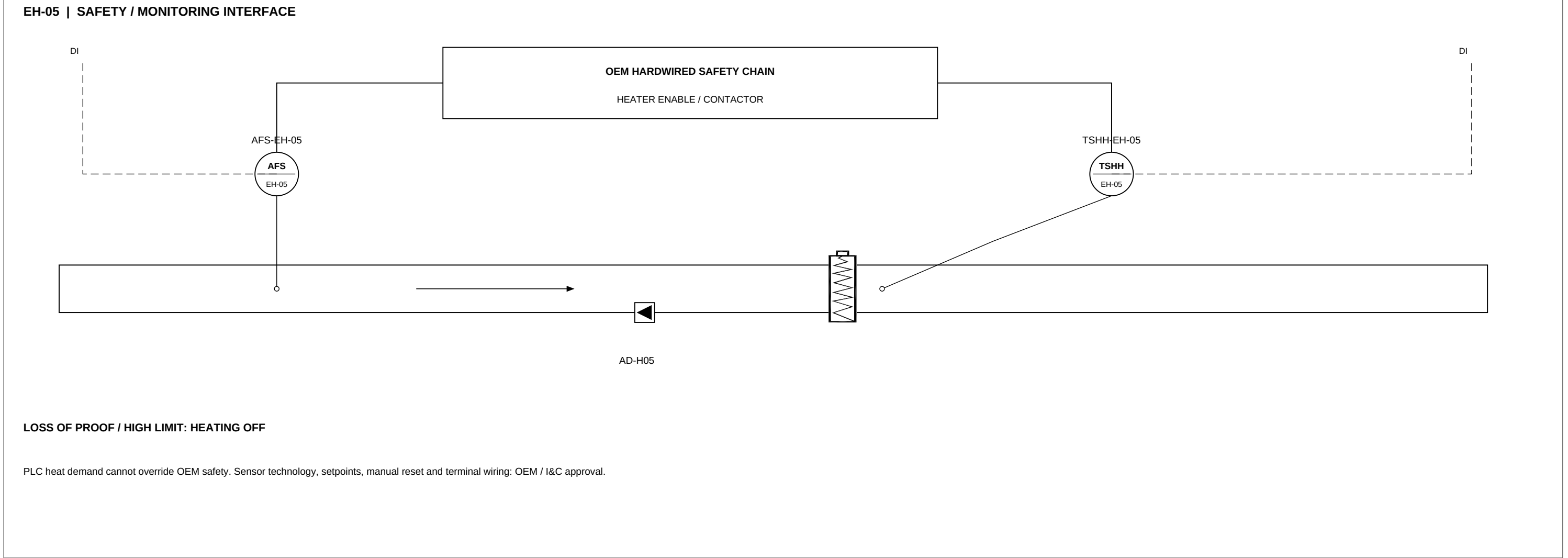
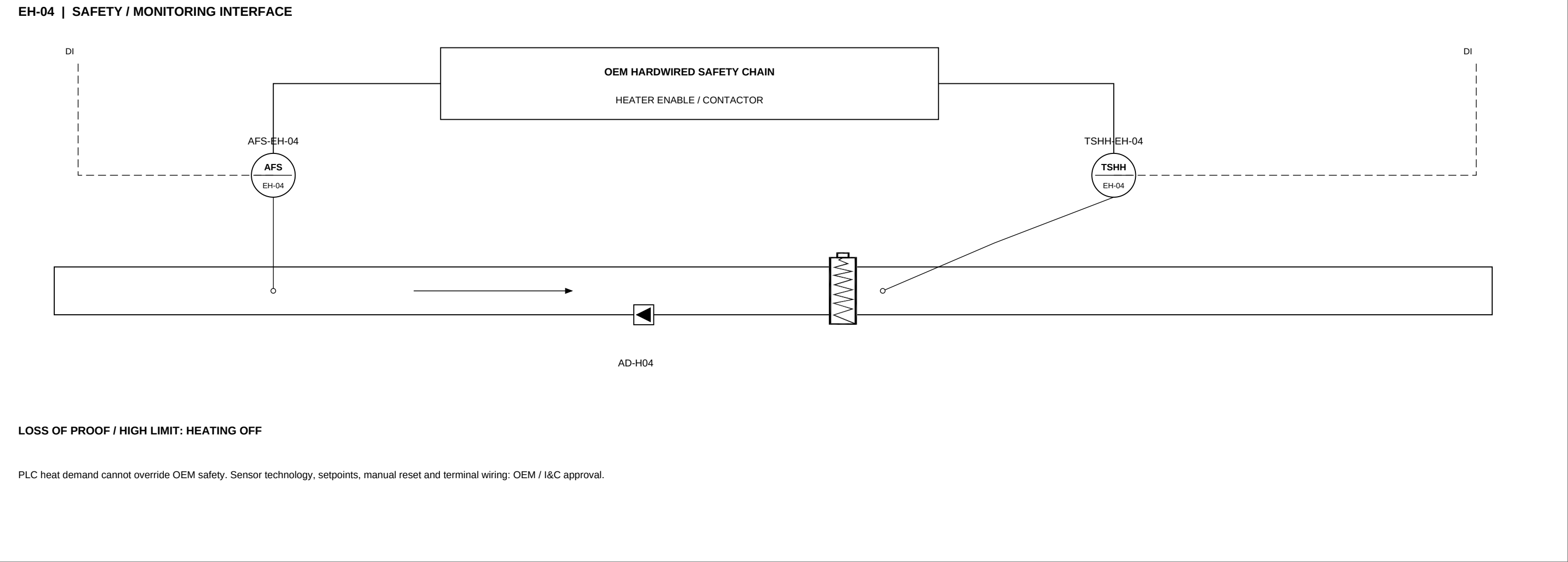
02

08 OCT 2026

SHEET 13 OF 50

MAIN DUCT AND RETURN - ELECTRIC HEATER PROTECTION AND CONTROL

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039



Functional interfaces, not certified wiring diagrams. Six airflow-proof DI and six high-limit DI tags are retained. Protection circuits must be independently implemented and approved by the heater manufacturer.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ELECTRIC HEATER PROTECTION AND CONTROL

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-014

SCALE: NTS AT A1

REVISION

02

08 OCT 2026

SHEET 14 OF 50

MAIN DUCT AND RETURN - FIELD SIGNALS, BASEMENT AND CONTROL INTERFACES

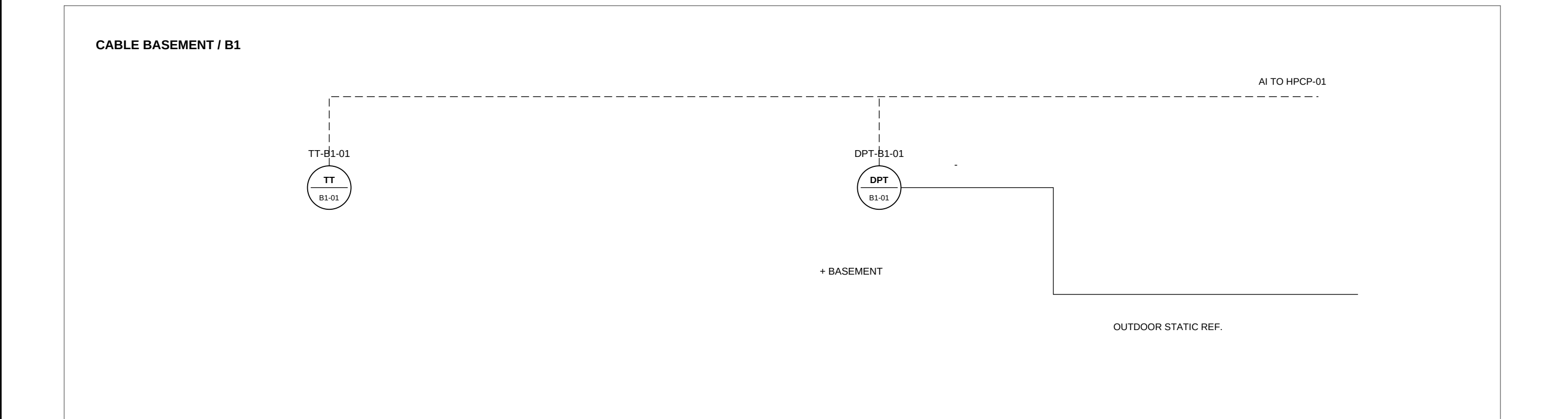
REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039



Sensing / physical interface (route schematic)

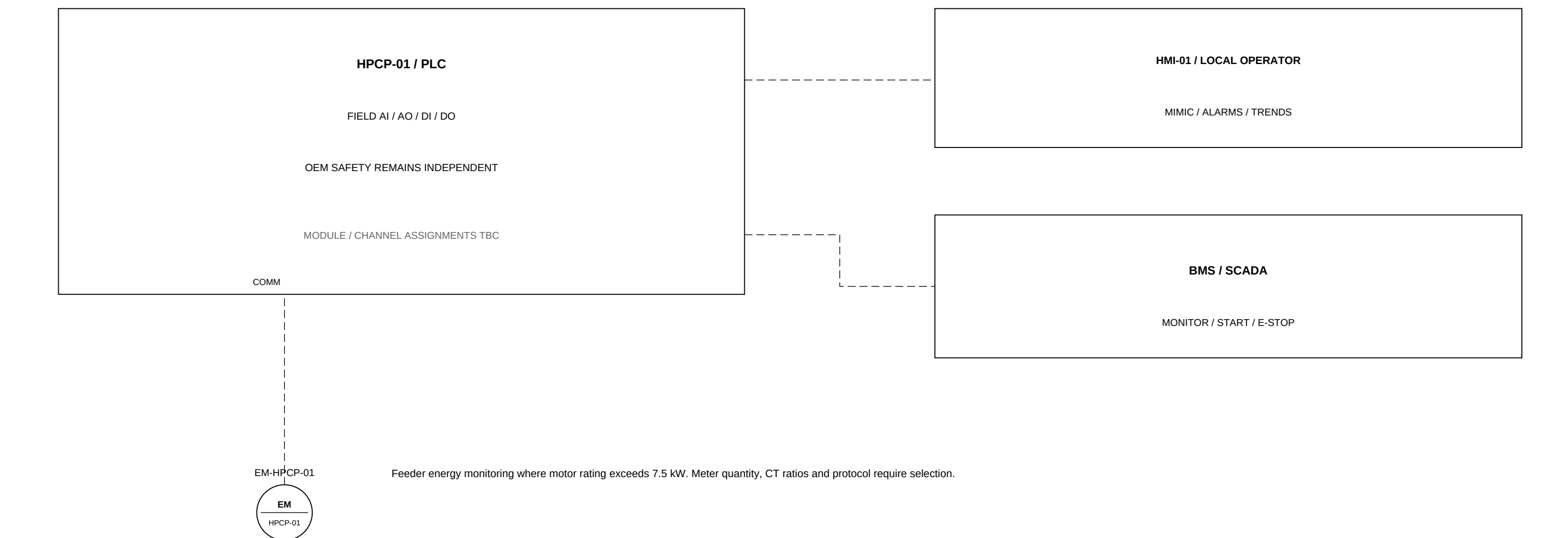
----- Electrical / communications signal (schematic)

AI / AO / DI / DO are point classes, not selected signal ranges. 4-20 mA, contacts, network protocol, cable types and terminal numbers are not inferred.



Basement temperature, volume, fan duties and actual pressure target remain open. Fan instrumentation is fully tagged on D-013; no basement architectural layout was supplied.

All controls shown are functional design intent. Approved OEM wiring and project I&C schedules govern implementation. D-016 provides an instrument-to-drawing cross-reference for every one of the 85 scheduled instruments.



EM-HPCP-01	Feeder energy monitoring where motor rating exceeds 7.5 kW. Meter quantity, CT ratios and protocol require selection.
------------	---

Class	Scheduled	20% spare	Min. channels
AI	65	13	78
AO	22	5	27
DI	141	29	170
DO	44	9	53

COMM interfaces are separate from hardwired channels. Local PDI instruments have no PLC channel. Additional proposals are identified in the instrument index, D-016.

Confirmed fire: stop every HVAC fan and heater. Restart after fire ESD requires local manual reset. Damper fire positions are governed by the approved cause-and-effect, not this drawing.

Normal operation: two duty 50% PAUs with an available alternate. Prove selected air paths before starting fans. Duty rotation, changeover, timing and final permissives require approved OEM logic.

TAQA TRANSMISSION	
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99	100

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

FIELD SIGNALS, BASEMENT AND CONTROL INTERFACES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-015

SCALE: NTS AT A1

REVISION

02

08 OCT 2026

SHEET 15 OF 50

MAIN DUCT AND RETURN - COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

Instrument tag	I/O	Service / location	Status	D-sheet
TT-GF-02	AI	GF-02 Elec. panel	Required	007
RHT-GF-02	AI	GF-02 Elec. panel	Required	007
DPT-GF-02	AI	GF-02 Elec. panel	Proposal	007
TT-GF-04	AI	GF-04 Meeting	Required	007
RHT-GF-04	AI	GF-04 Meeting	Required	007
DPT-GF-04	AI	GF-04 Meeting	Proposal	007
TSHH-EH-04	DI	EH-04	OEM safety	014
TT-GF-08	AI	GF-08 Telecom	Required	007
RHT-GF-08	AI	GF-08 Telecom	Required	007
DPT-GF-08	AI	GF-08 Telecom	Proposal	007
TT-GF-05	AI	GF-05 Operator	Required	007
RHT-GF-05	AI	GF-05 Operator	Required	007
DPT-GF-05	AI	GF-05 Operator	Proposal	007
TSHH-EH-05	DI	EH-05	OEM safety	014
TT-GF-06	AI	GF-06 Coridor	Required	007
RHT-GF-06	AI	GF-06 Coridor	Required	007
DPT-GF-06	AI	GF-06 Coridor	Required	007
TSHH-EH-06	DI	EH-06	OEM safety	014
TT-GF-09	AI	GF-09 Toilets	Required	007
RHT-GF-09	AI	GF-09 Toilets	Required	007
DPT-GF-09	AI	GF-09 Toilets	Proposal	007
TT-GF-07	AI	GF-07 Rest room	Required	007
RHT-GF-07	AI	GF-07 Rest room	Required	007
DPT-GF-07	AI	GF-07 Rest room	Proposal	007
TSHH-EH-07	DI	EH-07	OEM safety	014
TT-GF-03	AI	GF-03 Kitchen	Required	007
RHT-GF-03	AI	GF-03 Kitchen	Required	007
DPT-GF-03	AI	GF-03 Kitchen	Proposal	007
TSHH-EH-03	DI	EH-03	OEM safety	014
TT-GF-01	AI	GF-01 Clean agent	Required	007
RHT-GF-01	AI	GF-01 Clean agent	Proposal	007
DPT-GF-01	AI	GF-01 Clean agent	Proposal	007
TT-SA-01	AI	Common supply	Required	008 / 012
TT-RA-01	AI	Common return	Required	008 / 012
RHT-RA-01	AI	Common return	Required	008 / 012
FIT-SA-01	AI	Common SA	Required	008 / 012
FIT-RA-01	AI	Common RA	Required	008 / 012
PDT-OA-01	AI	F-OA-01	Proposal	012
PDI-OA-01	LOCAL	F-OA-01	Proposal	012
TSHH-EH-COM-01	DI	EH-COM-01	OEM safety	014
PDT-F1-A	AI	PAU-A filter 1	Required	012
PDI-F1-A	LOCAL	PAU-A filter 1	Required	012
PDT-F2-A	AI	PAU-A filter 2	Required	012

85 unique instruments; each has a tagged graphic and drawing reference. Proposal / OEM safety / conditional metering statuses are retained. D-sheet references are ADM-HVAC-D-nnn; local PDI instruments are not wired AI points.

Instrument tag	I/O	Service / location	Status	D-sheet
PDI-F2-A	LOCAL	PAU-A filter 2	Required	012
DPT-SF-A	AI	PAU-A fan	Required	012
VS-PAU-A	AI	PAU-A frame / fan bearing	Required	012
PDT-F1-B	AI	PAU-B filter 1	Required	012
PDI-F1-B	LOCAL	PAU-B filter 1	Required	012
PDT-F2-B	AI	PAU-B filter 2	Required	012
PDI-F2-B	LOCAL	PAU-B filter 2	Required	012
DPT-SF-B	AI	PAU-B fan	Required	012
VS-PAU-B	AI	PAU-B frame / fan bearing	Required	012
PDT-F1-C	AI	PAU-C filter 1	Required	012
PDI-F1-C	LOCAL	PAU-C filter 1	Required	012
PDT-F2-C	AI	PAU-C filter 2	Required	012
PDI-F2-C	LOCAL	PAU-C filter 2	Required	012
DPT-SF-C	AI	PAU-C fan	Required	012
VS-PAU-C	AI	PAU-C frame / fan bearing	Required	012
DPT-EF-03-01	AI	EF-03-01	Proposal	013
VS-EF-03-01	AI	EF-03-01	Required	013
DPT-EF-03-02	AI	EF-03-02	Proposal	013
VS-EF-03-02	AI	EF-03-02	Required	013
DPT-EF-09-01	AI	EF-09-01	Proposal	013
VS-EF-09-01	AI	EF-09-01	Required	013
DPT-EF-09-02	AI	EF-09-02	Proposal	013
VS-EF-09-02	AI	EF-09-02	Required	013
DPT-EF-01-01	AI	EF-01-01	Proposal	013
VS-EF-01-01	AI	EF-01-01	Required	013
DPT-SF-B1-01	AI	SF-B1-01	Proposal	013
VS-SF-B1-01	AI	SF-B1-01	Required	013
DPT-SF-B1-02	AI	SF-B1-02	Proposal	013
VS-SF-B1-02	AI	SF-B1-02	Required	013
DPT-EF-B1-01	AI	EF-B1-01	Proposal	013
VS-EF-B1-01	AI	EF-B1-01	Required	013
DPT-EF-B1-02	AI	EF-B1-02	Proposal	013
VS-EF-B1-02	AI	EF-B1-02	Required	013
TT-B1-01	AI	Cable basement	Required	015
DPT-B1-01	AI	Cable basement	Proposal	015
EM-HPCP-01	COMM	HPCP feeders >7.5 kW	Required*	015
AFS-EH-04	DI	Electric heater EH-04	Proposal	014
AFS-EH-05	DI	Electric heater EH-05	Proposal	014
AFS-EH-06	DI	Electric heater EH-06	Proposal	014
AFS-EH-07	DI	Electric heater EH-07	Proposal	014
AFS-EH-03	DI	Electric heater EH-03	Proposal	014
AFS-EH-COM-01	DI	Electric heater EH-COM-01	Proposal	014

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-016

SCALE: NTS AT A1

REVISION

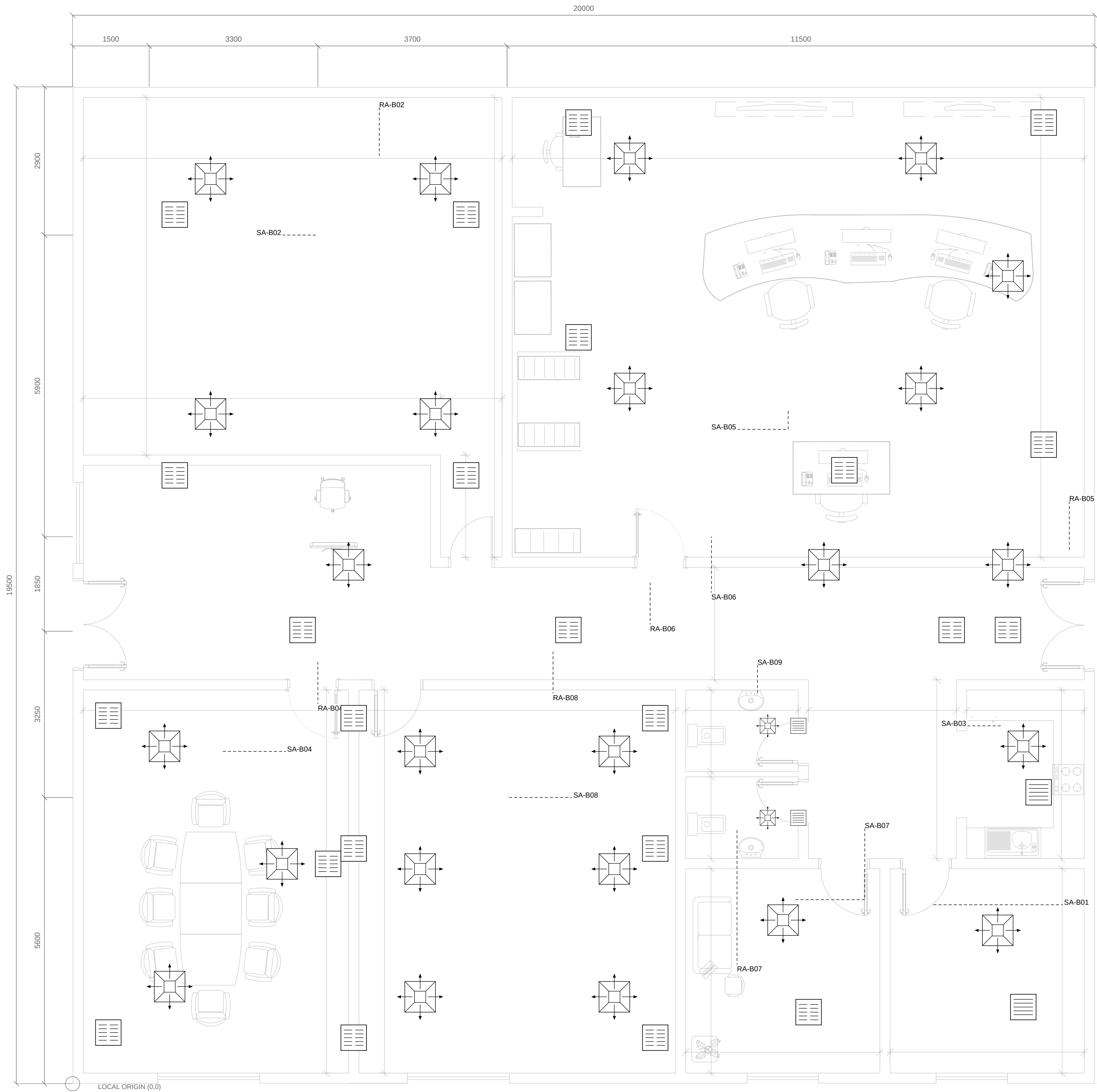
02

08 OCT 2026

SHEET 16 OF 50

MAIN DUCT AND RETURN - PLAN DIMENSIONS AND TERMINAL SETTING-OUT

REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039



Dimension chains are plan centerline offsets and architectural extents only. They are not approved fabrication, support or vertical-clearance dimensions. Terminal faces retain scheduled dimensions, final catalogue free areas and performance remain pending.

SOURCE PLAN COORDINATES / DATUM

Dimensions and coordinates are derived from the supplied architectural plan. Local (0,0) is a drawing origin, not a surveyed site benchmark. Confirm all setting-out against the coordinated ceiling and architecture.

GF	Terminal X range mm	Terminal Y range mm
02	2000 - 7700	11900 - 17700
05	9900 - 19000	12000 - 18800
04	700 - 5000	1000 - 7200
08	5500 - 11400	900 - 7150
06	4500 - 18300	8875 - 10150
07	13900 - 14400	1400 - 3200
09	13600 - 14200	5200 - 7000
03	18600 - 18900	5700 - 6600
01	18100 - 18600	1500 - 3000

Individual terminal coordinates: Terminals register. Room branch centerlines: Duct_sections register. Native CAD dimensions are provided for the known overall extents and room setting-out offsets.

User: structural underside +4.000 m; minimum ceiling +3.300 m. Proposed indoor bare BOD +3.400 m; insulation 50 mm assumed; deepest bare duct 400 mm. D-020 shows the single-layer envelope. Roof routing and penetrations: D-022, pending coordination.

INSPECTION / COORDINATION NOTES

1. Verify dimensions on site; do not scale NTS details.
2. Match device tags to registers before procurement.
3. Reserve access to dampers, heaters and filters.
4. Obtain selected-fitting losses and OEM duties.
5. Close pressure / outdoor-air design holds.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

PLAN DIMENSIONS AND TERMINAL SETTING-OUT

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-017

SCALE: 1:50 AT A1

REVISION

02

08 OCT 2026

SHEET 17 OF 50

Drawing	Title / subject
MDR-001	COORDINATED DUCT PLAN
MDR-002	SUPPLY AIR DUCT PLAN
MDR-003	RETURN AIR AND EXTRACT DUCT PLAN
MDR-004	NORTH ROOMS - DUCT ENLARGEMENT
MDR-005	SOUTH-WEST ROOMS - DUCT ENLARGEMENT
MDR-006	SOUTH-EAST ROOMS - DUCT ENLARGEMENT
MDR-007	ROOM INSTRUMENTATION AND CONTROL LOCATIONS
MDR-008	PAU BANK AND PROPOSED ROOF COMMON DISTRIBUTION
MDR-009	DUCT SYMBOLS AND INSTALLATION DETAILS
MDR-010	DUCT AND TERMINAL SCHEDULES
MDR-011	TAPERED FITTINGS AND MAINTENANCE ACCESS
MDR-012	PLANT INSTRUMENTATION AND HEATER PROTECTION
MDR-013	EXTRACT AND BASEMENT FAN INSTRUMENTATION
MDR-014	ELECTRIC HEATER PROTECTION AND CONTROL
MDR-015	FIELD SIGNALS, BASEMENT AND CONTROL INTERFACES
MDR-016	COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES
MDR-017	PLAN DIMENSIONS AND TERMINAL SETTING-OUT
MDR-018	COMPLETE DRAWING INDEX / DESIGN BASIS / ISSUE 02
MDR-019	ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS
MDR-020	3.30 m CEILING - SINGLE-LAYER INDOOR DUCT ENVELOPE
MDR-021	REV11 TERMINAL MOVES AND SHALLOW DUCT SIZE CHANGES
MDR-022	ROOF DISTRIBUTION AND ROOM RISER REGISTER
MDR-023	ROOF DUCT ROUTING PROPOSAL
MDR-024	ROOF CROSSING AND ROOM DROP SECTIONS
MDR-025	FITTING DEVELOPMENT AND FABRICATION RULES

Drawing	Title / subject
MDR-026	SYSTEM PRESSURE LOSS AND FAN DUTY BUDGET
MDR-027	AIR BALANCE AND TERMINAL SELECTION CRITERIA
MDR-028	INSTRUMENT MOUNTING AND PRESSURE TAP DETAILS
MDR-029	I&C WIRING AND HEATER SAFETY INTERFACES
MDR-030	ROOF CURB, SUPPORT AND WEATHERPROOFING
MDR-031	CEILING COMPONENT AND ACCESS ENVELOPES
MDR-032	ROOF SETTING-OUT AND ROOM DROP REGISTER
MDR-033	DX PAU BRANCH / HEADER TRANSITION DEVELOPMENT
MDR-034	ACTUAL ROOF SPLIT / REDUCER / ELBOW DEVELOPMENT
MDR-035	ROOM DROP / ROOF CURB / FIRESTOP INSTALLATION
MDR-036	COMMON HEADER INSTRUMENTS / CONDENSATE / ACCESS
MDR-037	DX / FAN / DAMPER / FIRE CAUSE AND EFFECT
MDR-038	DX OEM / HPCP FIELD WIRING AND I/O DEVELOPMENT
MDR-039	DX EQUIPMENT SELECTION / ROOM DUTIES / COMMISSIONING
MDR-040	WHOLE SYSTEM AIRFLOW METHOD
MDR-041	ALL ROOM DIFFUSERS, RETURN GRILLES AND EXTRACT PICKUPS
MDR-042	DEDICATED EXTRACTS AND RETURN AIR SEPARATION
MDR-043	DX PAU BANK, COMMON HEADERS AND AIR-SIDE INSTRUMENTS
MDR-044	DX REFRIGERANT CYCLE AND PROPOSED OEM PROTECTION
MDR-045	ROOM TEMPERATURE, HUMIDITY AND PRESSURE INSTRUMENTS
MDR-046	HEATER PERMISSIVES, ACCESS AND HARDWIRED SAFETY
MDR-047	ROOF CROSSING, ROOM DROPS AND CEILING ENVELOPES
MDR-048	BASEMENT, HPCP, ENERGY METERING AND CONTROL SEQUENCE
MDR-049	COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES
MDR-050	COMMON HEADER FLOW ACCOUNTING AND ENGINEERING BASIS

50 A1 sheets: full32-sheet engineering content, seven new DX/header details and11 supporting diagrams. First32 D-xxx references correspond to MDR-xxx. Original route geometry / duties retained. DX/OA/roof/structure/OEM/final approval remain pending.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

COMPLETE DRAWING INDEX / DESIGN BASIS / ISSUE 02

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-018

SCALE: NTS AT A1

DRAWING REVISION 02

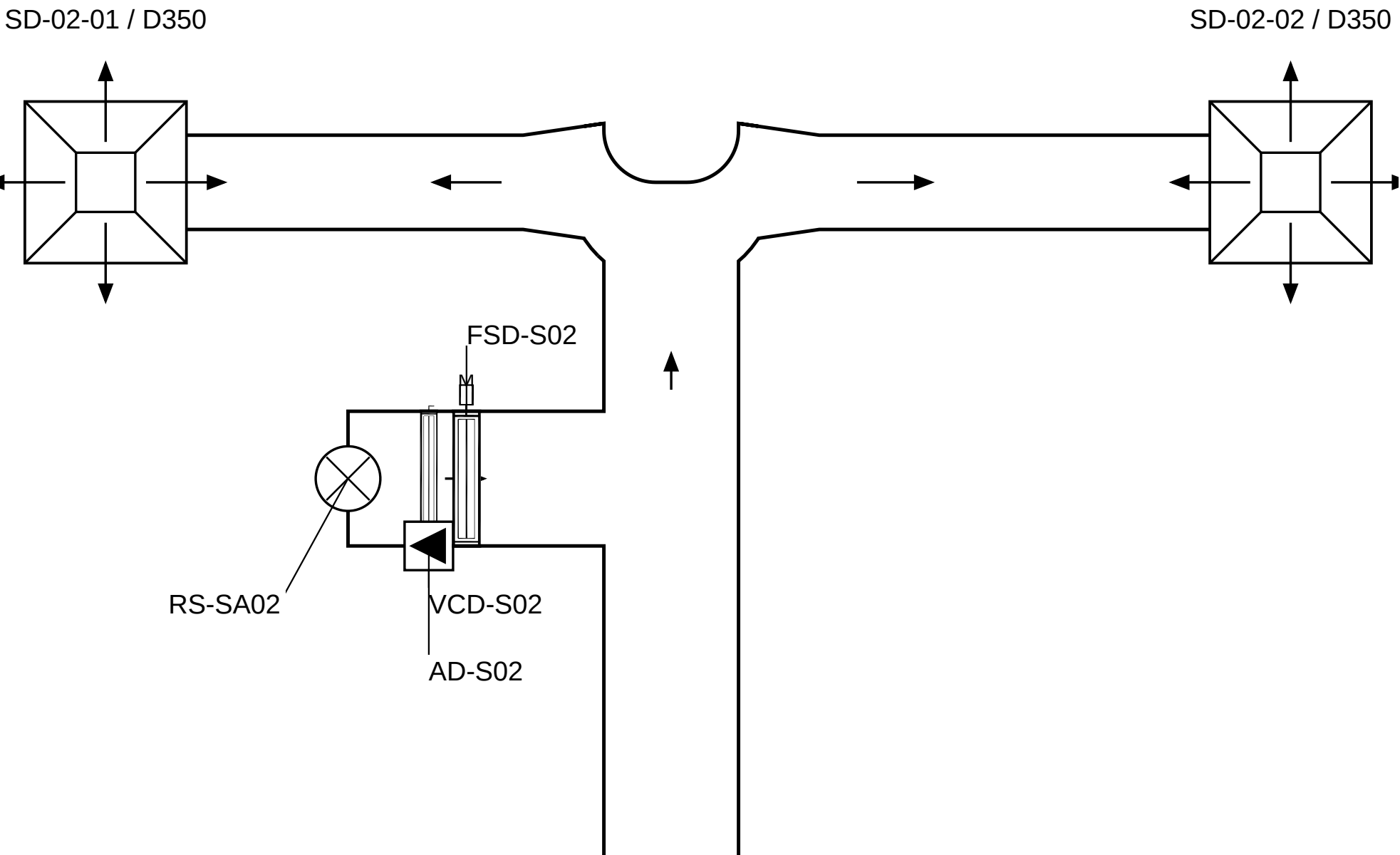
08 OCT 2026

SHEET 18 OF 50

MAIN DUCT AND RETURN - ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS

REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

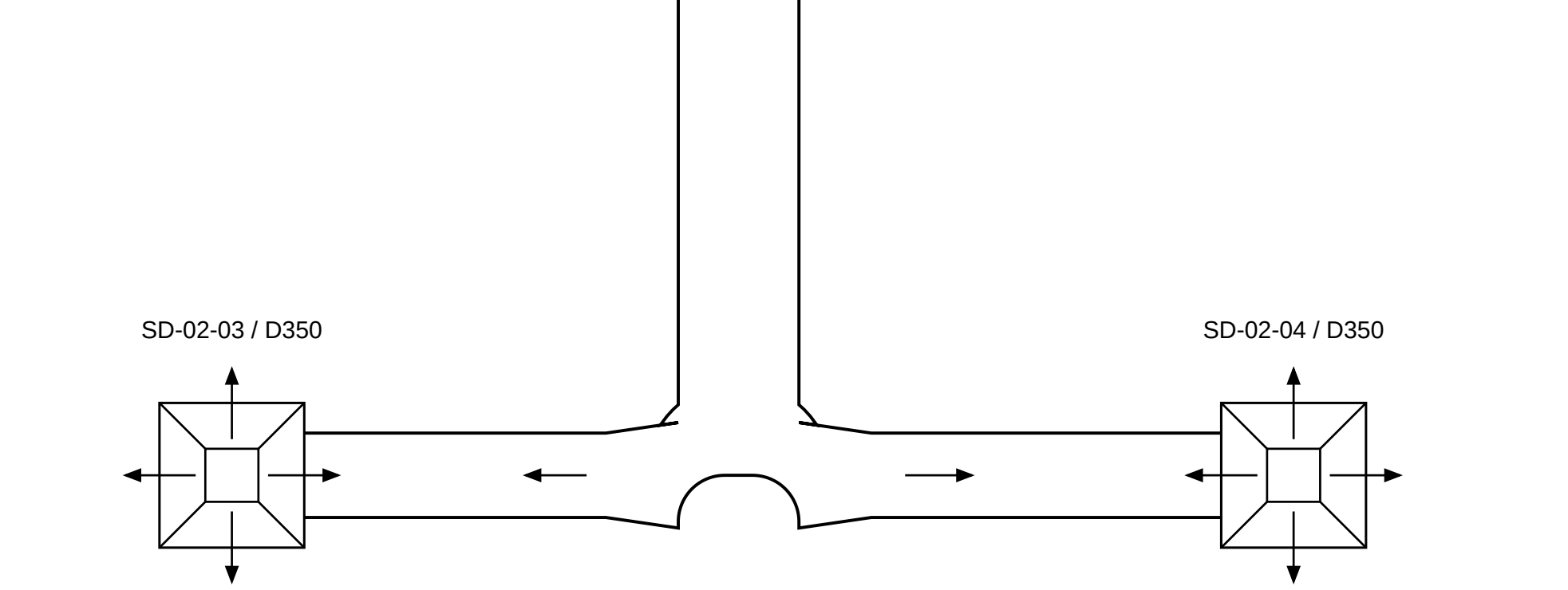
A GF-02 ROOF DROP / BRANCH / HEADER LEGS / 1:20



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

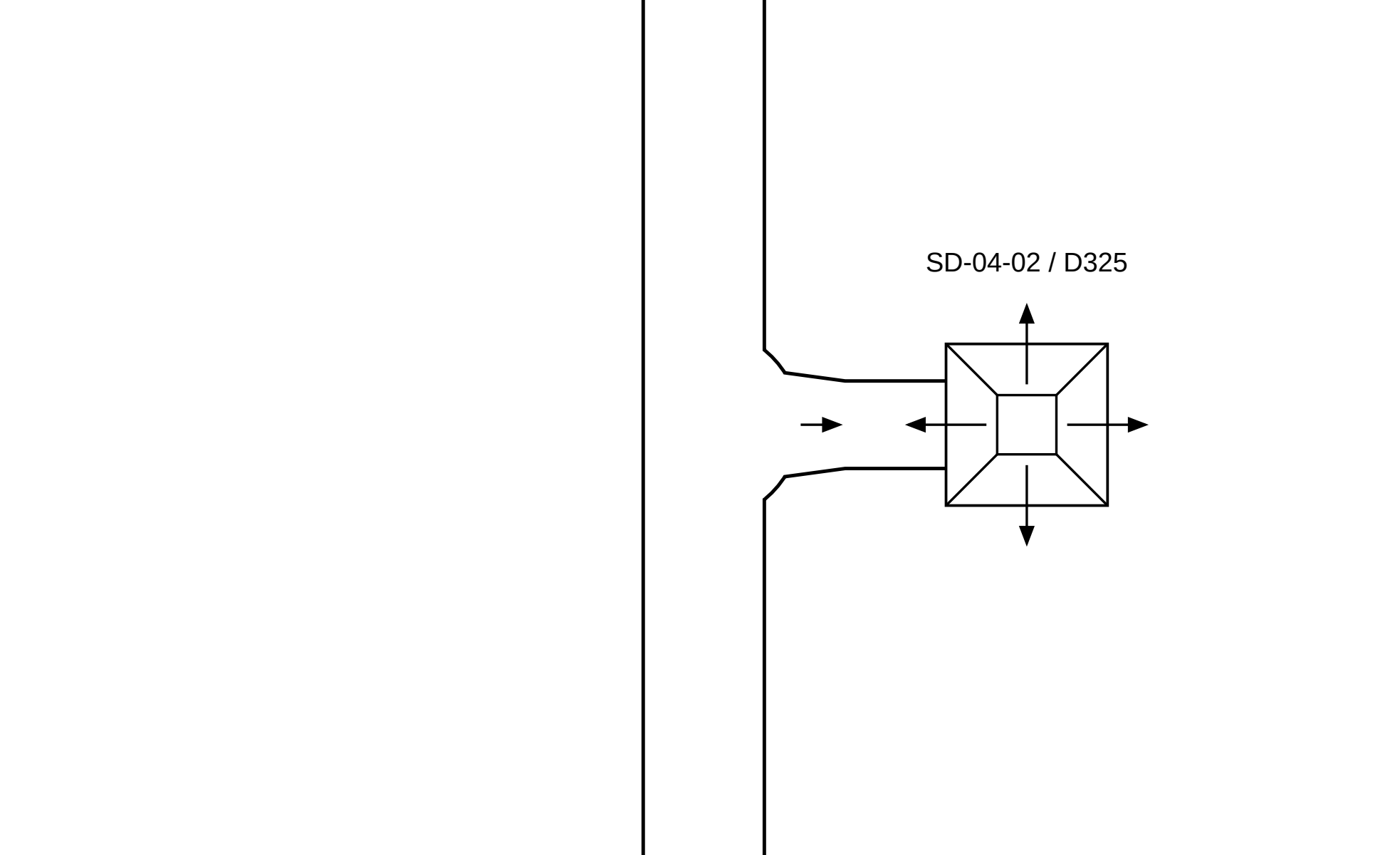
B GF-02 END TEE / TWO INDIVIDUAL NECKS / 1:25



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

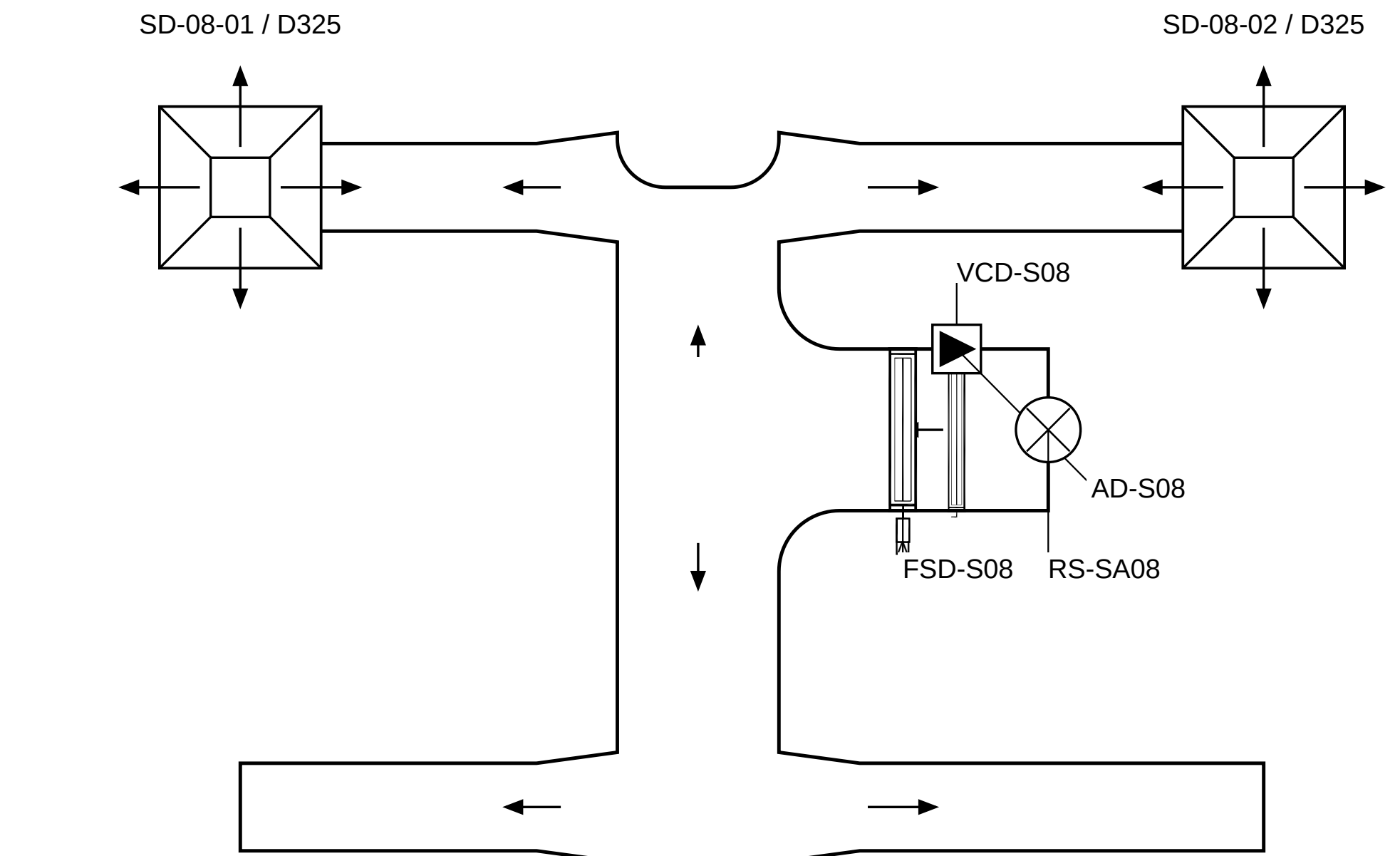
C MEETING CONTINUING HEADER / SIDE NECK / 1:20



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

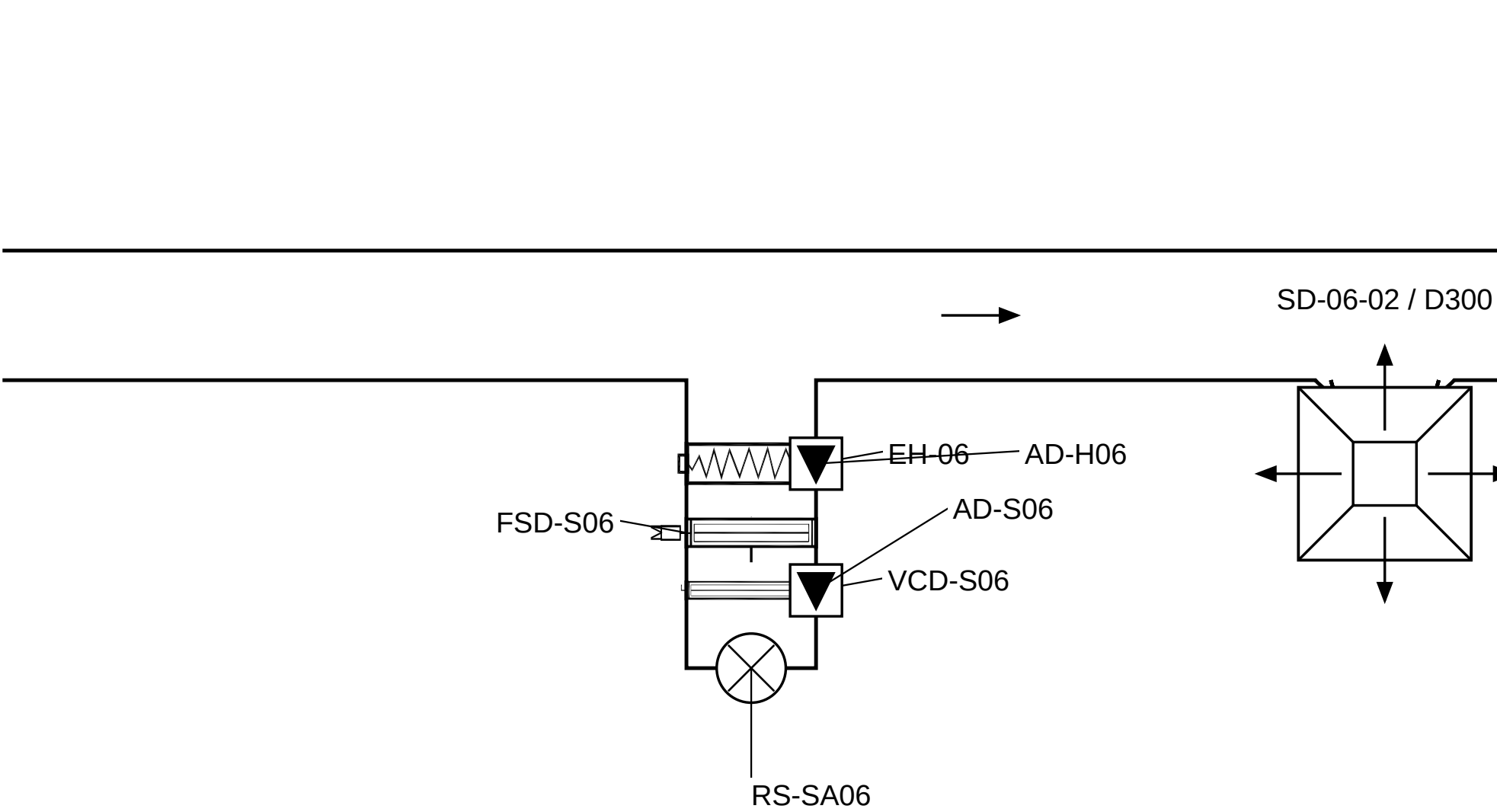
D TELECOM FEEDER / UP AND DOWN HEADERS / 1:20



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

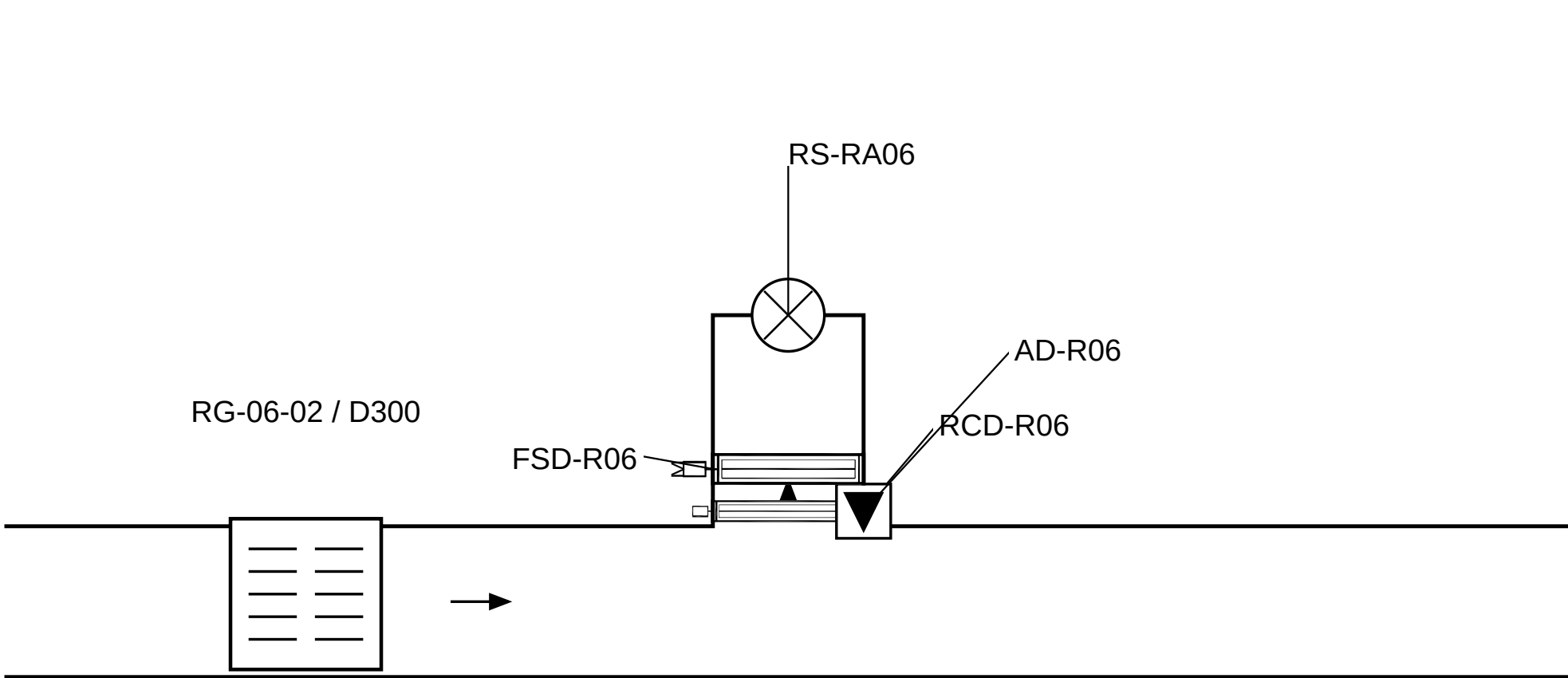
E CORRIDOR ROOM DROP BRANCH / HEATER / HEADER / 1:20



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

F CORRIDOR COLLECTION / RETURN DEVICE STATION / 1:20



Actual Rev11 duct envelope; connected inside seams removed. Individual clear sizes and neck duties: Duct_sections / Terminals registers.

Formed fitting / access / face-to-face dimensions require selected manufacturer confirmation. Other services omitted in this enlargement; single-layer envelope on D-020, roof drops on D-022.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ACTUAL DUCT JUNCTIONS - ENLARGED PLAN DETAILS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-019

SCALE: AS SHOWN AT A1

REVISION

02

08 OCT 2026

SHEET 19 OF 50

MAIN DUCT AND RETURN - 3.30 m CEILING - SINGLE-LAYER INDOOR DUCT ENVELOPE

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

USER DATUMS AND PROPOSED INDOOR LEVELS

Item	Value	Basis / status
Structural underside	+4.000 m AFFL	User confirmed slab / roof
Minimum ceiling / clear height	+3.300 m AFFL	User requirement
Available total space	700 mm	Between confirmed datums
Proposed bare duct bottom	+3.400 m AFFL	All indoor trees; not installed
Indoor insulation	50 mm each face	Assumed; project specification TBC
Largest bare duct depth	400 mm	Round necks max D350 retained
Lowest insulated bottom	+3.350 m AFFL	50 mm above required ceiling
Highest insulated top	+3.850 m AFFL	150 mm below slab for deepest duct

WHAT THE REVISED ROUTES RESOLVE

Deep common mains move to a proposed roof distribution system. Fifteen independent room supply / return drops replace the indoor common-main feeders.

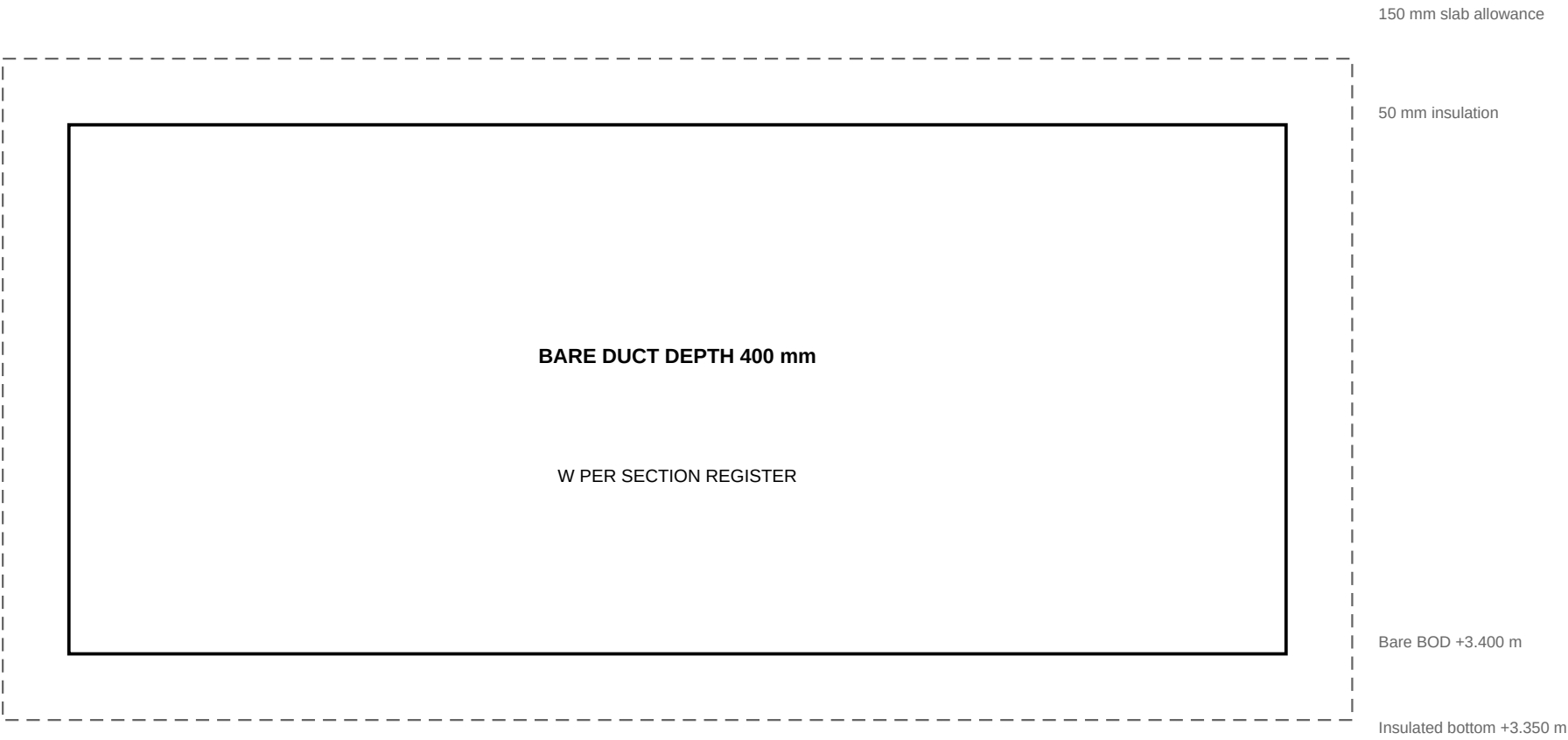
Electrical and telecom returns collect at room perimeters. Every room tree is connected, every collection node conserves airflow, and distinct room trees stay separate.

No indoor bare duct crossings or insulated separate-service overlaps. Continuous duct walls, formed junctions and individual terminal necks are generated from the real revised plan geometry.

Rev10 CX-08 had its lowest insulated bottom at +2.050 m, 1250 mm below this requirement. The Rev10 stacked arrangement is superseded by this roof option; it must not be installed at a 3.30 m ceiling.

DEEPEST DUCT ENVELOPE - NTS

SLAB / ROOF UNDERSIDE +4.000 m



CEILING / CLEAR HEIGHT +3.300 m

Stack: 150 mm slab / support allowance + 50 mm insulation + 400 mm bare duct + 50 mm insulation + 50 mm ceiling gap = 700 mm.

This checks scheduled duct envelopes only. Beam downstands, other services, flanges / stiffeners, damper actuators, heater casing, supports, access hatches and terminal plenums are not manufacturer-selected or surveyed.

The 50 mm gap below insulation is not a general maintenance clearance or a selected plenum dimension. Terminal drops / plenums must use the available connection space within the duct envelope; selected assemblies may require a local redesign.

User selected roof distribution / room drops. Do not cut the slab from these drawings. New penetrations, curbs, weatherproofing, roof loads and fire boundaries require structural / architectural / fire coordination.

PROPOSED LEVELS, NOT INSTALLATION APPROVAL. AFFL datum requires site confirmation. 50 mm insulation is an indoor assumption, not a verified UAE standard. D-022 identifies all proposed roof drops; actual roof routing and equipment envelopes remain open.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

3.30 m CEILING - SINGLE-LAYER INDOOR DUCT ENVELOPE

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-020

SCALE: NTS AT A1

REVISION

02

08 OCT 2026

SHEET 20 OF 50

MAIN DUCT AND RETURN - REV11 TERMINAL MOVES AND SHALLOW DUCT SIZE CHANGES

REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

PROPOSED TERMINAL MOVES FROM REV10

Terminal	Rev10 X, Y mm	Rev11 X, Y mm	L/s
SD-02-01	2700, 18000	2700, 17700	166.2
SD-02-02	7100, 18000	7100, 17700	166.2
RG-02-01	5400, 17000	2000, 17000	151.36584740090174
RG-02-03	5400, 11900	2000, 11900	151.36584740090174
SD-08-01	6500, 6500	6800, 6500	147.43333333333334
SD-08-03	6500, 4200	6800, 4200	147.43333333333334
SD-08-05	6500, 1700	6800, 1700	147.43333333333334
RG-08-01	6400, 7150	5500, 7150	137.54389826726785
RG-08-02	11500, 7150	11400, 7150	137.54389826726785
RG-08-03	5900, 4600	5500, 4600	137.54389826726785
RG-08-04	11500, 4600	11400, 4600	137.54389826726785
RG-08-05	5900, 900	5500, 900	137.54389826726785
RG-08-06	11500, 900	11400, 900	137.54389826726785

Individual duties, neck diameters and face sizes are retained. Moves make room for perimeter return trees and insulation; coordinate reflected ceiling, throw / noise, lighting and maintenance access before adoption.

TRACEABLE ROUTING / AIRFLOW

116 indoor section paths; 15 proposed roof drops. Local branches, headers and necks remain connected to their allocated room riser. Roof-main sizes / flows are separately listed as a functional basis; actual roof lengths and levels are TBC.

All indoor section velocities, plan centerline lengths and straight-duct friction are recomputed. Wider shallow sections replace the deep returns. Selected elbow / riser / transition losses and revised total fan ESP remain pending; straight friction is not a system pressure calculation.

Rev11 is a roof-distribution proposal developed for the new 3.30 m clear-height requirement. Rev10 remains preserved as superseded for this ceiling constraint. These are engineering coordination proposals; final ceiling / OEM / pressure / structural approvals are pending.

CLEAR SIZE REVISIONS FROM REV10

Section	Rev10 mm	Rev11 mm
RA-B02	600x400	650x400
RA-H02-U01	600x400	450x350
RA-H02-D01	600x400	450x350
RA-B08	500x650	850x400
RA-H08-U01	500x650	500x350
RA-H08-D01	500x650	500x350
RA-H08-D02	500x650	500x350
RA-B05	500x525	700x400
RA-H05-U01	500x525	700x400
RA-H05-U02	500x525	700x400
RA-H05-U03	500x525	700x400
RA-H05-D01	500x525	700x400
RA-B06	350x550	500x400
RA-H06-D01	350x550	500x400
RA-H05-L01	500x525	700x400
RA-H05-N01	500x525	700x400
RA-H05-E01	500x525	700x400
RA-H06-L01	350x550	500x400
RA-H06-L02	350x550	500x400
RA-H06-E01	350x550	500x400

Largest indoor bare depth: 400 mm. Outdoor common mains retain their flow / clear-size basis; weather insulation, actual roof routes and fitting / level selection are pending.

85 instrument requirements and all existing I/O duties are retained. Common heater / probes / conditional humidifier move to the proposed roof plant, with weather-protected equipment and mounting still to be selected.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

REV11 TERMINAL MOVES AND SHALLOW DUCT SIZE CHANGES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-021

SCALE: NTS AT A1

REVISION

02

08 OCT 2026

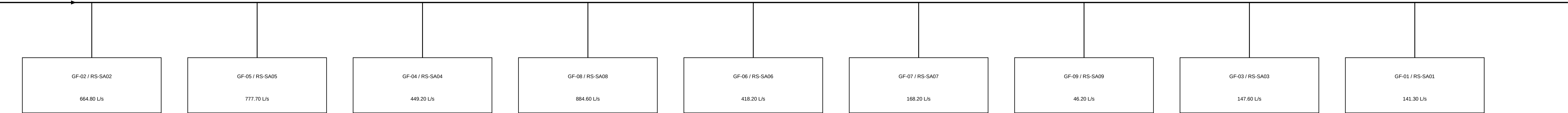
SHEET 21 OF 50

MAIN DUCT AND RETURN - ROOF DISTRIBUTION AND ROOM RISER REGISTER

REISSUED DX MAIN / RETURN OPTION / ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

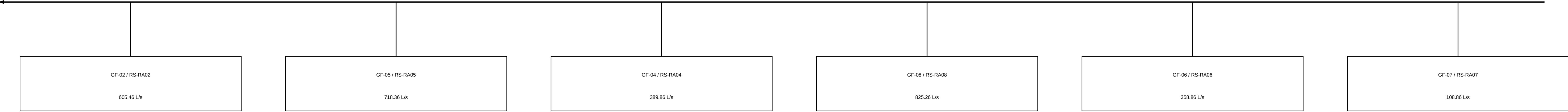
FUNCTIONAL ROOF CONNECTIONS - NTS / ACTUAL ROOF ROUTING TBC

SUPPLY FROM ROOF COMMON PLANT D-008 / 3697.80 L/s



Separate supply drops connect to each room VCD / conditional MFD / heater station; no ceiling common main.

RETURN TO ROOF COMMON PLANT D-008 / 3006.68 L/s ASSUMED



PROPOSED ROOF DROP CENTERS - SOURCE PLAN X / Y, mm

Riser	Branch / GF	X, Y mm	Clear W x H	L/s	Indoor bare BOD	Roof level / vertical length
RS-SA02	SA-B02 / 02	3600, 16600	500x350	664.80	+3.400 m	TBC / TBC
RS-RA02	RA-B02 / 02	6000, 18150	650x400	605.46	+3.400 m	TBC / TBC
RS-SA04	SA-B04 / 04	4200, 6500	350x400	449.20	+3.400 m	TBC / TBC
RS-RA04	RA-B04 / 04	4800, 8250	500x350	389.86	+3.400 m	TBC / TBC
RS-SA08	SA-B08 / 08	9800, 5600	600x400	884.60	+3.400 m	TBC / TBC
RS-RA08	RA-B08 / 08	9400, 8450	850x400	825.26	+3.400 m	TBC / TBC
RS-SA05	SA-B05 / 05	12500, 12800	600x350	777.70	+3.400 m	TBC / TBC
RS-RA05	RA-B05 / 05	19500, 10400	700x400	718.36	+3.400 m	TBC / TBC
RS-SA06	SA-B06 / 06	12500, 9475	450x300	418.20	+3.400 m	TBC / TBC
RS-RA06	RA-B06 / 06	11300, 9800	500x400	358.86	+3.400 m	TBC / TBC
RS-SA09	SA-B09 / 09	13400, 8200	200x150	46.20	+3.400 m	TBC / TBC
RS-SA07	SA-B07 / 07	15500, 5000	300x250	168.20	+3.400 m	TBC / TBC
RS-RA07	RA-B07 / 07	13000, 5000	350x250	108.86	+3.400 m	TBC / TBC
RS-SA03	SA-B03 / 03	17000, 7000	300x200	147.60	+3.400 m	TBC / TBC
RS-SA01	SA-B01 / 01	19400, 3500	300x200	141.30	+3.400 m	TBC / TBC

Roof flow/size basis is functional, not a fabricated plan: no roof survey, beam map, selected curb / sleeve / fire assembly or common heater weather enclosure was supplied. All 15 penetrations are proposals; coordinate spacing, framing, access and roof drainage before setting them out.

USER SELECTED ROOF DISTRIBUTION / ROOM DROPS. Do not treat this functional schematic as a physical roof routing / clash clearance drawing. The indoor plan is geometrically checked; roof branches, levels, insulation, lengths, losses, support loads and OEM clearances remain to be designed from site information.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOF DISTRIBUTION AND ROOM RISER REGISTER

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-022

SCALE: NTS AT A1

REVISION

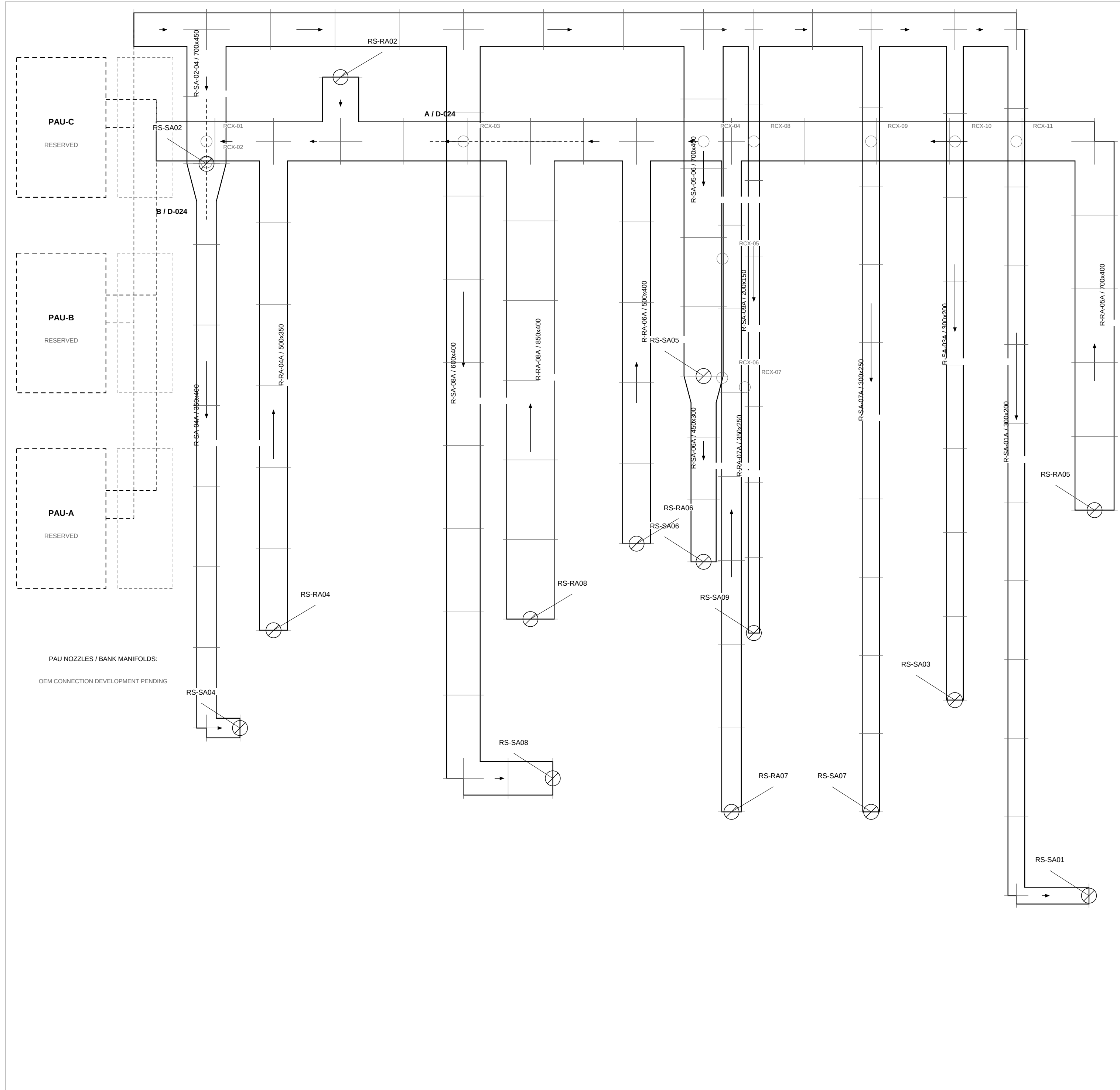
02

08 OCT 2026

SHEET 22 OF 50

MAIN DUCT AND RETURN - ROOF DUCT ROUTING PROPOSAL

REISSUED DX MAIN / RETURN OPTION || ROUTES / DUTIES RETAINED FROM REV12 || D-xxx REFERENCES = MDR-xxx || DX DETAILS MDR-033..035



ROOF ROUTES / LEVELS

31 connected route segments feed all 15 room supply / return drops. Coordinates use the supplied floor plan origin. Roof footprint and PAU reservation zones must be surveyed.

GREEN SUPPLY: bare BOD +4.800 m; main 600 x 1300 mm. BLUE RETURN: bare BOD +5.550 m; main 700 x 900 mm. Tall roof mains limit plan width; wind / parapet screening and support stability require structural design.

75 mm outdoor insulation each face assumed. Eleven roof plan overlaps have separated insulation envelopes; minimum calculated separation 150 mm. No upper return riser intersects lower supply insulation in this proposed geometry.

PAU-A / B / C reservations: 1600 x 2500 mm each, 1000 mm east service strip. Dashed nozzle stubs and bank manifolds are schematic interfaces awaiting manufacturer dimensions.

RCX tags: Roof_crossings_Rev12 register. A and B sections: D-024. Curbs and supports: D-030. Native centerline setting-out dimensions: D-032.

Service	Main clear mm	Bare BOD	Insulated top
SA	600 x 1300	+4.800	+6.175
RA	700 x 900	+5.550	+6.525

ACCESS / LOAD HOLD

Service strips are reserved for PAUs only. Walking platforms, safe roof access, guarding, filter pull-out, common heater / conditional humidifier weather enclosure, banks and lifting path require selected equipment layout. Roof proposal is not an approved installation.

Roof levels use provisional roof surface +4.200 m: user underside +4.000 m plus assumed 200 mm slab. Route lengths are measured proposals, not surveyed as-built lengths. All sizes are clear internal mm.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOF DUCT ROUTING PROPOSAL

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-023

SCALE: 1:50 AT A1

REVISION

02

08 OCT 2026

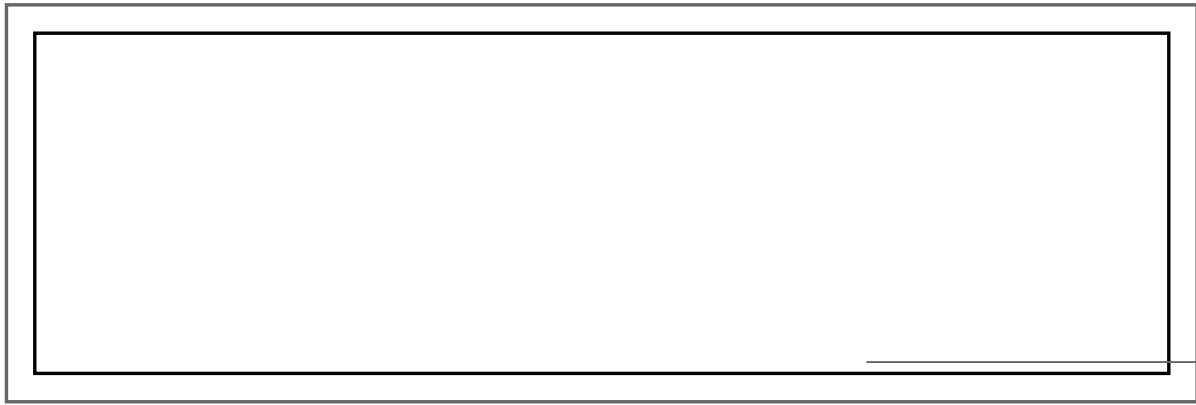
SHEET 23 OF 50

MAIN DUCT AND RETURN - ROOF CROSSING AND ROOM DROP SECTIONS

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

A RCX-03 / SA08 x RA MAIN | 1:20

SA 600 x 400 / RA 700 x 900



RA BOD +5.550
200 BETWEEN INSULATION AT RCX-03

SA BOD +4.800

ROOF +4.200*

SLAB U/S +4.000

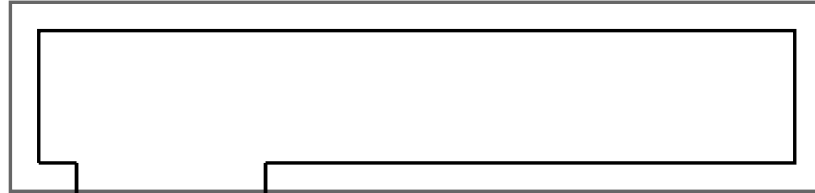
MINIMUM CEILING +3.300

* Proposed dimensions / levels. Structural slab thickness, beam downstands, curb opening, firestop, weatherproof flashing, pipe / cable routes and support loads require coordination. No roof penetration may be set out from this proposal alone.

B RS-SA02 DROP | VERTICAL 1:20 / HORIZONTAL NTS

TWO ELBOWS / TRANSITION AT ROOF

BARE 500 x 350; RISE 1.400 m*



ROOF BOD +4.800*

CURB 300* / WEATHER SEAL

ROOF +4.200*

SLAB U/S +4.000

INDOOR BOD +3.400

MINIMUM CEILING +3.300

* Proposed dimensions / levels. Structural slab thickness, beam downstands, curb opening, firestop, weatherproof flashing, pipe / cable routes and support loads require coordination. No roof penetration may be set out from this proposal alone.

VERTICAL DEVELOPMENT / CONNECTION RULES

Riser straight-length schedules use proposed bare BOD differences at equal-size horizontal ends. Elbow development and differing roof header sizes are separate fitting allowances. Maintain 3.300 m finished ceiling; do not lower a room duct to clear a roof route.

Use shop-developed mitred elbows with turning vanes or selected radiused elbows. Square plan junctions on the roof sheet denote fitted tees; they are not four-way crosses. Every room drop has an independent riser ID and reserved curb opening.

Sections are coordination details, 1:20 at A1 where stated; equipment and curb symbols are NTS. Clear opening schedules: Roof_curb, Rev12. Structural capacity, waterproofing and fire ratings have not been approved.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOF CROSSING AND ROOM DROP SECTIONS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-024

SCALE: AS SHOWN AT A1

REVISION

02

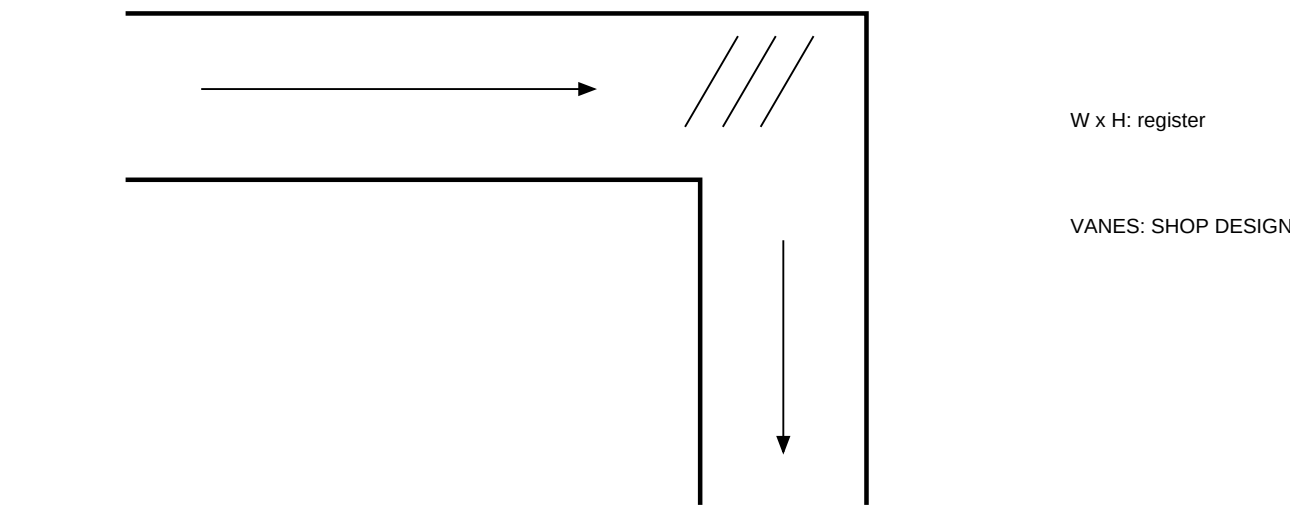
08 OCT 2026

SHEET 24 OF 50

MAIN DUCT AND RETURN - FITTING DEVELOPMENT AND FABRICATION RULES

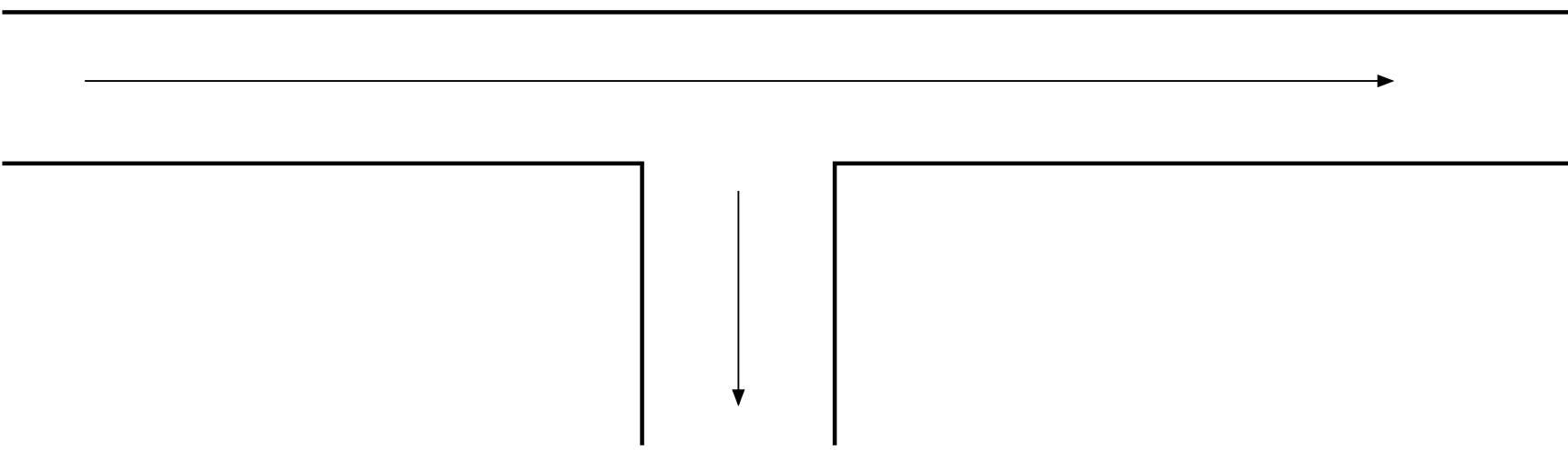
REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

01 MITRED ELBOW / VANES



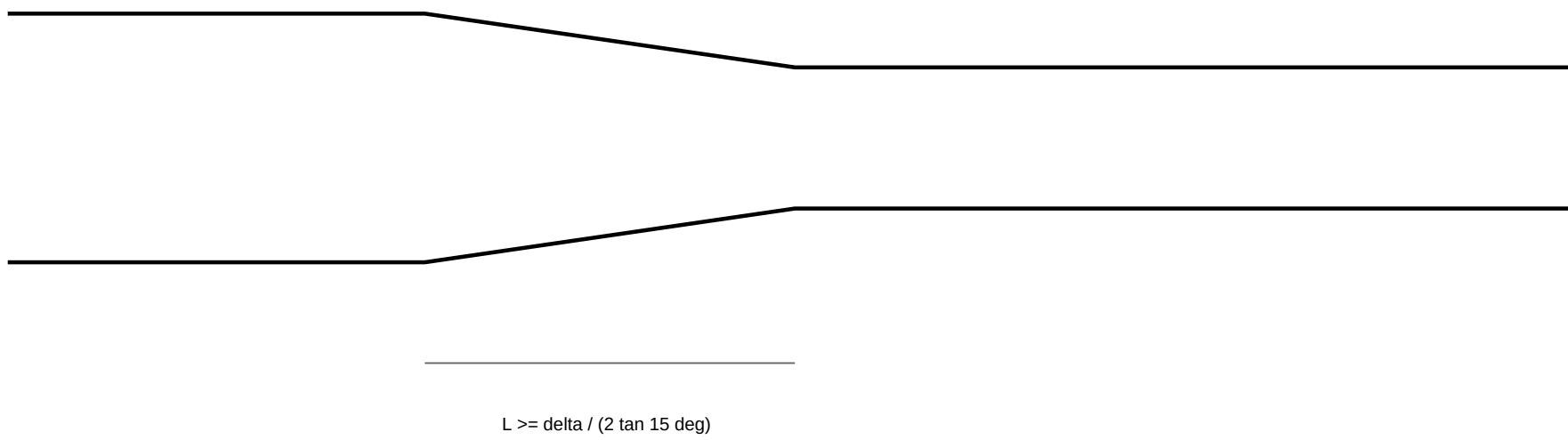
SQUARE THROAT / 45 DEG MITRES

02 THREE-LEG TEE



THREE LEGS ONLY; OPEN BRANCH THROAT

03 NECK / RISER TRANSITION



MAX 30 DEG INCLUDED ANGLE PROPOSED

ALL FITTINGS / DIMENSION SCHEDULE

Fitting_design_Rev12 contains 236 tagged loss / geometry allowances covering indoor path bends, joints, terminal adapters, balancing / conditional fire dampers and roof / riser connections. It is a design development register, not an OEM bill of materials.

For an actual two-dimensional reducer, proposed L = max(300 mm, max(abs(W1-W2), abs(H1-H2)) / (2 tan 15 deg)). Hold the bottom level only if the selected terminal / device height permits; otherwise show the offset in the shop section.

Roof main sizes are constant along each service spine, so no unshown step reductions occur. Branch width / height change at the tee is developed within a shop boot. Keep main and branch walls open at the connected throat.

Bend / tee lengths must fit the reserved straight station before moving any VCD, RCD, MFD, EH or access door. Reference CAD glyph dimensions are symbols; actual face-to-face lengths and actuator envelopes require an OEM selection.

Fitting	Riser	Clear W x H	Elbows / K each
F12-207	RS-SA02	500.0 x 350.0	2 / 0.50 assumed
F12-209	RS-RA02	650.0 x 400.0	2 / 0.50 assumed
F12-211	RS-SA04	350.0 x 400.0	2 / 0.50 assumed
F12-213	RS-RA04	500.0 x 350.0	2 / 0.50 assumed
F12-215	RS-SA08	600.0 x 400.0	2 / 0.50 assumed
F12-217	RS-RA08	850.0 x 400.0	2 / 0.50 assumed
F12-219	RS-SA05	600.0 x 350.0	2 / 0.50 assumed
F12-221	RS-RA05	700.0 x 400.0	2 / 0.50 assumed
F12-223	RS-SA06	450.0 x 300.0	2 / 0.50 assumed
F12-225	RS-RA06	500.0 x 400.0	2 / 0.50 assumed
F12-227	RS-SA09	200.0 x 150.0	2 / 0.50 assumed
F12-229	RS-SA07	300.0 x 250.0	2 / 0.50 assumed
F12-231	RS-RA07	350.0 x 250.0	2 / 0.50 assumed
F12-233	RS-SA03	300.0 x 200.0	2 / 0.50 assumed
F12-235	RS-SA01	300.0 x 200.0	2 / 0.50 assumed

ACTUAL ROOF BRANCH TAPERS - mm

Route	Inlet	Outlet	Developed L
R-SA-04A	700x450	350.0x400.0	675
R-SA-06A	700x400	450.0x300.0	475

All K values on this issue are explicit preliminary allowances, not selected catalogue coefficients. Adapter_details_Rev12 retains the actual connected terminal adaptor coordinates / proposed lengths. Fabrication approval remains open.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

FITTING DEVELOPMENT AND FABRICATION RULES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-025

SCALE: NTS AT A1

REVISION

02

08 OCT 2026

SHEET 25 OF 50

MAIN DUCT AND RETURN - SYSTEM PRESSURE LOSS AND FAN DUTY BUDGET

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

System	Duty each L/s	Estimated external Pa	Budget Pa +20%	Low / high sensitivity Pa
PAU bank - combined external SA + RA path	1848.90	329.9	400	169.7 / 490.1
EF-03 duty / standby	147.60	30.7	40	15.7 / 45.7
EF-09 duty / standby	64.89	33.6	50	17.6 / 49.6

CRITICAL TERMINAL PATHS - ALL 52 IN REGISTER

Terminal	Air	Straight Pa	Fittings Pa*	Heater Pa*	Common Pa*	Terminal Pa*	Total Pa*
SD-05-01	SA	5.802	55.411	25	150	30	266.212
SD-05-02	SA	5.728	55.411	25	150	30	266.138
SD-05-03	SA	5.827	55.147	25	150	30	265.974
SD-05-04	SA	5.347	54.555	25	150	30	264.901
SD-05-05	SA	5.273	54.555	25	150	30	264.828
SD-06-02	SA	5.962	43.929	25	150	30	254.892
SD-04-02	SA	5.911	43.476	25	150	30	254.387
SD-06-03	SA	6.124	42.891	25	150	30	254.014
SD-04-03	SA	6.005	42.646	25	150	30	253.651
SD-06-01	SA	5.939	40.843	25	150	30	251.782
SD-04-01	SA	5.514	40.284	25	150	30	250.798
SD-08-06	SA	7.099	54.359	0	150	30	241.458

METHOD / BUDGET LIMITS

Straight friction: Darcy-Weisbach / Colebrook, rho 1.20 kg/m3, mu 1.81e-5 Pa.s, galvanized roughness 0.09 mm, hydraulic diameter 2WH/(W+H). Loss = L x friction + sum(K rho V2/2) + component allowance.

Assumed K: elbows 0.50; tee branch 1.00; tee straight run 0.20; joint bend 0.50; adapter 0.25; balancing damper 2.00; conditional fire damper 1.00. Terminal 30 Pa, room heater 25 Pa, supply common external equipment 150 Pa. Sensitivity scales all assumed local / equipment losses to 50% and 150%.

PAU budget adds worst SA and RA terminal paths. Unit internal coils / dirty filters belong in the OEM fan calculation with explicit external-pressure convention. PAU nozzle / bank manifolds, OA intake, selected dampers, silencers and exhaust discharge are not measured in this model; no fan is selected.

Pressure paths include measured proposed roof plan segments and provisional riser rises, plus retained indoor centerlines. Budget values are screening estimates; fan curves, dirty-filter losses and rotating-duty bank cases require OEM confirmation.

TAQA TRANSMISSION N-19082-RO-012 PUMPING STATION STANDARDIZATION ADMINISTRATION HVAC DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE	DRAWING TITLE SYSTEM PRESSURE LOSS AND FAN DUTY BUDGET FOR COORDINATION CHECKED / APPROVED: PENDING	DRAWING NUMBER ADM-HVAC-MDR-026 SCALE: NTS AT A1	REVISION 02 08 OCT 2026 SHEET 26 OF 50
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MAIN DUCT AND RETURN - AIR BALANCE AND TERMINAL SELECTION CRITERIA

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

GF	Supply L/s	Return L/s	Known extract L/s	Net transfer / leakage L/s	Pressure target / verification
02	664.80	605.46	0.00	59.34	50 Pa / not verified
04	449.20	389.86	0.00	59.34	50 Pa / not verified
08	884.60	825.26	0.00	59.34	50 Pa / not verified
05	777.70	718.36	0.00	59.34	50 Pa / not verified
06	418.20	358.86	0.00	59.34	50 Pa / not verified
09	46.20	0.00	64.89	-18.69	25 Pa / not verified
07	168.20	108.86	0.00	59.34	50 Pa / not verified
03	147.60	0.00	147.60	0.00	25 Pa / not verified
01	141.30	0.00	0.00	141.30	25 Pa / not verified

OUTDOOR AIR / PRESSURE ACTION

Mass-balance minimum OA at the retained SA / RA duty is 691.12 L/s, before any code or process ventilation requirement. Current OA reference 420.70 L/s is short by 270.42 L/s. This is a required design reconciliation, not a selected OA duty.

Known extract is 212.49 L/s; residual exfiltration / transfer is 478.63 L/s less the unresolved clean-agent extract. Verify each door / transfer path and exterior leakage before asserting +50 / +25 Pa.

Toilets have -18.69 L/s net supply minus extract; kitchen has zero net. Positive pressure targets need owner / process / hygiene reconciliation and a measured leakage basis. Do not reduce extract simply to obtain a pressure label.

SUPPLY NECK / COVERAGE SCREEN - 53 ROW REGISTER

Representative	L/s	Neck	m/s	Half spacing m*
SD-02-01	166.20	350	1.727	2.2
SD-05-01	155.54	325	1.875	2.25
SD-04-01	149.73	325	1.805	1.626
SD-08-01	147.43	325	1.777	1.15
SD-06-01	139.40	300	1.972	4.65
SD-07-01	168.20	350	1.748	Room coverage required
SD-09-01	23.10	150	1.307	0.9
SD-03-01	147.60	325	1.779	Room coverage required
SD-01-01	141.30	300	1.999	Room coverage required

* Half spacing is a coverage screening distance, not certified throw. Require OEM pressure loss, NC and isothermal / cooling throw at 0.25 m/s terminal velocity; check wall coverage, induction, ADPI and occupied-zone draught.

Proposed noise brief. NC <=30 meeting / rest / operator rooms; NC <=35 other occupied rooms, subject to owner / acoustic confirmation. No catalogue model, throw or NC has been invented.

Duty, neck velocities and same-type spacing are calculated from scheduled terminal coordinates. Air_balance_Rev12 and Terminal_performance_Rev12 are editable calculation / selection registers; existing duties are retained pending engineering reconciliation.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

AIR BALANCE AND TERMINAL SELECTION CRITERIA

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-027

SCALE: NTS AT A1

REVISION

02

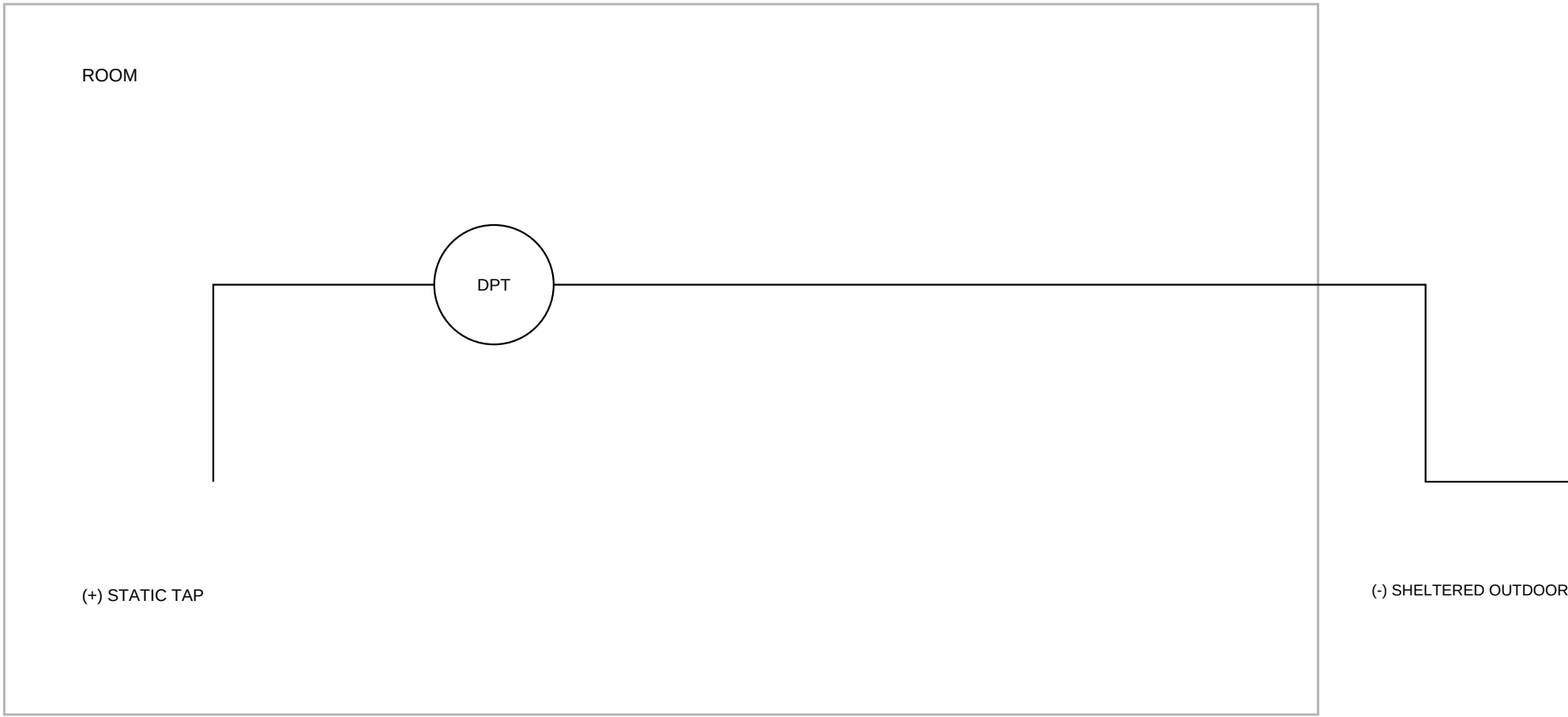
08 OCT 2026

SHEET 27 OF 50

MAIN DUCT AND RETURN - INSTRUMENT MOUNTING AND PRESSURE TAP DETAILS

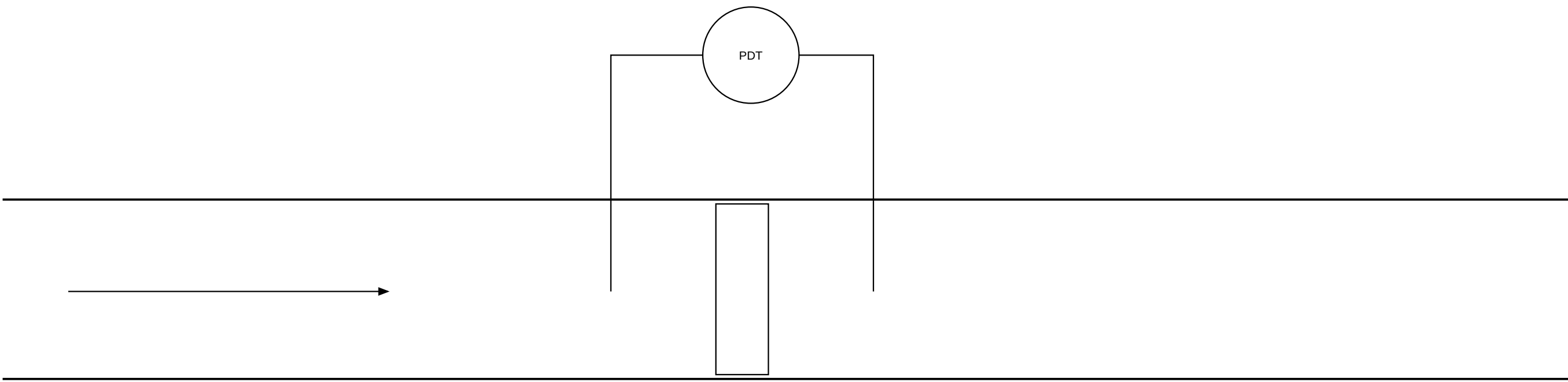
REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

A ROOM STATIC PRESSURE



ROOM TT / RHT: +1.500 m AFFL PROPOSED

B FILTER / FAN DIFFERENTIAL



PDI: SEPARATE LOCAL GAUGE

INSTALLATION SPECIFICATION - ALL 85 TAGS

Instrument_installation_Rev12 gives mounting, provisional ranges, pressure connections, wiring, access and commissioning requirements for every instrument. Tag requirement and I/O duties are retained from the approved input register, not replaced with generic symbols.

Proposed room DPT range -100..+100 Pa; fan / filter differential 0..1000 Pa; TT 0..50 degC; RH 0..100%; vibration 0..25 mm/s RMS. Ranges are selection proposals, not final alarm / trip settings.

Use labelled static pressure pickups and equal reference datums. Protect outdoor reference from wind / rain; avoid jet impingement, condensate traps and unserviceable tubing. Provide isolation and test ports per manufacturer.

COMMON-DUCT AIRFLOW / SENSOR ACCESS

FIT-SA-01 / FIT-RA-01: reserve 5Dh upstream and 3Dh downstream as a screening space allowance. Actual manufacturer measurement station / traversing method governs; no accuracy claim at the provisional roof position.

Main SA Dh = 0.821 m: screen 4.11 m upstream / 2.46 m downstream. Main RA Dh = 0.788 m: screen 3.94 m upstream / 2.36 m downstream. The current short bank-to-first-branch space does not prove these lengths; measurement station integration is an open hold.

Room temperature / humidity sensors avoid direct supply jets, exterior solar gain and heaters. VS installs on approved bearing / frame position. TSHH and AFS remain accessible for test / manual reset; selected heaters govern technology and settings.

Installation detail references D-028 / D-029 supplement D-012 to D-016 and the 85-row index. No final cable route, panel terminal number, approved sensor accuracy or OEM setpoint is asserted.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

INSTRUMENT MOUNTING AND PRESSURE TAP DETAILS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-028

SCALE: NTS AT A1

REVISION

02

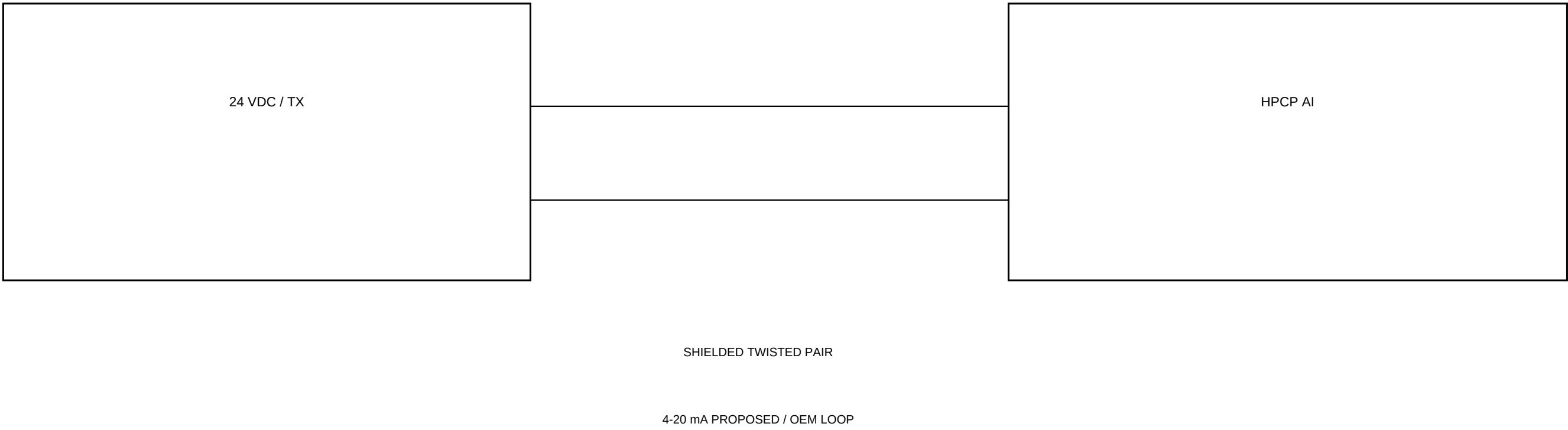
08 OCT 2026

SHEET 28 OF 50

MAIN DUCT AND RETURN - I&C WIRING AND HEATER SAFETY INTERFACES

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

A ANALOG TRANSMITTER / AI



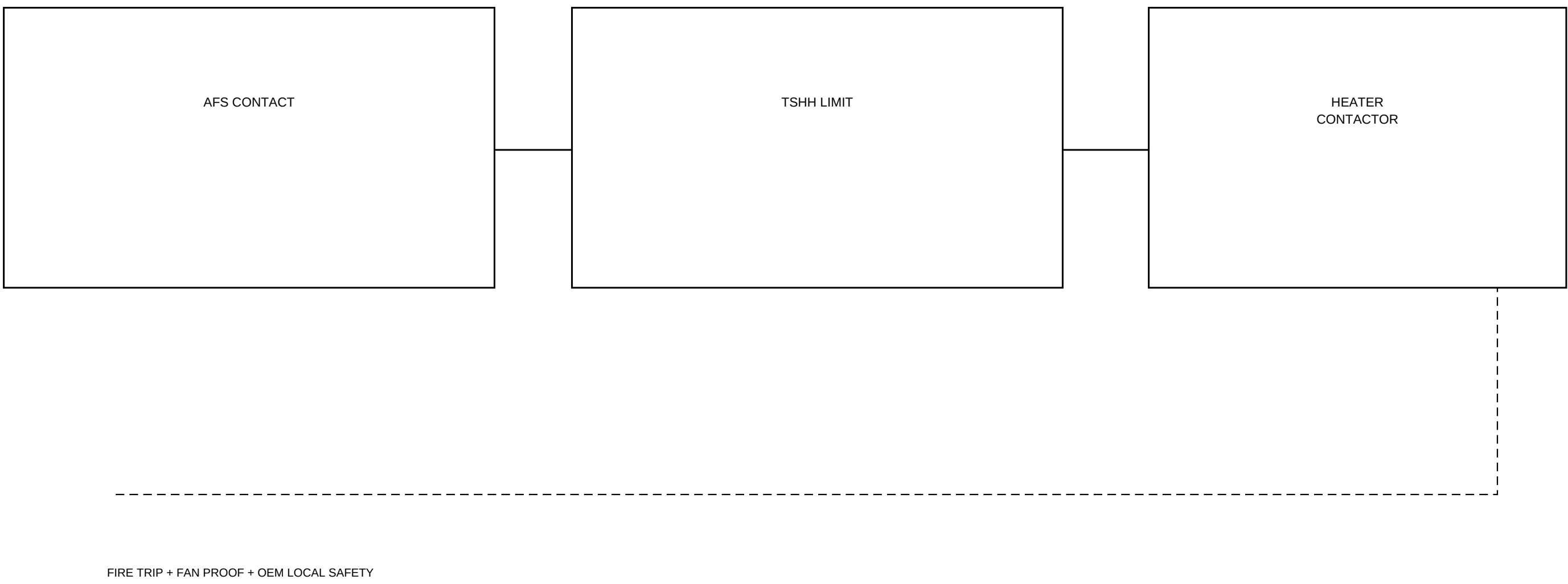
SIGNAL / PANEL REQUIREMENTS

AI proposals use isolated 4-20 mA channels and shielded twisted pairs. Confirm transmitter supply, two / three / four wire circuit, loop burden and earthing with OEM. Earth the cable shield at the panel end per project EMC design; segregate from mains / VFD cables.

DI proposals are dry contacts. Contact state, wetting voltage, galvanic isolation and fail-state indication require manufacturer wiring. Local PDI is not an AI. COMM energy meter protocol, CT ratio, address and termination remain pending.

Every tag maps to the unchanged 275-row I/O register. Final channel numbers, junction box locations and cable lengths await panel design and coordinated routes. Commission with traceable point-to-point checks and calibrated simulated signals.

B HARDWIRED HEATER PERMISSIVES



SAFETY / CAUSE AND EFFECT

Six AFS and six independent TSHH heater provisions remain present. Loss of airflow or high-limit trip inhibits heat through the OEM hardwired circuit. PLC heating demand cannot bypass either permissive. Alarm feedback reports to HPCP.

Confirmed fire stops all HVAC fans and heaters. No automatic fire extraction / purge is added. Fire damper rating, fail position, reset procedure and boundary quantity follow approved fire strategy. Manual reset is required after confirmed fire.

This is an interface diagram, not an electrical terminal drawing. Complete the heater OEM wiring, upstream electrical protection, safety relay architecture and fire interface review before procurement.

Input requirement and I/O assignments are retained. Final wiring and approved safety settings require selected hardware. Instrument_Installation_Rev12 is the 85-row specification companion.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

I&C WIRING AND HEATER SAFETY INTERFACES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-029

SCALE: NTS AT A1

REVISION

02

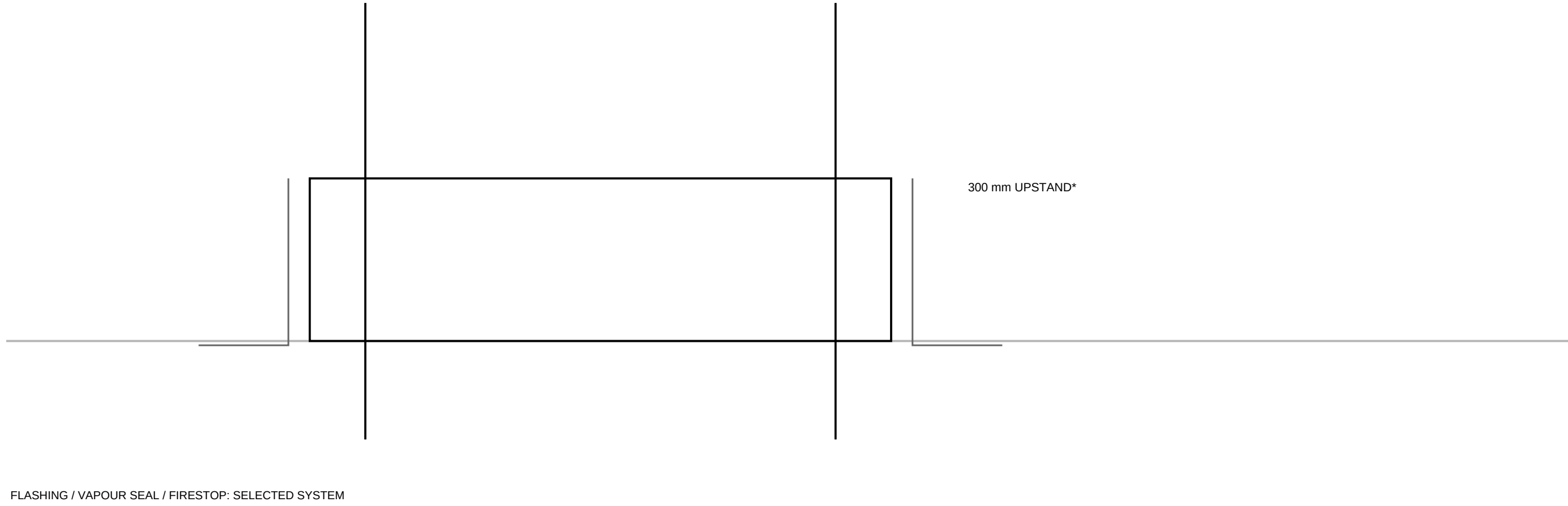
08 OCT 2026

SHEET 29 OF 50

MAIN DUCT AND RETURN - ROOF CURB, SUPPORT AND WEATHERPROOFING

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

A PROPOSED CURB SECTION - NTS



CURBS / PENETRATIONS

15 curb reservations have X / Y coordinates and opening envelopes: clear duct plus 75 mm insulation and 100 mm installation allowance each side. Upstand 300 mm above proposed roof surface; confirm finished waterproofing datum and local exposure.

Coordinate opening clearances, sleeves, firestop and insulated riser separation with structural and fire disciplines. Keep weatherproofing continuous; selected flashing / vapour seal and cladding joints must shed rain. Do not cut a slab or beam from this drawing.

Outdoor insulation 75 mm is provisional. Specify thermal conductivity, vapour barrier, sealed cladding, weather / UV resistance and fire classification from approved project materials. It is not a declared UAE average.

B ADJUSTABLE SUPPORT FRAME - NTS



SUPPORT / ACCESS PROCUREMENT HOLD

157 support station reservations, maximum nominal 1500 mm route spacing. Deduplicate junction stations and adjust fitting / equipment supports in shop drawings. Geometry alone does not select a bracket or roof anchor.

Support load table remains TBC until gauge, fitting weight, insulation / cladding, actuator, wind, seismic requirements and frame self-weight are known. Tall roof mains need wind stability / bracing and maintenance platform design.

Verify structure and membrane bearing / protection before choosing nonpenetrating ballast or anchors. PAU lifting, coil / filter withdrawal and heater / humidifier service envelopes require a complete manufacturer footprint.

* Proposal only, Roof_curbs_Rev12 and Roof_supports_Rev12 provide setting-out reservations, not anchor design or certified structural loads. Roof surface +4.200 m assumes 200 mm structural slab above the confirmed +4.000 m underside.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOF CURB, SUPPORT AND WEATHERPROOFING

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-030

SCALE: NTS AT A1

REVISION

02

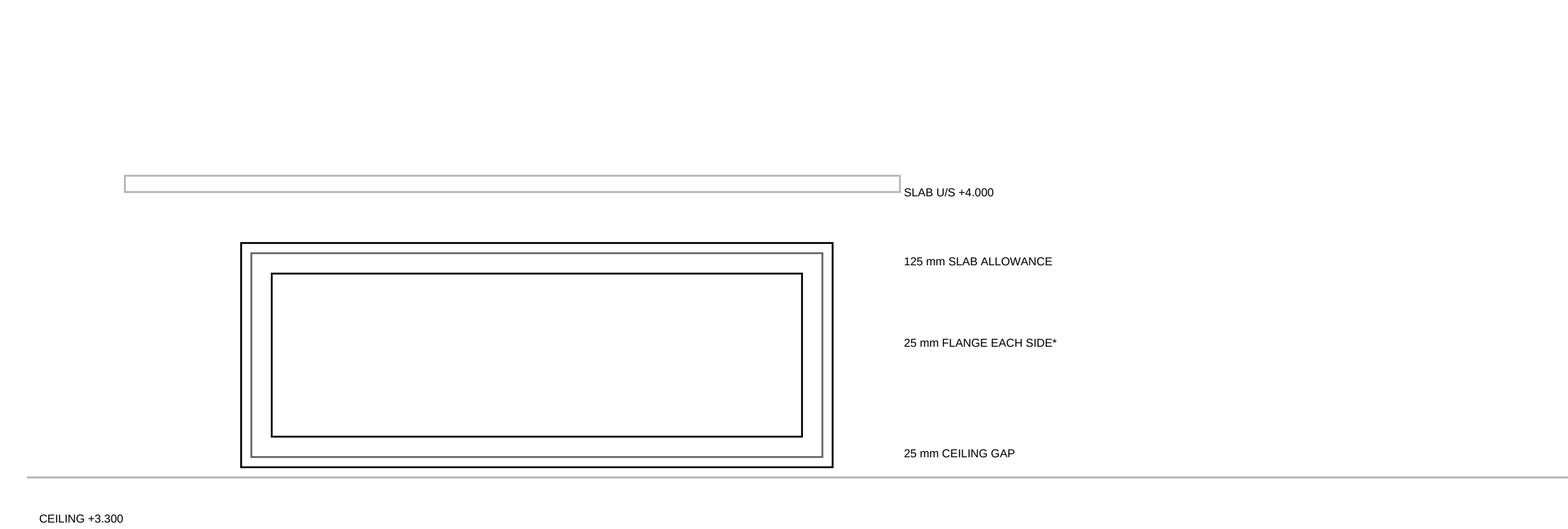
08 OCT 2026

SHEET 30 OF 50

MAIN DUCT AND RETURN - CEILING COMPONENT AND ACCESS ENVELOPES

REISSUED DX MAIN / RETURN OPTION | ROUTES / DUTIES RETAINED FROM REV12 | D-xxx REFERENCES = MDR-xxx | DX DETAILS MDR-033_039

A DEEPEST DUCT / FLANGE | 1:10



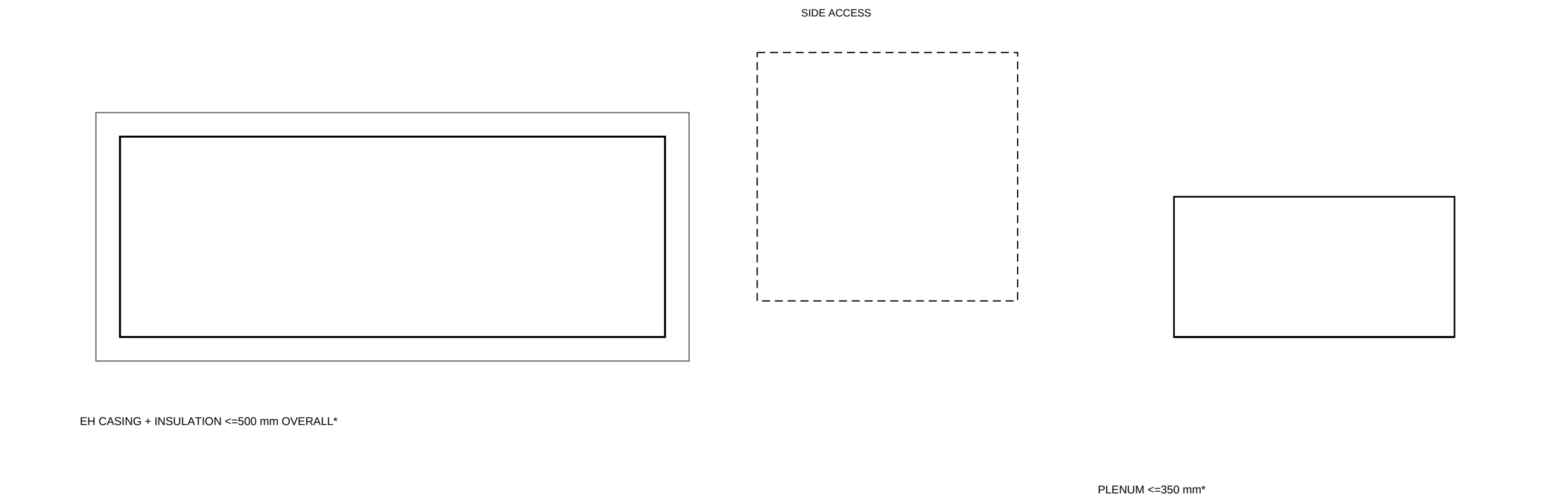
DUCT + FLANGES

Deepest bare depth 400 mm, BOD +3.400 m, 50 mm insulation each face, proposed 25 mm flange allowance beyond insulation: overall reserved depth 550 mm. Lowest +3.325 m; highest +3.875 m. Ceiling gap reduces to 25 mm; slab allowance reduces to 125 mm.

This is a fit screen, not a maintenance clearance. Hanger / beam / other-service and flange shape / vapour seal require shop coordination. An actuator above the duct can exceed this envelope: select a side actuator / lower-profile assembly or locally reroute.

Installation_envelopes_Rev12 checks every indoor duct and specifies room-heater procurement limits. No selected component envelope is represented as verified.

B ROOM HEATER / PLENUM | NTS



HEATER / TERMINAL PROCUREMENT LIMITS

Room heater casing plus insulation must fit between +3.350 and +3.850 m (500 mm overall) unless separately coordinated. Reserve side service access and an aligned ceiling hatch for reset / removal. OEM isolation and thermal clearances govern.

A 350 mm overall terminal plenum above ceiling +3.300 m reaches +3.650 m. Provide an offset / vertical neck to the branch; verify the actual plenum, collar and flex radius in the 700 mm ceiling void. A 600 mm face is not a 600 mm-deep plenum.

All access doors and actuators need a removal path. Final hatch dimensions, heater face-to-face length and damper / actuator position cannot be approved from symbolic plan blocks. If OEM envelopes exceed limits, revise the local branch before release.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

CEILING COMPONENT AND ACCESS ENVELOPES

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-031

SCALE: AS SHOWN AT A1

REVISION

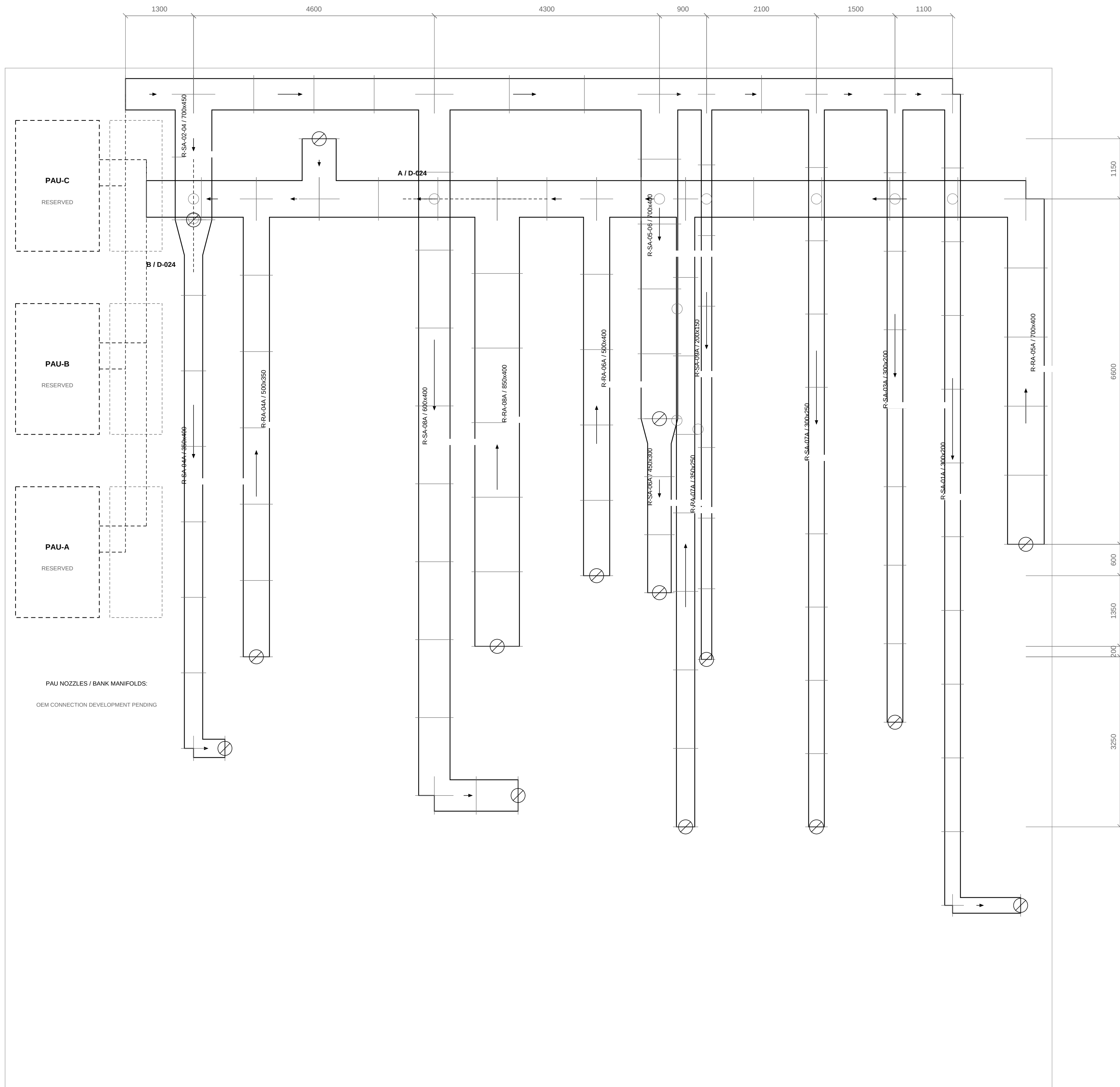
02

08 OCT 2026

SHEET 31 OF 50

MAIN DUCT AND RETURN - ROOF SETTING-OUT AND ROOM DROP REGISTER

REISSUED DX MAIN / RETURN OPTION || ROUTES / DUTIES RETAINED FROM REV12 || D-xxx REFERENCES = MDR-xxx || DX DETAILS MDR-033..035



PROPOSED ROOF DROP COORDINATES - mm

Riser	X	Y	Clear mm	Rise m*
RS-SA02	3600	16600	500 x 350	1.4
RS-RA02	6000	18150	650 x 400	2.15
RS-SA04	4200	6500	350 x 400	1.4
RS-RA04	4800	8250	500 x 350	2.15
RS-SA08	9800	5600	600 x 400	1.4
RS-RA08	9400	8450	850 x 400	2.15
RS-SA05	12500	12800	600 x 350	1.4
RS-RA05	19500	10400	700 x 400	2.15
RS-SA06	12500	9475	450 x 300	1.4
RS-RA06	11300	9800	500 x 400	2.15
RS-SA09	13400	8200	200 x 150	1.4
RS-SA07	15500	5000	300 x 250	1.4
RS-RA07	13000	5000	350 x 250	2.15
RS-SA03	17000	7000	300 x 200	1.4
RS-SA01	19400	3500	300 x 200	1.4

SETTING-OUT / REGISTER LINKS

13 native roof dimension segments supplement the 11 retained architectural dimensions. Individual route end coordinates and lengths are in Roof_routes_Rev12; curbs use the same XY origin.

Local origin is the supplied floor drawing (0,0), not a surveyed benchmark. Verify parapets, equipment plinths, openings, wind screening, beam grids and vertical datums before establishing roof setting-out.

* Rises use provisional roof levels; bend / reducer development and selected PAU nozzle connections are separate. Bank interface stubs are schematic, not dimensioned equipment outlets.

Source footprint 20000 x 19500 mm. Dimensions indicate proposed route centerline coordinates. Nominal units mm / m AFFL; proposed heights and fabrication details require professional review and coordinated OEM submissions.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOF SETTING-OUT AND ROOM DROP REGISTER

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-MDR-032

SCALE: 1:50 AT A1

REVISION

02

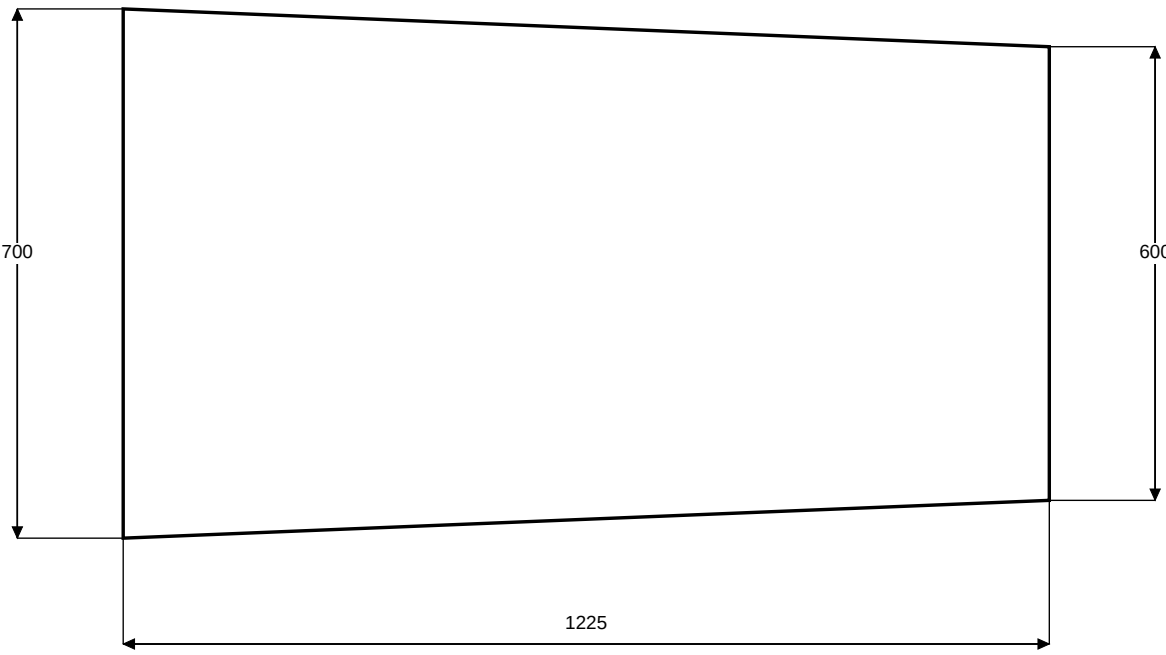
08 OCT 2026

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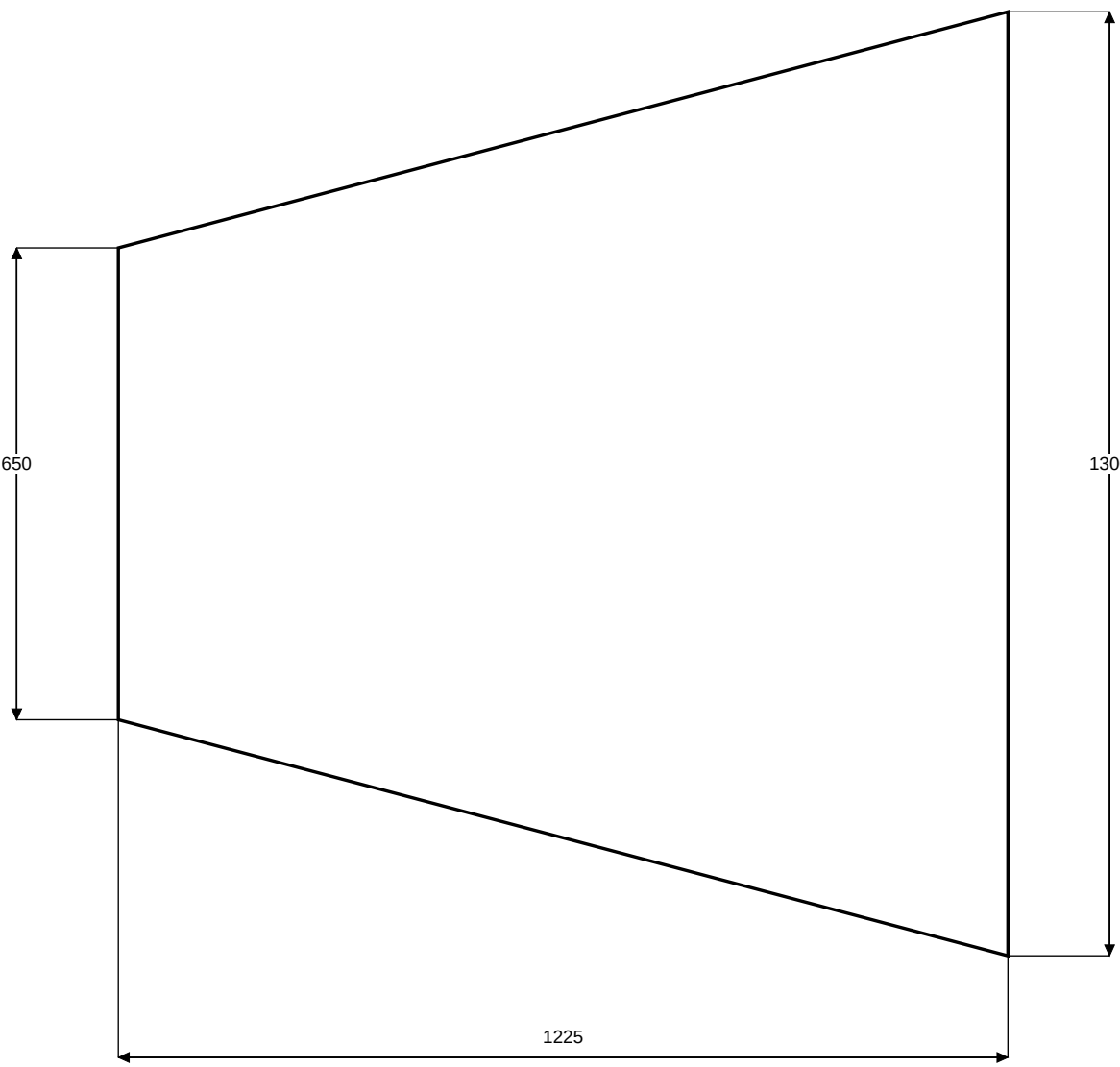
MAIN DUCT AND RETURN - DX PAU BRANCH / HEADER TRANSITION DEVELOPMENT

SA / PER DUTY UNIT / COORDINATION PROPOSAL

PLAN - CLEAR WIDTHS



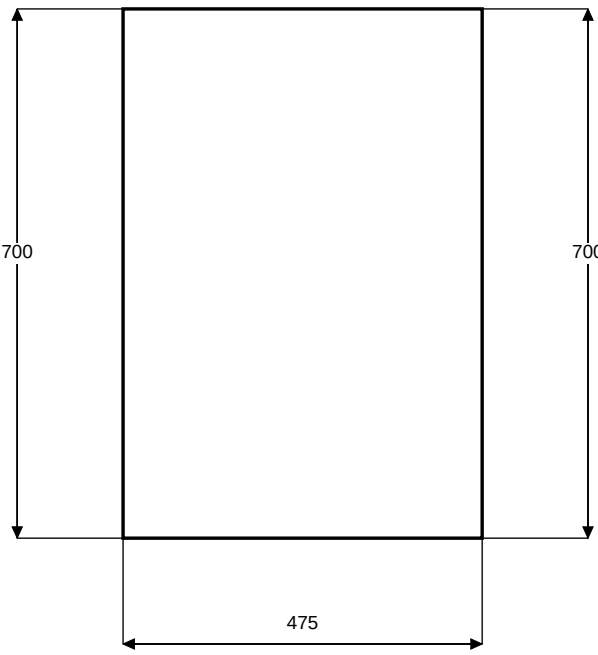
ELEVATION - CLEAR DEPTHS



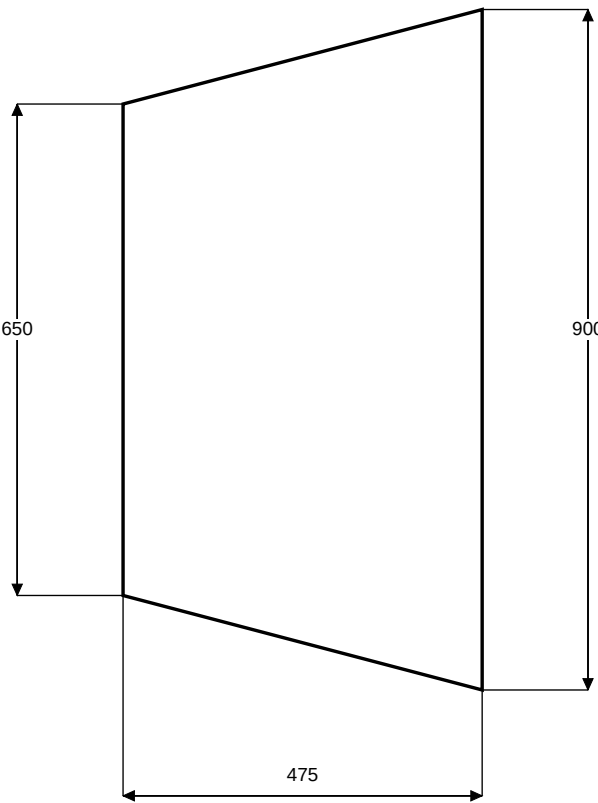
NOZZLE / FLEX / DAMPER / TEE BODY: OEM DEVELOPMENT

RA / PER DUTY UNIT / COORDINATION PROPOSAL

PLAN - CLEAR WIDTHS



ELEVATION - CLEAR DEPTHS



NOZZLE / FLEX / DAMPER / TEE BODY: OEM DEVELOPMENT

Connection	Duty L/s	Branch W x H	Header W x H	m/s	Taper L*
DX-SA-A	1848.90	700 x 650	600 x 1300	4.064	1225
DX-RA-A	1503.34	700 x 650	700 x 900	3.304	475
DX-SA-B	1848.90	700 x 650	600 x 1300	4.064	1225
DX-RA-B	1503.34	700 x 650	700 x 900	3.304	475
DX-SA-C	1848.90	700 x 650	600 x 1300	4.064	1225
DX-RA-C	1503.34	700 x 650	700 x 900	3.304	475

* Concentric development L >= max(300, max(delta W,delta H)/(2tan15deg)), rounded up to25 mm. Dimensions are proposed spool development, not selected PAU ports or fabrication release. Three branches; maximum two duty units. Header/branch joining and pressure losses remain OEM/shop coordination.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

DX PAU BRANCH / HEADER TRANSITION DEVELOPMENT

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-033

SCALE: 1:10 DETAILS AT A1

DRAWING REVISION 02

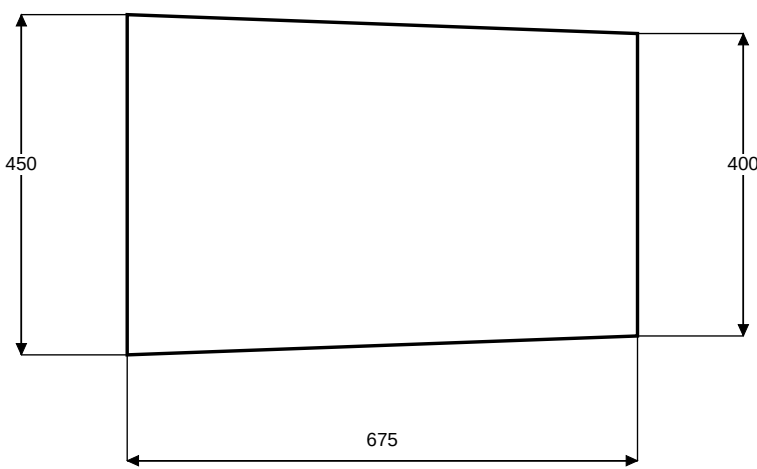
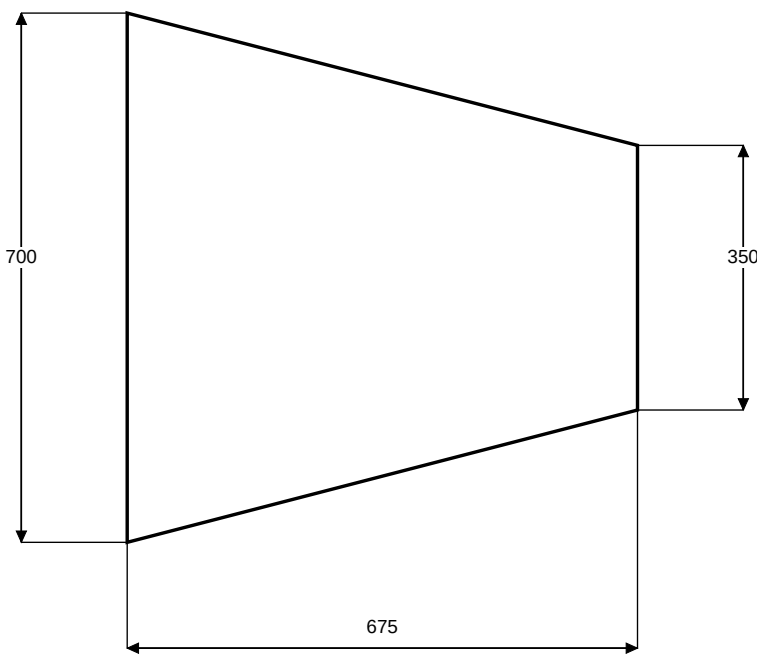
08 OCT 2026

SHEET 33 OF 50

MAIN DUCT AND RETURN - ACTUAL ROOF SPLIT / REDUCER / ELBOW DEVELOPMENT

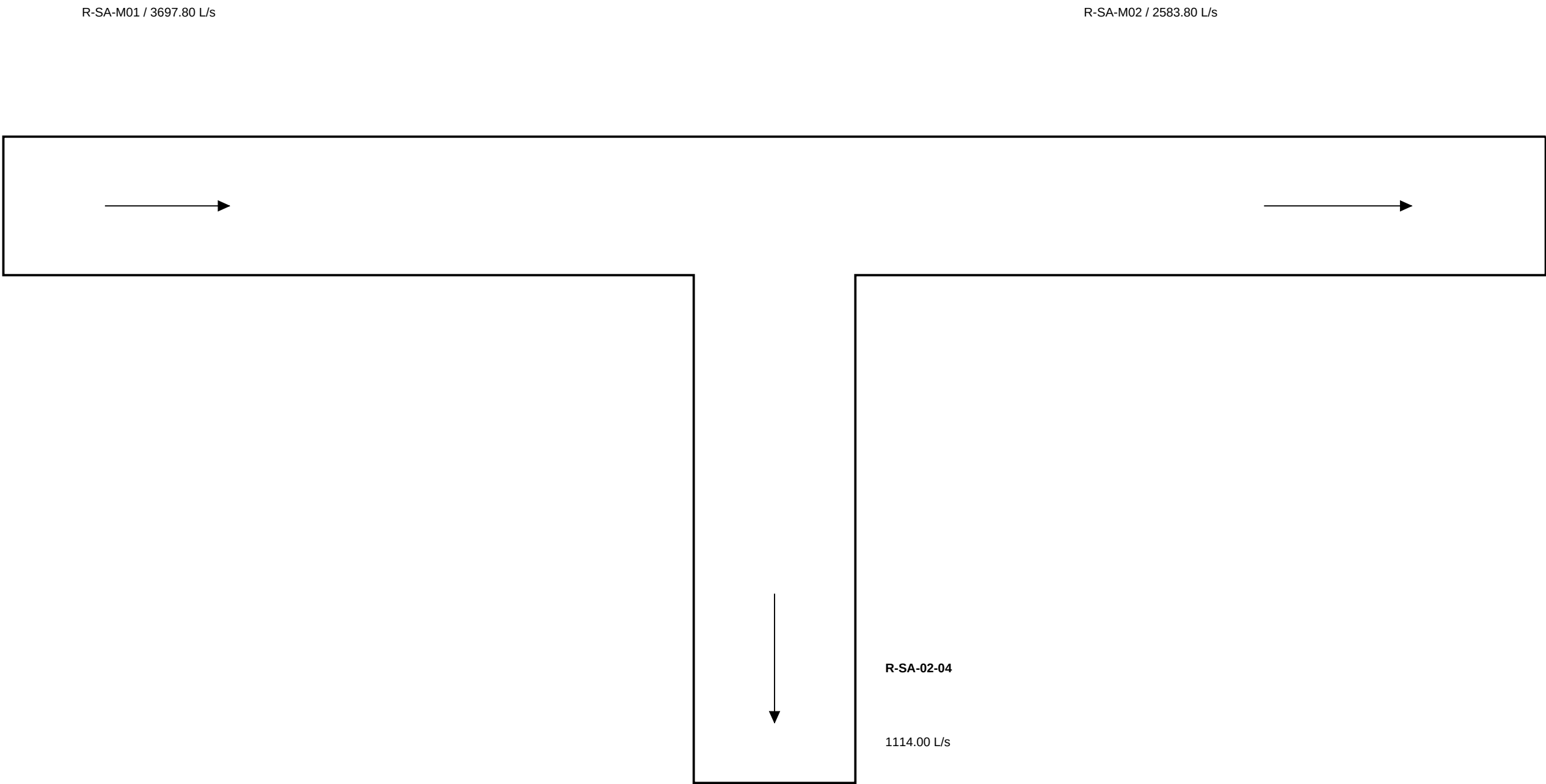
A R-SA-04A: ACTUAL REGISTERED TAPER

700 x 450 -> 350 x 400 / L675 / 449.20 L/s



PLAN ABOVE / ELEVATION BELOW / CLEAR INTERNAL mm

B ROOF SPLIT AT X3600 / Y19000



Open connected throat shown. SA spine600x1300; split branch700x450. Plan boot/radius and depth change are shop proposals. Keep bare BOD+4.800 m; coordinate every insulated offset with MDR-023/024 before release.

ELBOW / TRANSITION / ACCESS DEVELOPMENT

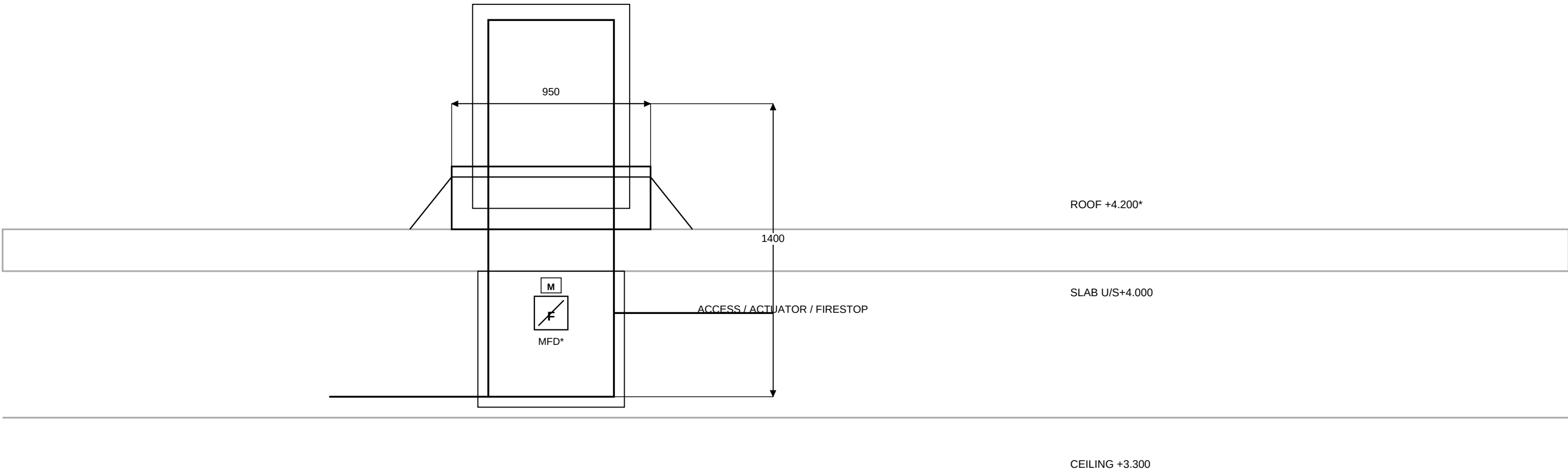
Item	Engineering requirement	Coordination evidence
Reducer	Concentric half-angle <=15deg proposed; actual space and fitted K checked	R-SA-04A taper675 retained; other route tapers in Roof_routes
Radius elbow	Proposed centerline radius >= clear width; vanes per selected construction	Actual bend quantities / radii in Fitting_design_Rev12
Boot / tee	Keep connected openings; no line across throat; detail all three elevations	Original actual room junction enlargements MDR-019
Damper / access	Device body + actuator + removal space; inspection hatch reachable	Components / Device_locations / MDR-028,031
Construction	Pressure class, gauge, reinforcement, seam and leakage class to approved spec	OEM/SMACNA-based shop design, project approval pending

A uses the retained physical route dimensions. B is a development diagram of the registered split, not a new roof route. Actual duct profiles, independent-service separation and riser locations remain as MDR-001/023.

ADMINISTRATION HVAC / DX PAU OPTION	DRAWING TITLE	ADM-HVAC-MDR-034	DRAWING REVISION 02
COMMON SUPPLY MAIN + COMMON RETURN MAIN	ACTUAL ROOF SPLIT / REDUCER / ELBOW DEVELOPMENT		08 OCT 2026
FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION	ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING	SCALE: 1:10 REDUCER / NTS BOOT AT A1	SHEET 34 OF 50

MAIN DUCT AND RETURN - ROOM DROP / ROOF CURB / FIRESTOP INSTALLATION

RS-SA08 / TELECOM / 600x400



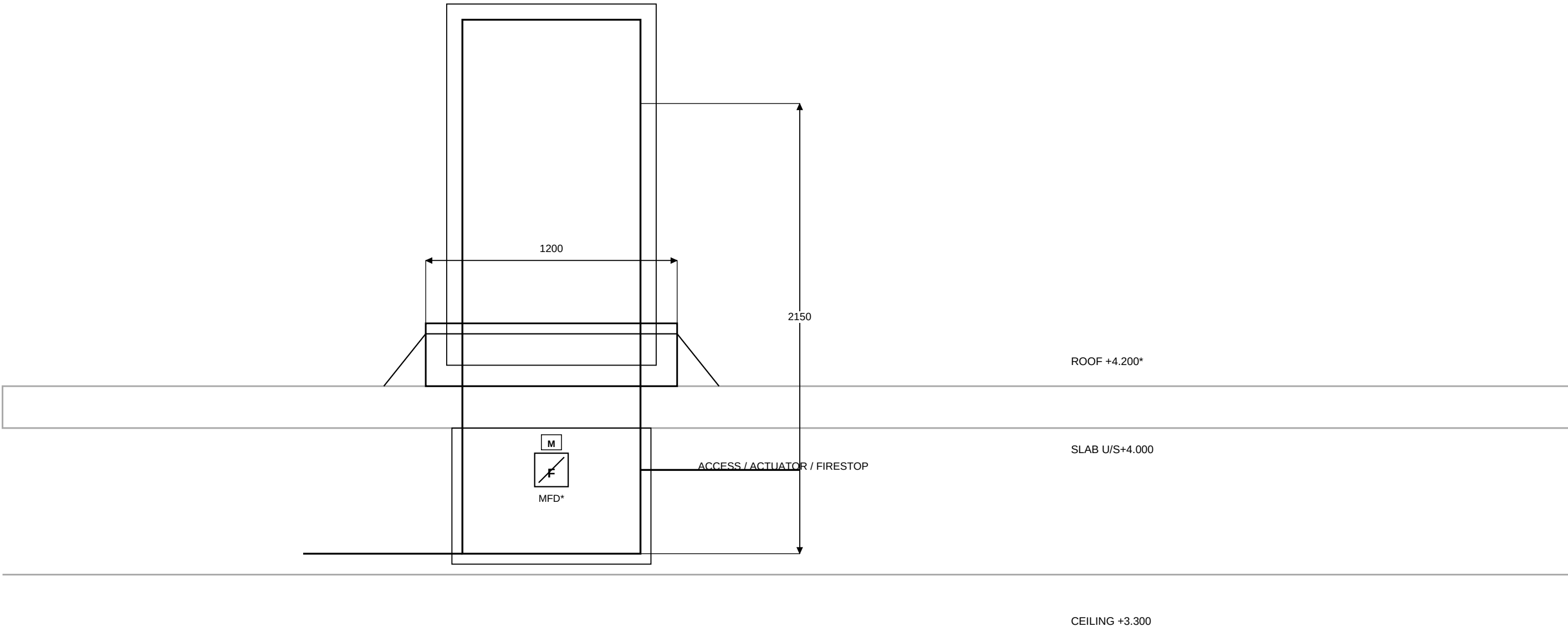
CURB OPENING 950.0 x 750.0 mm*

Diagram locates the reserved shaft, curb300 mm upstand, weather flashing and conditional fire damper. Duct cross-section width / level rise are scaled; exact 3D elbow, vertical-transition/roof-to-room insulation joint, sleeve/firestop and selected damper body must be shop-developed.

For all15 shaft coordinates / openings use MDR-022/032 and the Roof_risers / Roof_curbs tables. Local depth400 with50 mm insulation +25 mm flange allowance leaves25 mm ceiling gap and125 mm slab allowance; actual components/beam access remain open.

* Roof+4.200 assumes200 mm slab; outdoor75 / indoor50 mm insulation and curb/opening allowances are proposals. Installed elevations not assigned. RA05 curb extends25 mm beyond nominal footprint: resolve roof/parapet/shaft setting-out before approval.

RS-RA08 / TELECOM / 850x400



CURB OPENING 1200.0 x 750.0 mm*

Diagram locates the reserved shaft, curb300 mm upstand, weather flashing and conditional fire damper. Duct cross-section width / level rise are scaled; exact 3D elbow, vertical-transition/roof-to-room insulation joint, sleeve/firestop and selected damper body must be shop-developed.

For all15 shaft coordinates / openings use MDR-022/032 and the Roof_risers / Roof_curbs tables. Local depth400 with50 mm insulation +25 mm flange allowance leaves25 mm ceiling gap and125 mm slab allowance; actual components/beam access remain open.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

ROOM DROP / ROOF CURB / FIRESTOP INSTALLATION

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-035

SCALE: VERTICAL 1:20 AT A1

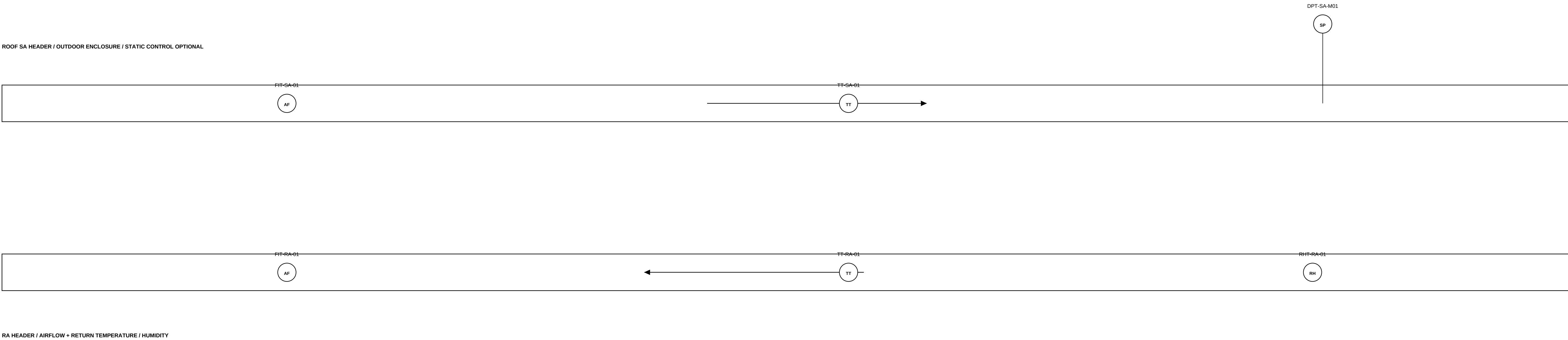
DRAWING REVISION 02

08 OCT 2026

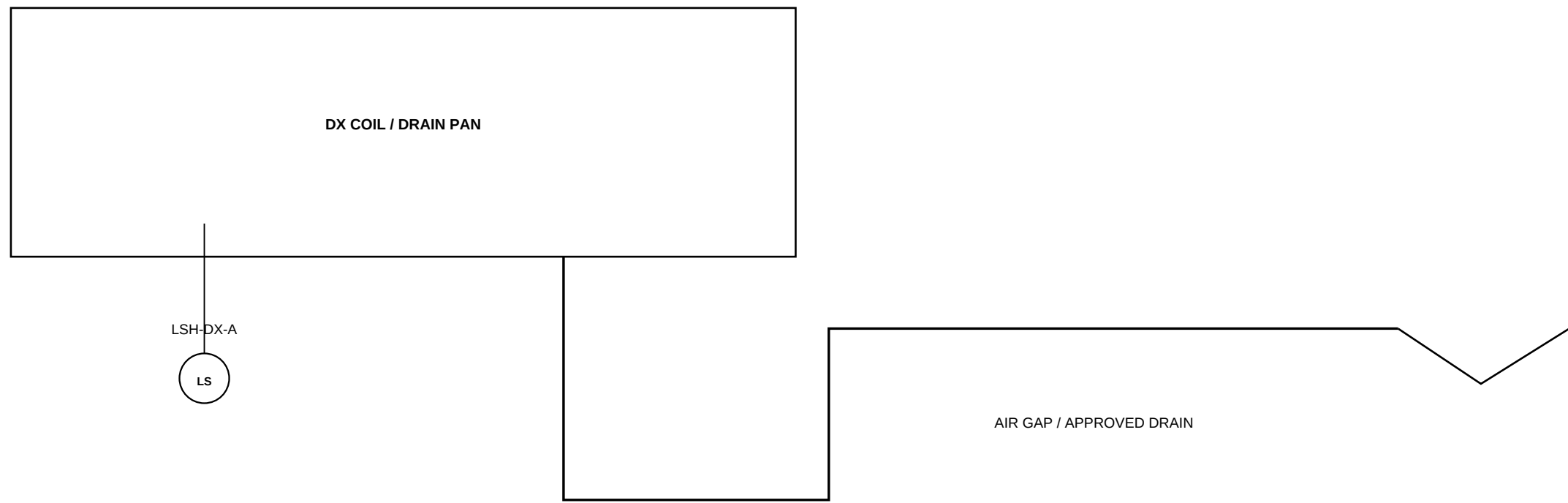
SHEET 35 OF 50

MAIN DUCT AND RETURN - COMMON HEADER INSTRUMENTS / CONDENSATE / ACCESS

A COMMON AIR STATIONS AND SENSORS



B NEGATIVE-PRESSURE DRAIN-PAN TRAP / SERVICE



Trap seal / leg dimensions selected from the actual maximum negative pan pressure and OEM instructions; do not select from the provisional fan budget. Prime, provide cleaning access, slope continuous drain, prevent overflow and protect insulation/vapour seals. Apply separately to PAU-A/B/C.

C MOUNTING / MEASUREMENT / ACCESS RULES

Point	Detailed requirement
FIT	Selected station straight runs / calibration and min-flow check; bank elbows can invalidate station
TT/RHT	Serviceable insertion, representative air; no heater radiant bias / condensate contact
Static DPT	Flush static pickup, sealed labelled tubing, weather enclosure; station/setpoint OEM review
Pressure taps	DPT-SF across fan; filter PDT/PTDI tap pairs; identify H/L, isolation/testing access
DX refrigerant	Pressure taps / transducers only at approved factory service points
Maintenance	Filter pull-out, coil face, trap, compressor, damper and electrical access to OEM drawings

AIR85 installation specifications are restored in the registers and MDR-028/029. This sheet adds the DX/common-header requirements. LSH example uses PAU-A tag; B/C have the same factory-reviewed function. No trap dimensions or refrigerant service taps are fabricated.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

COMMON HEADER INSTRUMENTS / CONDENSATE / ACCESS

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-036

SCALE: NTS AT A1

DRAWING REVISION 02

08 OCT 2026

SHEET 36 OF 50

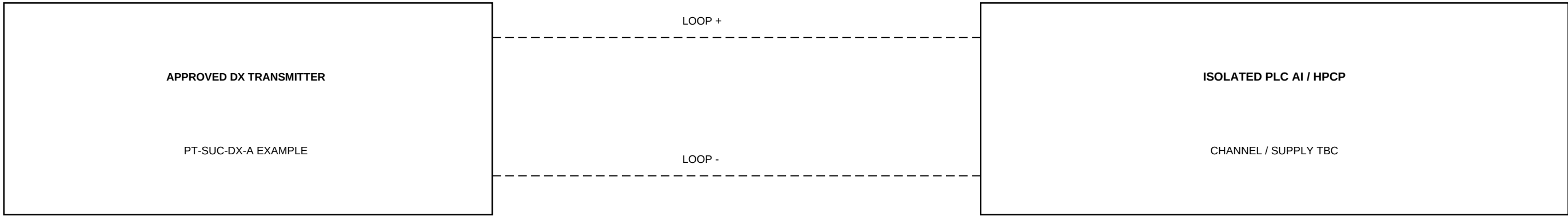
MAIN DUCT AND RETURN - DX / FAN / DAMPER / FIRE CAUSE AND EFFECT

Event / references	DX / heating	Fans / maximum duty	Dampers / isolation	Reset
Confirmed fire / E-stop FIRE/ESD	Stop all affected DX equipment through approved fire/OEM interface	Stop ALL HVAC fans; heaters OFF	Approved fire positions; not inferred from normal isolation	Manual fire reset after clearance
Compressor high pressure PSH-DX-A / PSH-DX-B / PSH-DX-C	OEM hardwired cutout	OEM safe response; failed unit/changeover as approved	Isolate failed unit if stopped	OEM latch/reset threshold TBC
Low pressure PSL-DX-A / PSL-DX-B / PSL-DX-C	OEM operating-envelope protection	OEM routine; no bypass	OEM safe state	OEM reset / delay TBC
Evaporator frost TSL-L-DX-A / TSL-L-DX-B / TSL-L-DX-C	Inhibit / stage through OEM	OEM defrost/anti-freeze routine	Approved OEM positions	OEM anti-freeze strategy TBC
Pan overflow LSH-DX-A / LSH-DX-B / LSH-DX-C	Proposed inhibit + alarm	OEM safe response	Maintain safe air path	Inspect drain; OEM/manual policy TBC
Compressor overload OL-CMP-DX-A / OL-CMP-DX-B / OL-CMP-DX-C	Local electrical trip	OEM unit fault/changeover	Isolate stopped unit	Local/OEM reset; fault diagnosis
DX pressure monitoring PT-SUC-DX-A / PT-SUC-DX-B / PT-SUC-DX-C / PT-LIQ-DX-A / PT-LIQ-DX-B / PT-LIQ-DX-C	Monitoring only; OEM safeties govern	No new safety bypass	No independent command	OEM alarm thresholds TBC
Supply static demand DPT-SA-M01	OEM cooling/staging logic	Optional fan/VFD control; max TWO duty units	Maintain required room flows	Control range/station/setpoint TBC
Fan proof lost DPT-SF-A / DPT-SF-B / DPT-SF-C	Compressor/heat permissive inhibited via OEM	Stop failed unit; open/prove standby before start	Close stopped unit to avoid reverse flow	Fault clear / approved changeover
Filter DP high PDT-F1-A / PDT-F2-A	Alarm / OEM protection as selected	No arbitrary automatic shutdown	Normal duty unless approved protection	Service filter; threshold OEM
Heater airflow/high limit AFS-EH-COM-01 / TSHH-EH-COM-01	DX cooling follows OEM; heater OFF independently	Approved ventilation mode	No safety bypass	Independent heater high-limit reset
Standby / rotation PAU-A / PAU-B / PAU-C	Stopped unit isolated; duty pair selected	Max two running; prove new air path then controlled transfer	Stopped SA/RA/OA closed/proved	Prove normal duty after transfer

Proposal for final OEM / fire review. Factory hardwired protections remain local and cannot be bypassed by HPCP. AM22 new tags covered in the matrix. Original full-system cause/effect and275 I/O points retained. Fire is not used to initiate extraction/purge.

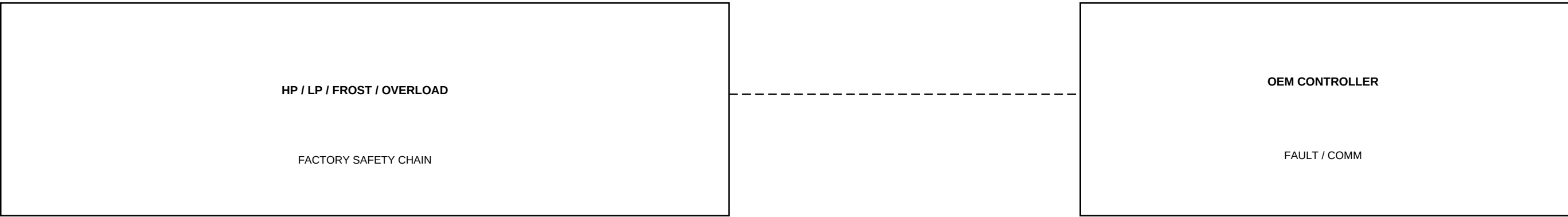
MAIN DUCT AND RETURN - DX OEM / HPCP FIELD WIRING AND I/O DEVELOPMENT

A ANALOG LOOP / OEM DATA ALTERNATIVE



Two-wire illustration only. Sensor supply, isolation and module wiring follow OEM. Route shielded twisted pair apart from motor/VFD power; shield earthing per approved panel design. Factory communication can replace extra transducers if approved.

B OEM LOCAL SAFETY / APPROVED FAULT INTERFACE



HPCP receives approved status; it does not bridge or modify factory compressor protection. Pan overflow candidate contact convention, fail-safe design and isolation require OEM review. Actual terminals and cable numbers are unassigned.

New tag	Category	Factory / HPCP interface	Panel assignment
PSH-DX-A	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
PSL-DX-A	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
TSLL-DX-A	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
LSH-DX-A	DI candidate	Approved volt-free contact; contact convention TBC	UNASSIGNED
PT-SUC-DX-A	AI candidate	Approved factory data or 4-20mA isolated input	UNASSIGNED
PT-LIQ-DX-A	AI candidate	Approved factory data or 4-20mA isolated input	UNASSIGNED
OL-CMP-DX-A	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
PSH-DX-B	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
PSL-DX-B	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
TSLL-DX-B	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
LSH-DX-B	DI candidate	Approved volt-free contact; contact convention TBC	UNASSIGNED
PT-SUC-DX-B	AI candidate	Approved factory data or 4-20mA isolated input	UNASSIGNED
PT-LIQ-DX-B	AI candidate	Approved factory data or 4-20mA isolated input	UNASSIGNED
OL-CMP-DX-B	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
PSH-DX-C	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
PSL-DX-C	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
TSLL-DX-C	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
LSH-DX-C	DI candidate	Approved volt-free contact; contact convention TBC	UNASSIGNED
PT-SUC-DX-C	AI candidate	Approved factory data or 4-20mA isolated input	UNASSIGNED
PT-LIQ-DX-C	AI candidate	Approved factory data or 4-20mA isolated input	UNASSIGNED
OL-CMP-DX-C	OEM LOCAL	Local protection; use approved factory status/communications	UNASSIGNED
DPT-SA-M01	AI candidate	Approved factory data or 4-20mA isolated input	UNASSIGNED

22 candidates =7 AI +3 DI +12 OEM local protections. These are not added silently to the275 baseline I/O rows. Selected factory communications / reused fault contacts can change the final additional count. Wiring is an interface proposal, not a signed terminal schedule.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

DX OEM / HPCP FIELD WIRING AND I/O DEVELOPMENT

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-038

SCALE: NTS AT A1

DRAWING REVISION 02

08 OCT 2026

SHEET 38 OF 50

MAIN DUCT AND RETURN - DX EQUIPMENT SELECTION / ROOM DUTIES / COMMISSIONING

Unit	Operation	SA L/s	RA L/s	OA lower bound*	DX / fan selection
PAU-A	50% / two duty / one standby	1848.90	1503.34	345.56	Capacity, refrigerant, nozzle and ESP TBC
PAU-B	50% / two duty / one standby	1848.90	1503.34	345.56	Capacity, refrigerant, nozzle and ESP TBC
PAU-C	50% / two duty / one standby	1848.90	1503.34	345.56	Capacity, refrigerant, nozzle and ESP TBC

ROOM AIR DUTIES / MEASURED TAB RECORD REQUIRED

GF / room	SA L/s	RA L/s	EA L/s	Recorded pressure target	Acceptance / unresolved item
02 / Electrical panel	664.80	605.46	-	50 Pa / unverified	OA / leakage / comfort / field TAB and specified tolerance
04 / Meeting	449.20	389.86	-	50 Pa / unverified	OA / leakage / comfort / field TAB and specified tolerance
08 / Telecom panel	884.60	825.26	-	50 Pa / unverified	OA / leakage / comfort / field TAB and specified tolerance
05 / Operator console	777.70	718.36	-	50 Pa / unverified	OA / leakage / comfort / field TAB and specified tolerance
06 / Reception / corridor	418.20	358.86	-	50 Pa / unverified	OA / leakage / comfort / field TAB and specified tolerance
09 / Toilet group: two cubicles	46.20	-	64.89	25 Pa / unverified	OA / leakage / comfort / field TAB and specified tolerance
07 / Bedroom / Resting Room	168.20	108.86	-	50 Pa / unverified	OA / leakage / comfort / field TAB and specified tolerance
03 / Kitchen / kitchenette	147.60	-	147.60	25 Pa / unverified	OA / leakage / comfort / field TAB and specified tolerance
01 / Clean-agent cylinder room	141.30	-	TBC	25 Pa / unverified	OA / leakage / comfort / field TAB and specified tolerance

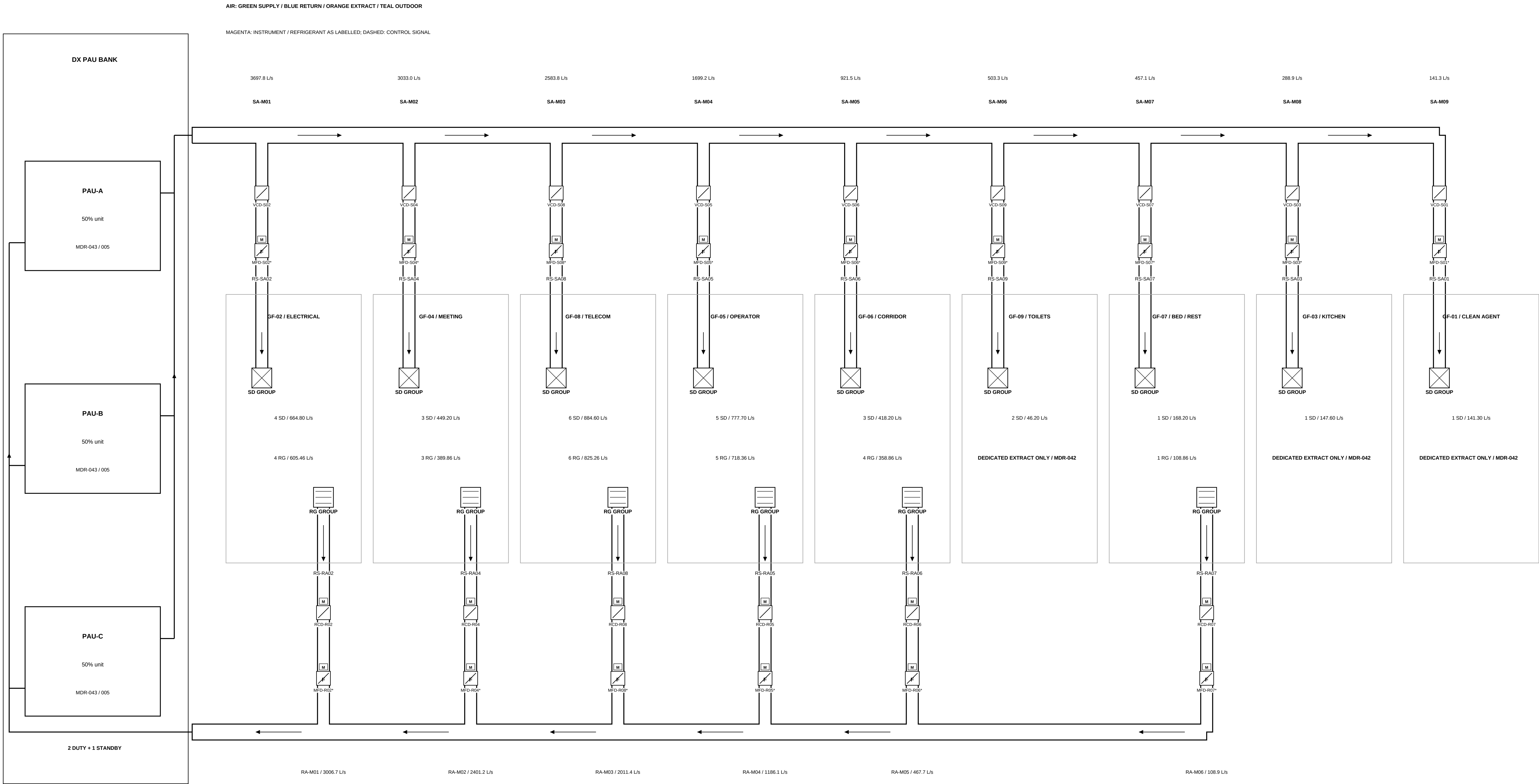
PRE-START / COMMISSIONING / PROCUREMENT HOLD

Confirm roof supports/penetrations/firestops, leakage class/pressure/gauge, weather and vapour seals; inspect fan rotation, duct cleanliness, damper end switches, drain/trap prime and every instrument scale. Prove fire stop and independent heater/DX safeties. Balance all53 terminals; record each room SA/RA/EA and pressure with sheltered outdoor reference. Confirm comfort/humidity against revised thermal loads.

* Total makeup lower bound691.12 L/s exceeds earlier OA420.70 by270.42. Unknown CAG extract remains open; toilet net-18.69 and kitchen0 do not establish+25 Pa. The52 Rev12 pressure paths are preliminary allowances excluding bank/nozzle/OA details; they do not select a DX fan. Complete OEM coils/dirty filters, actual accessories and full fan curve before procurement.

No capacities, refrigerant safety settings, airflow tolerances, leakage measurements or certified diffuser throwNC are invented. All room duties / necks / faces remain in the full retained registers. Installed levels and final engineering approval remain open.

MAIN DUCT AND RETURN - WHOLE SYSTEM AIRFLOW METHOD



HOW THE METHOD WORKS

DX cooling takes place inside the PAUs. Refrigerant stays within the selected factory DX system; supply and return mains carry AIR. Common SUPPLY and RETURN mains stay on the ROOF. Nine supply drops and six return risers connect room branches to the shared headers; every PAU connects to the same header bank. Terminal groups and all instruments are expanded on the following sheets.

Supply 3697.80 L/s; assumed return 3006.68 L/s; minimum mass-balance makeup 691.12 L/s. DX cooling capacity / outside air / actual pressures remain unselected. Rooms with dedicated extract have no common return branch.

NTS functional air diagram: room order and sequential flow steps are illustrative; actual grouped roof takeoffs / coordinates on MDR-023. MFD* conditional on approved fire boundaries; RCD pressure modulation is a proposal. All room flows retained. Main sizes: MDR-050.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

WHOLE SYSTEM AIRFLOW METHOD

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-040

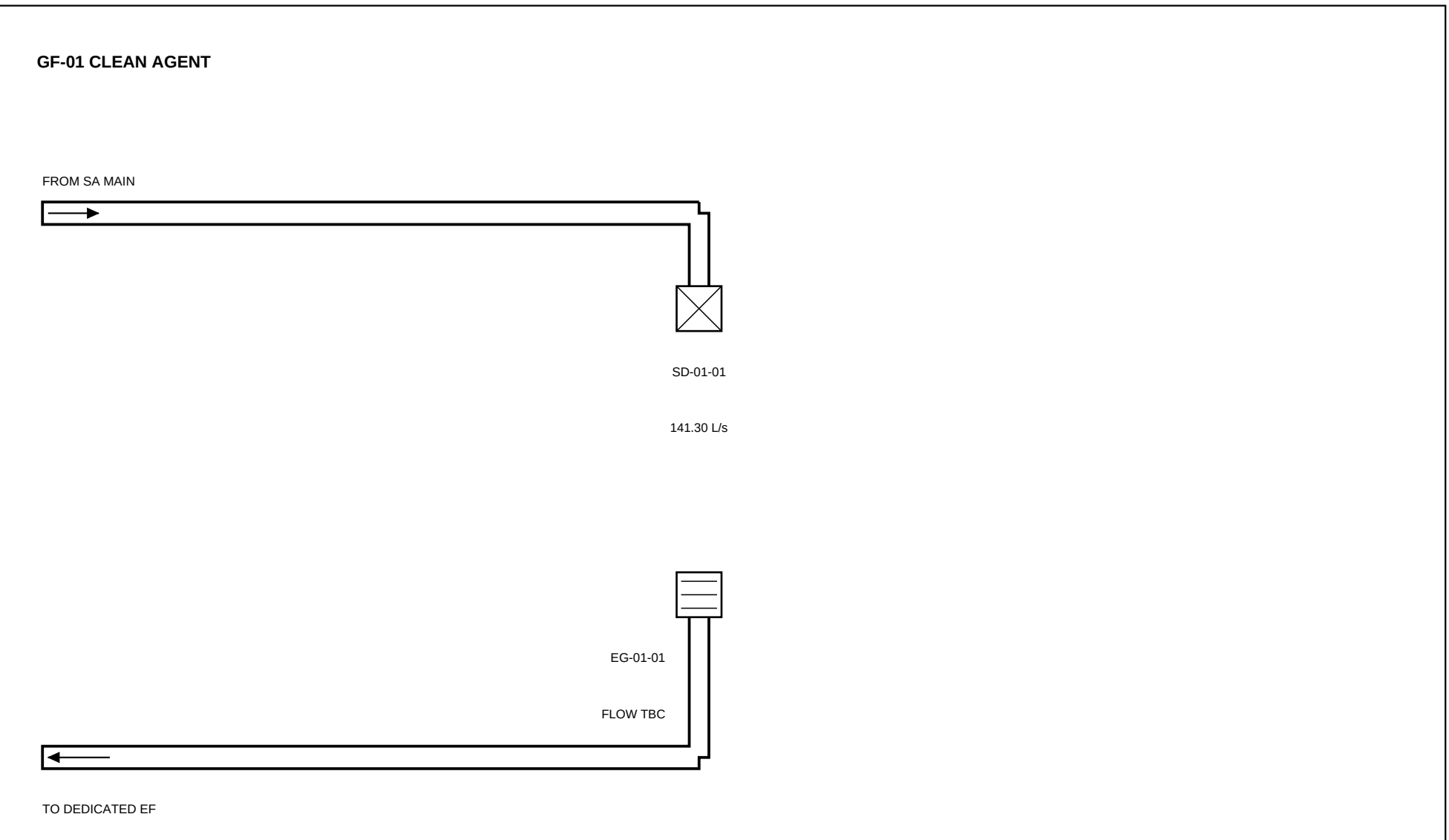
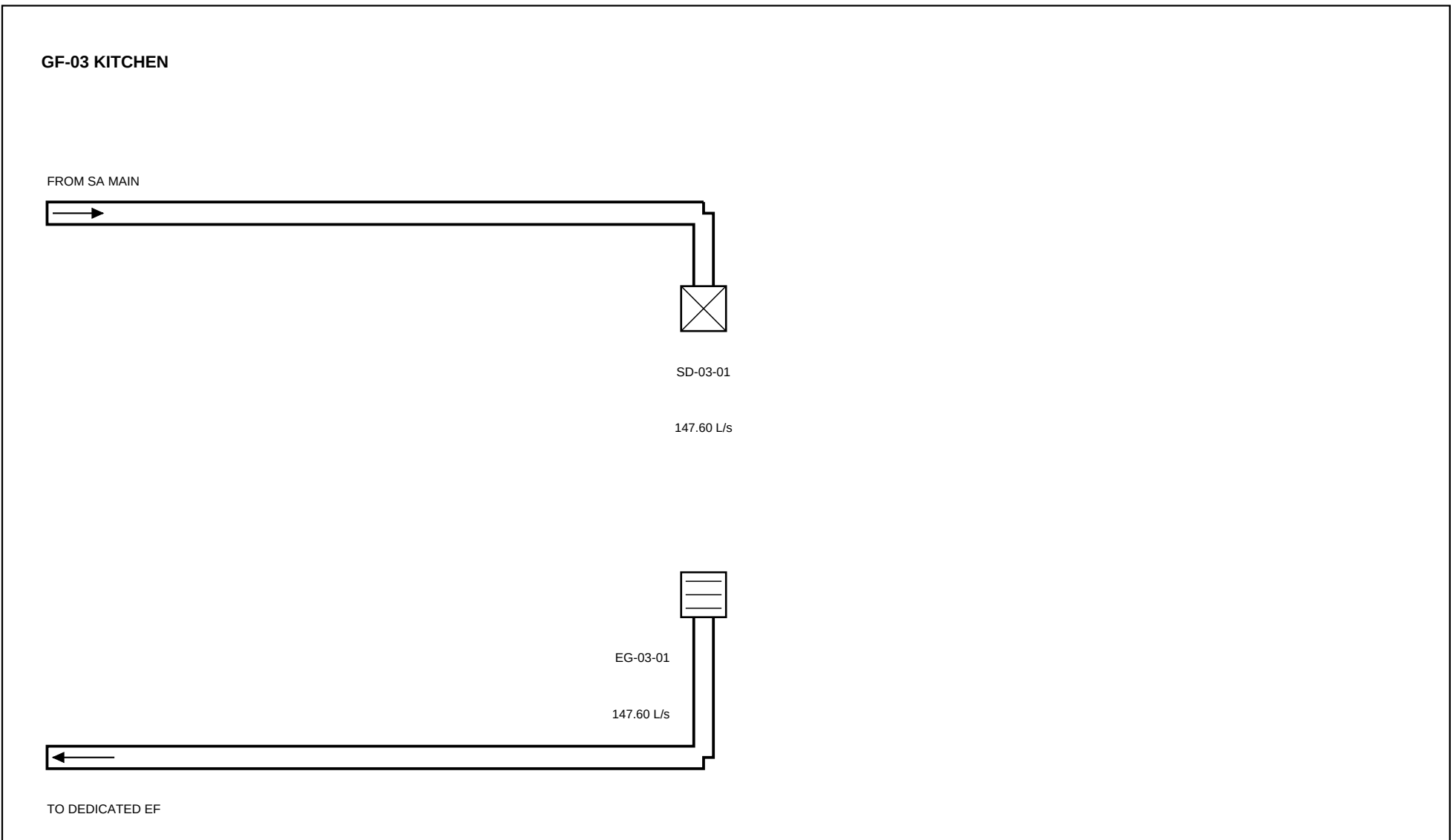
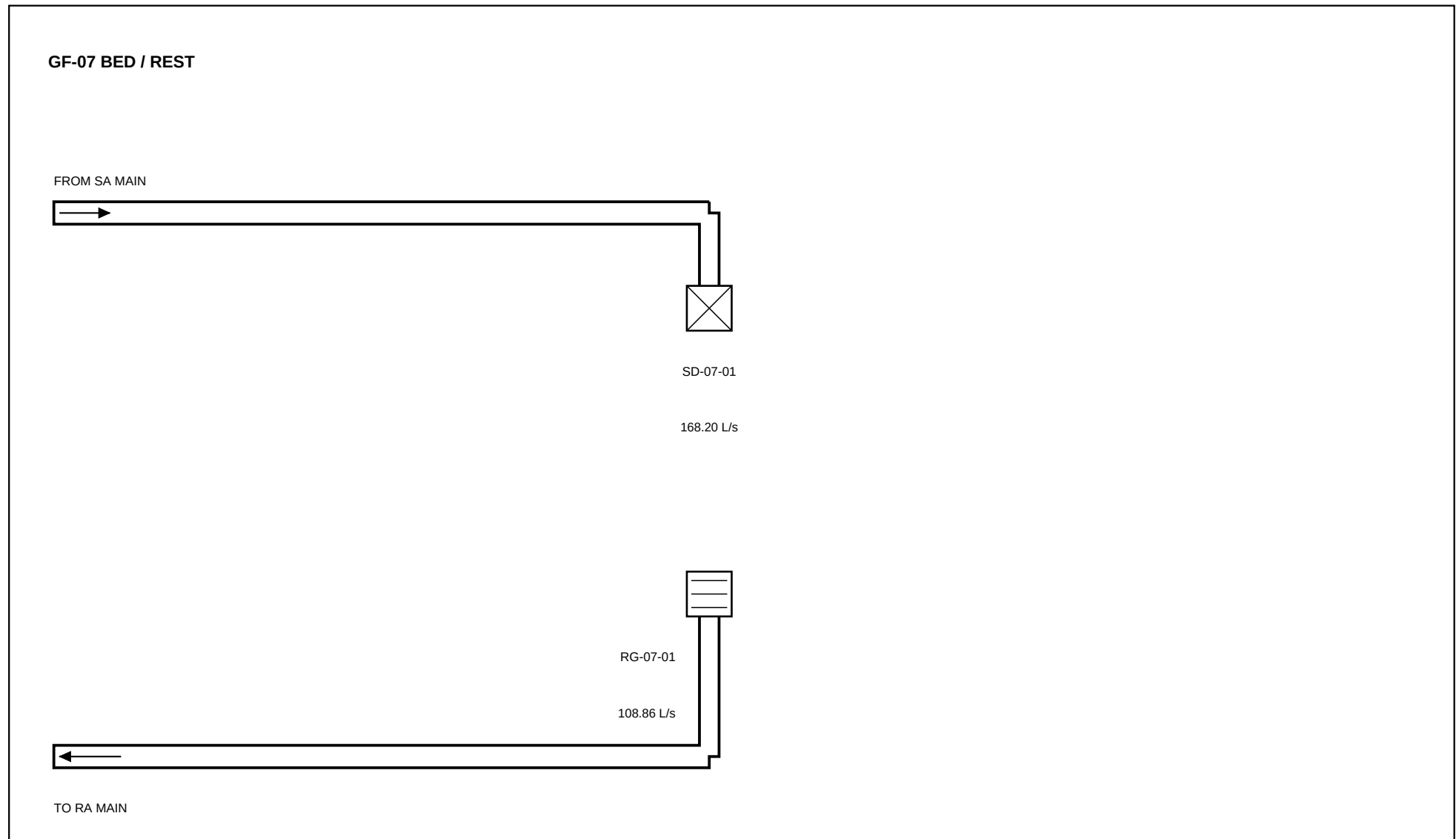
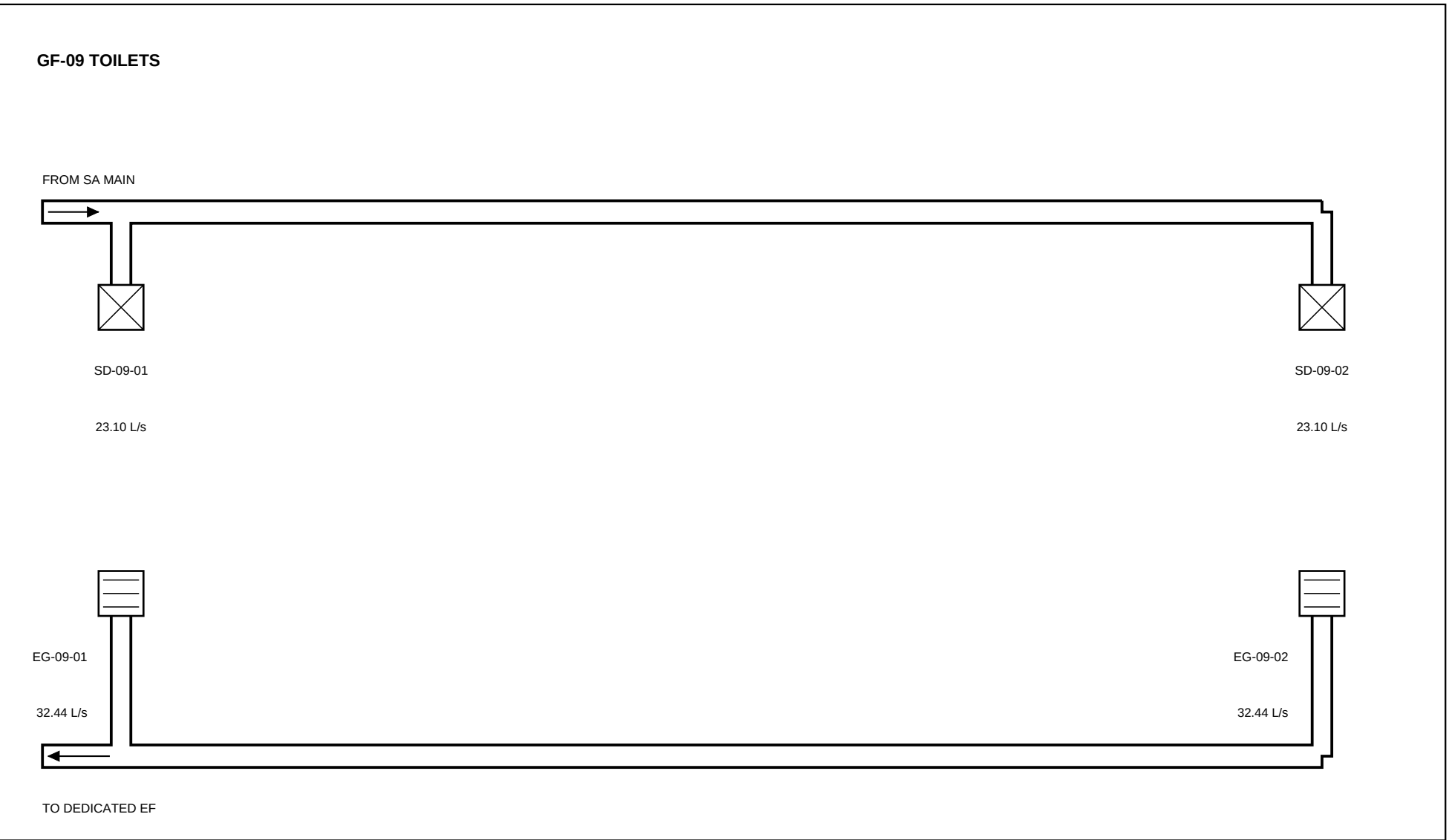
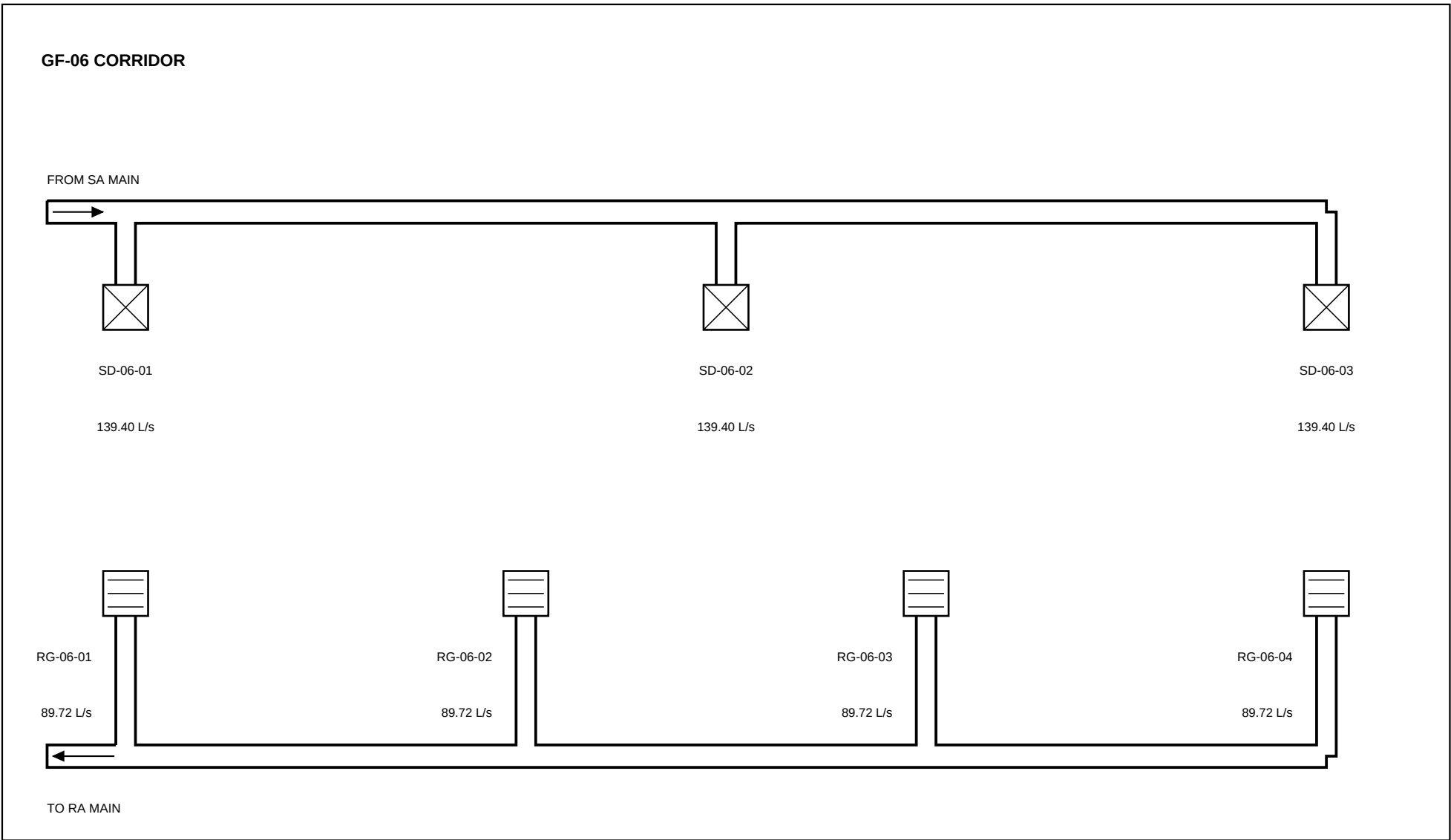
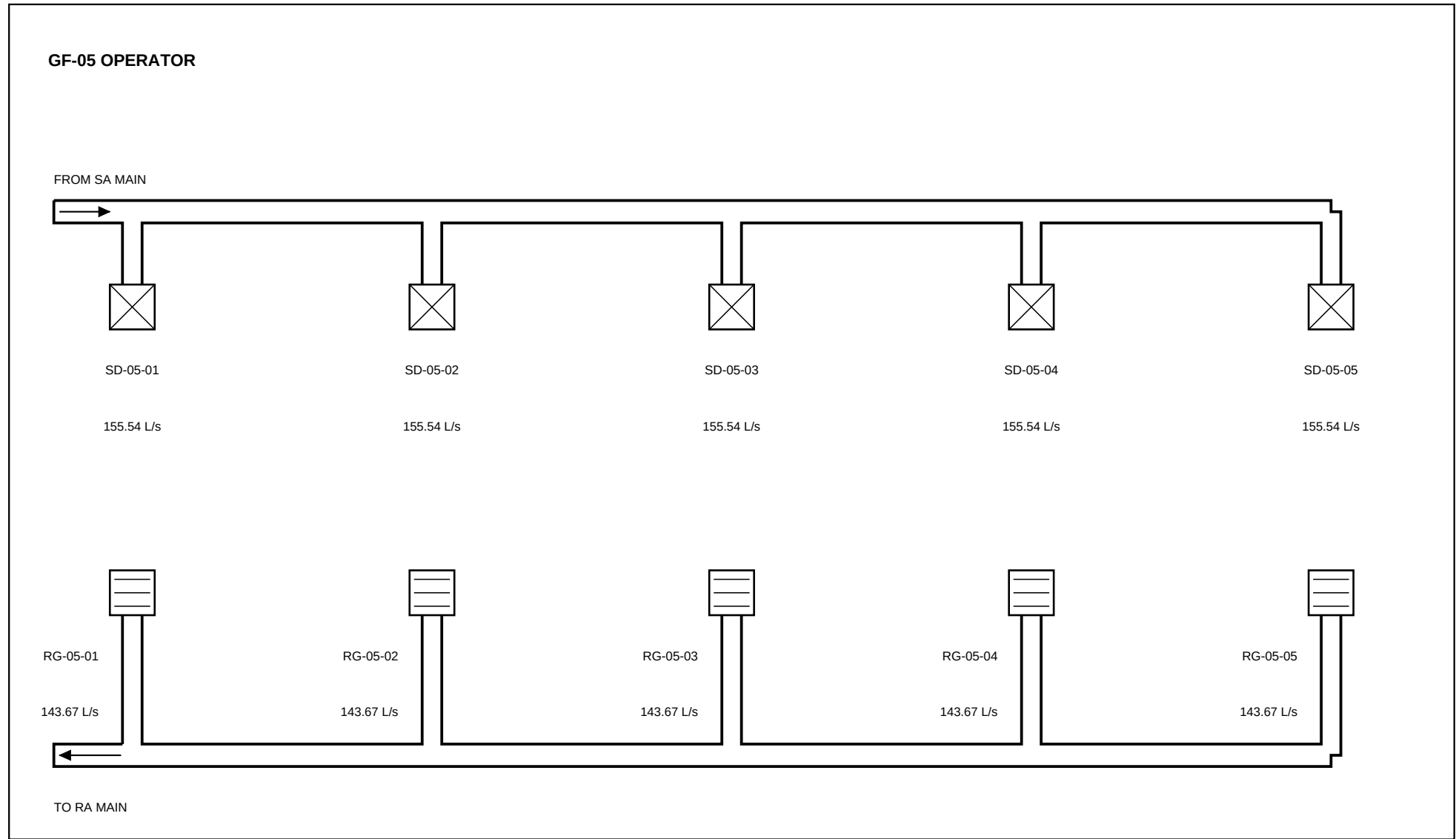
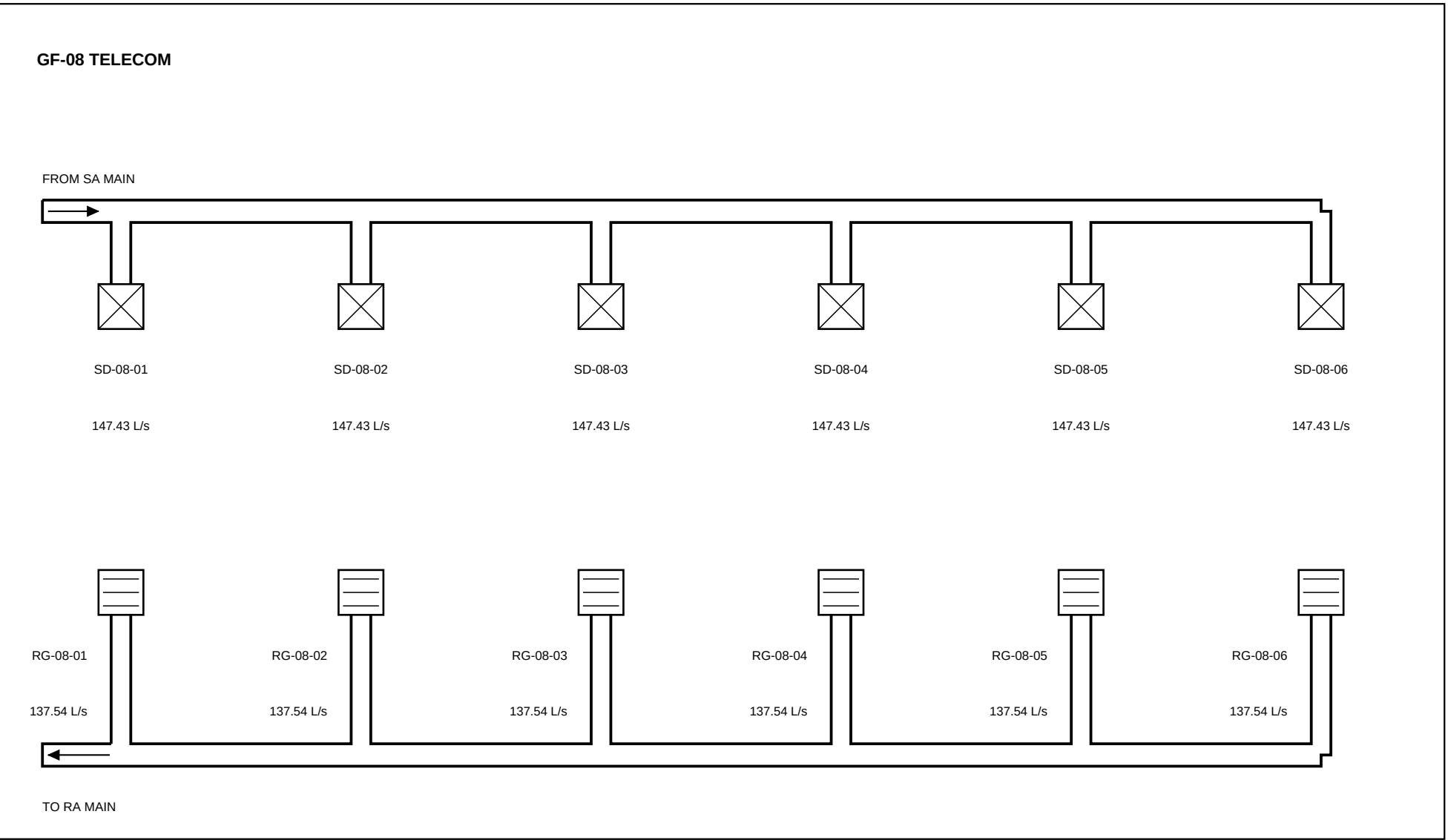
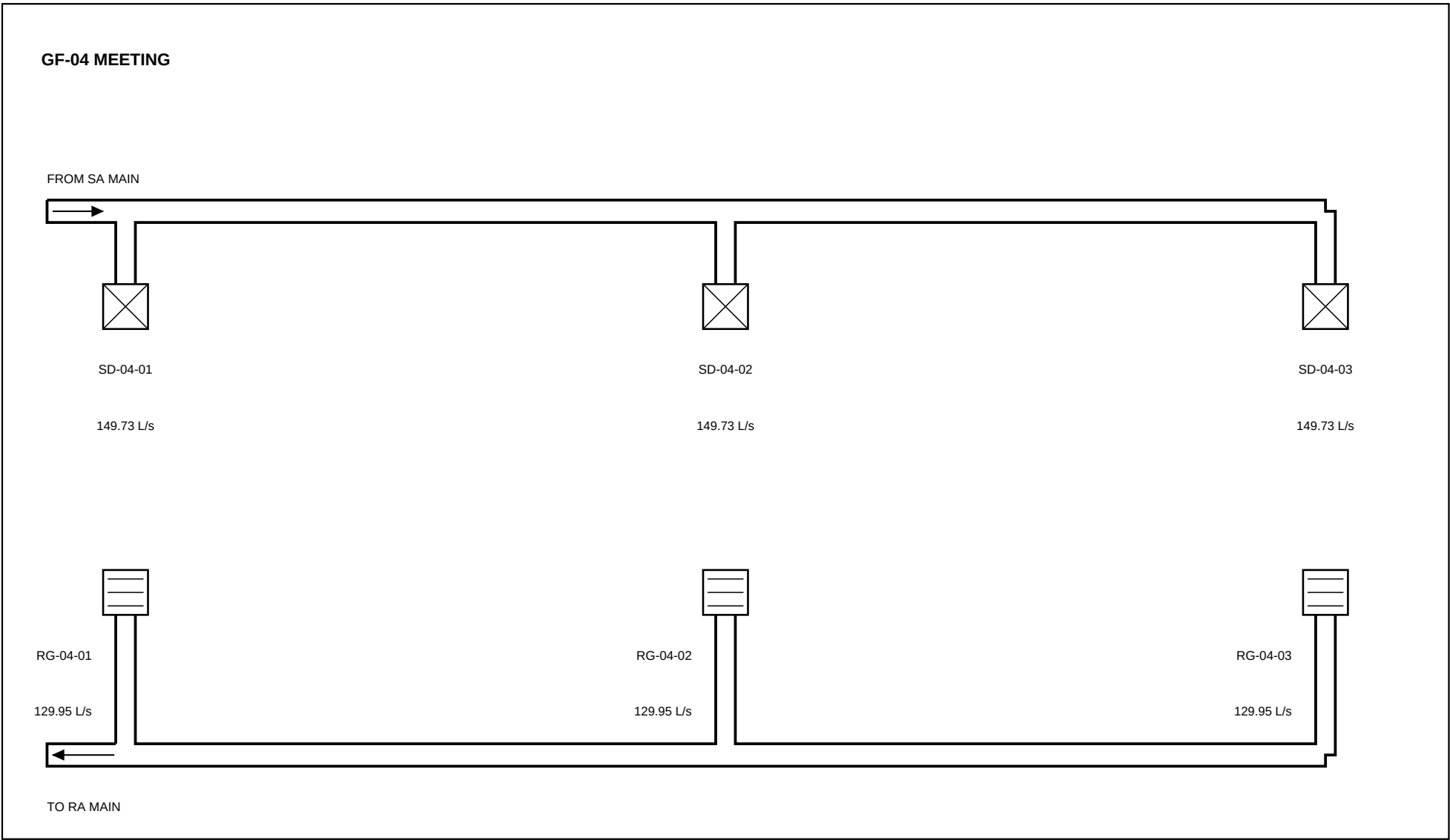
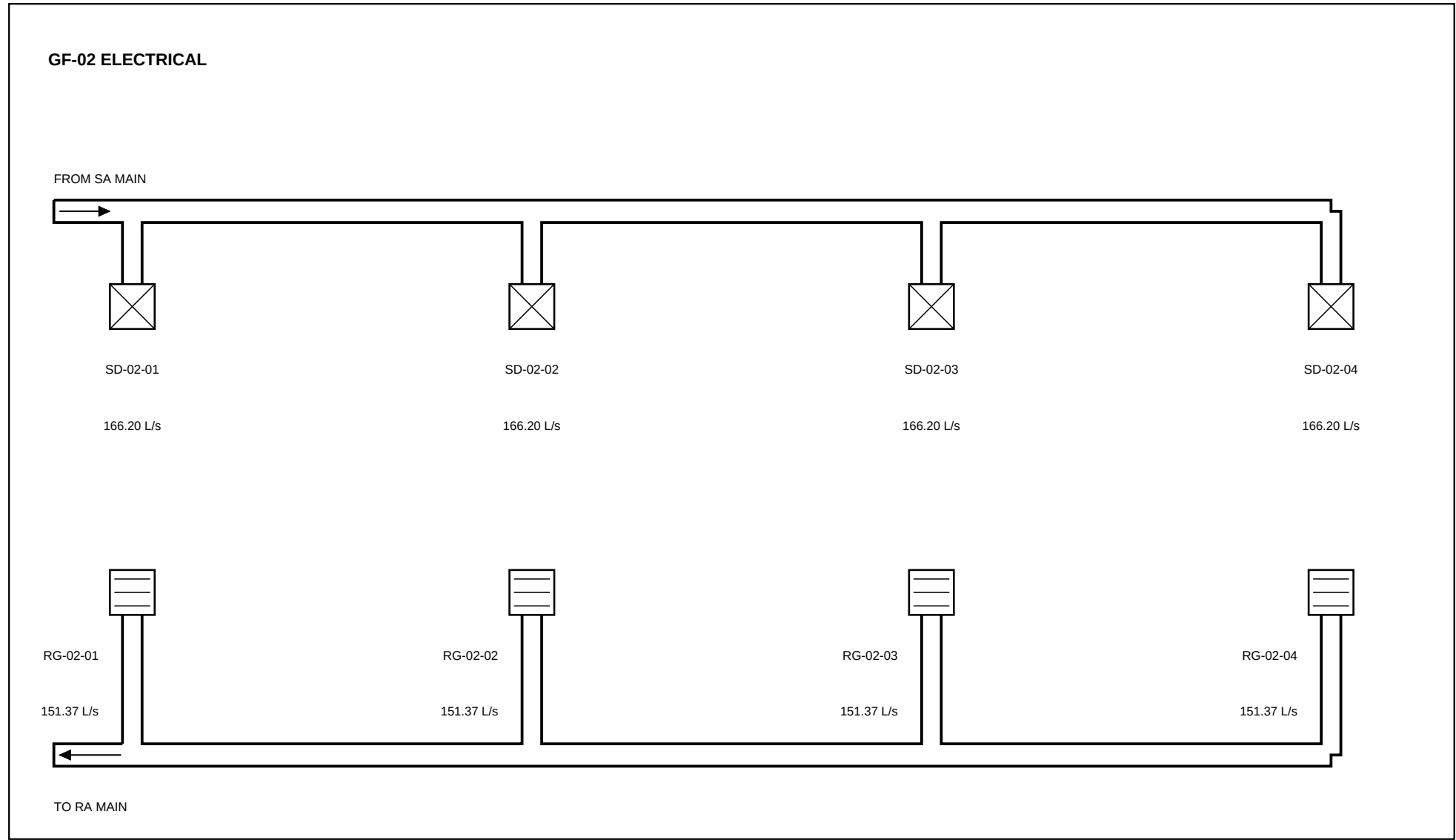
SCALE: NTS AT A1

DRAWING REVISION 02

08 OCT 2026

SHEET 40 OF 50

MAIN DUCT AND RETURN - ALL ROOM DIFFUSERS, RETURN GRILLES AND EXTRACT PICKUPS



All S3 terminal tags / duties are retained. Glyph represents the registered terminal face; this sheet is a functional room header diagram, not ceiling setting-out. Terminal neck / face sizes and coordinates retained in the workbook.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

ALL ROOM DIFFUSERS, RETURN GRILLES AND EXTRACT PICKUPS

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-041

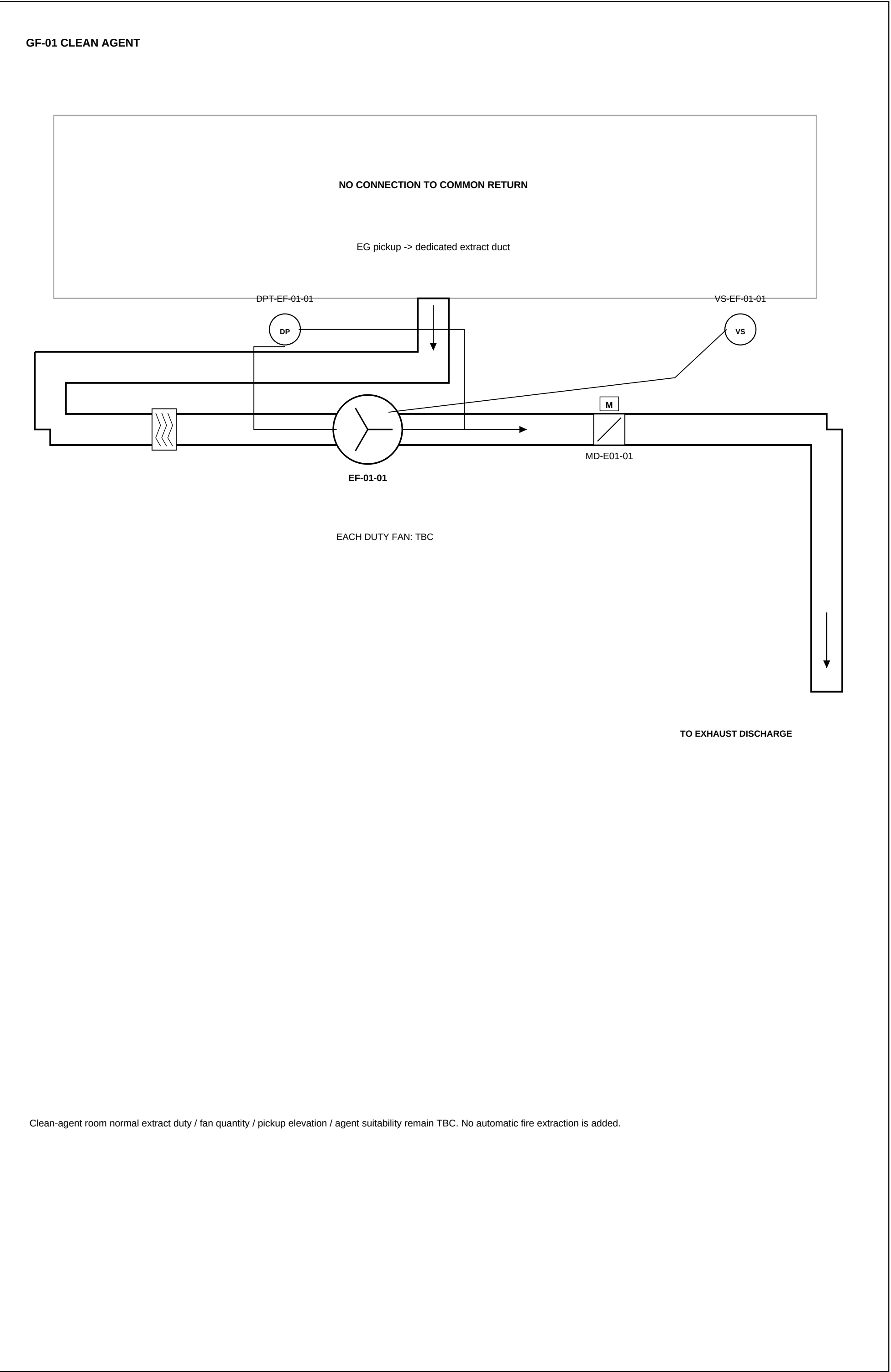
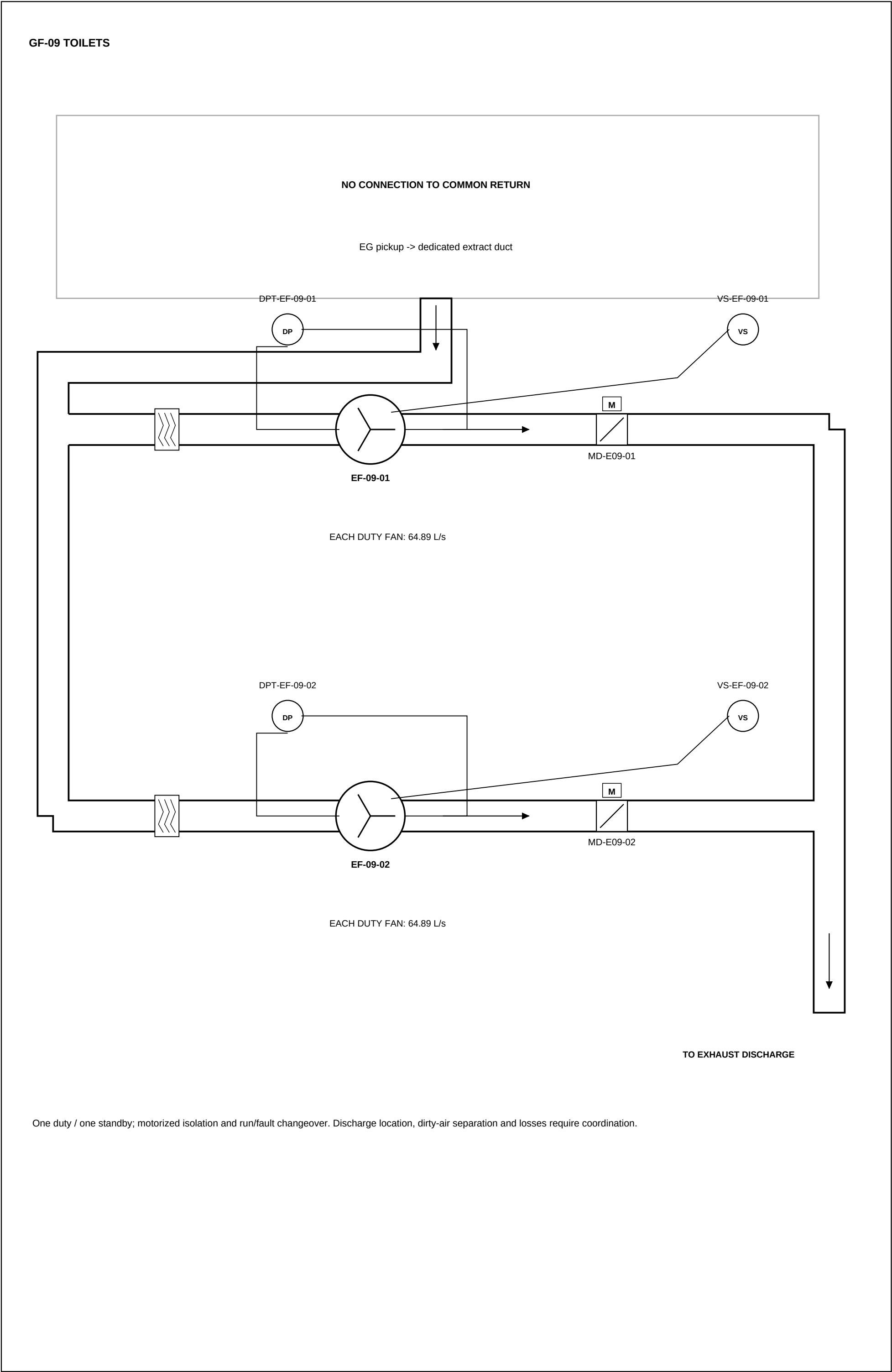
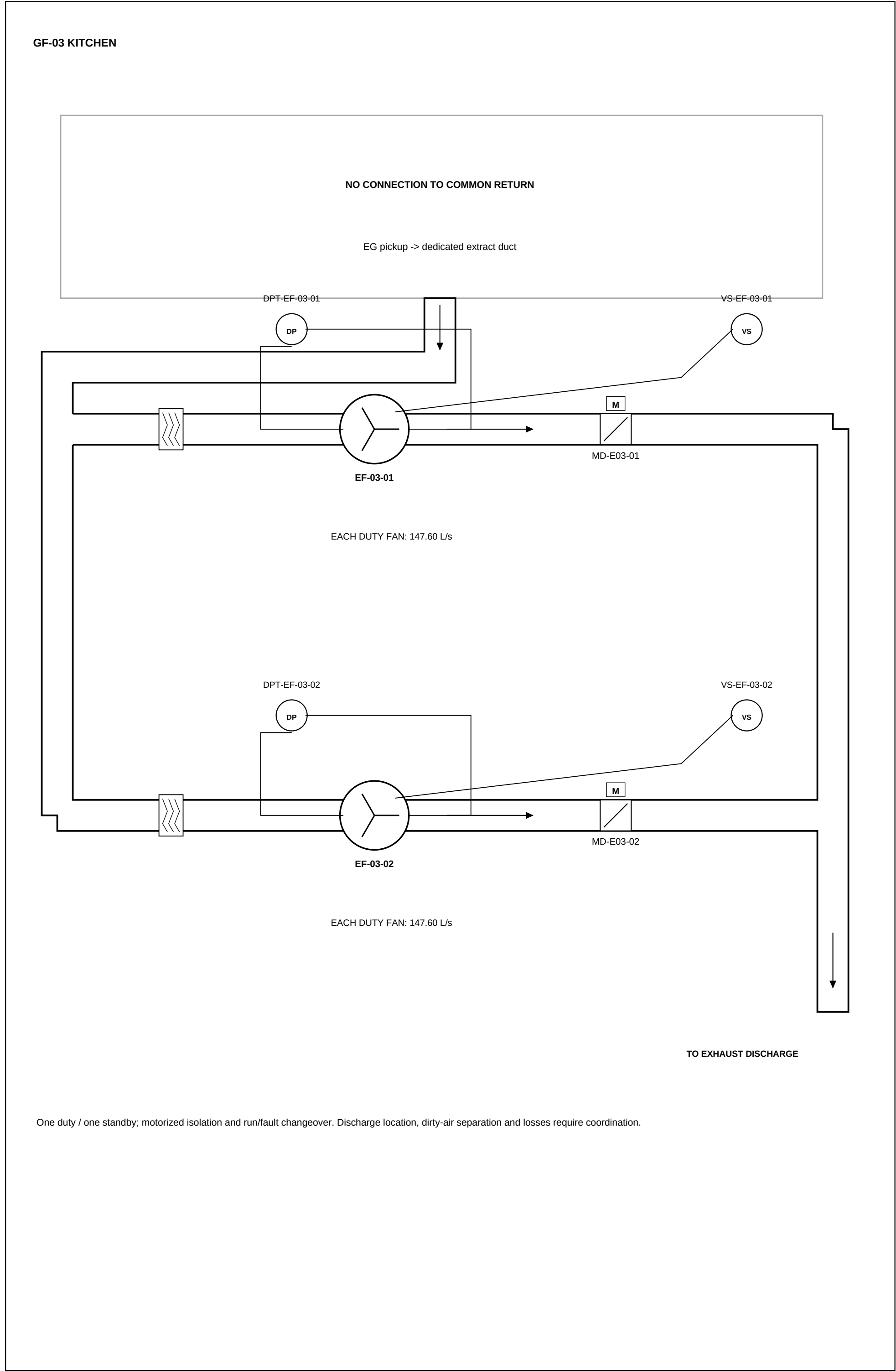
SCALE: NTS AT A1

DRAWING REVISION 02

08 OCT 2026

SHEET 41 OF 50

MAIN DUCT AND RETURN - DEDICATED EXTRACTS AND RETURN AIR SEPARATION



AIR BALANCE / FIRE REQUIREMENTS

Toilet SA 46.20 minus EA 64.89 = -18.69 L/s; kitchen SA 147.60 minus EA 147.60 = zero. These do not establish the previous +25 Pa targets. Agree transfer / hygiene / pressure strategy using measured leakage and project requirements.

Confirmed fire stops every HVAC fan and heater; fire reset is manual. No automatic smoke extraction or post-fire purge is implied. Dedicated extract fan outlet / discharge duty and CAG selection remain open.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

DEDICATED EXTRACTS AND RETURN AIR SEPARATION

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-042

SCALE: NTS AT A1

DRAWING REVISION 02

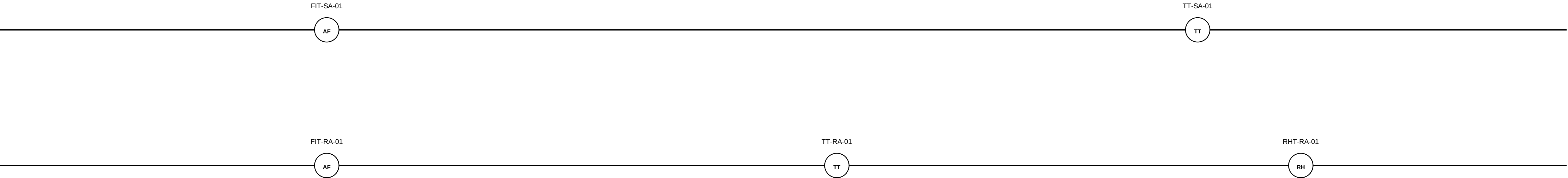
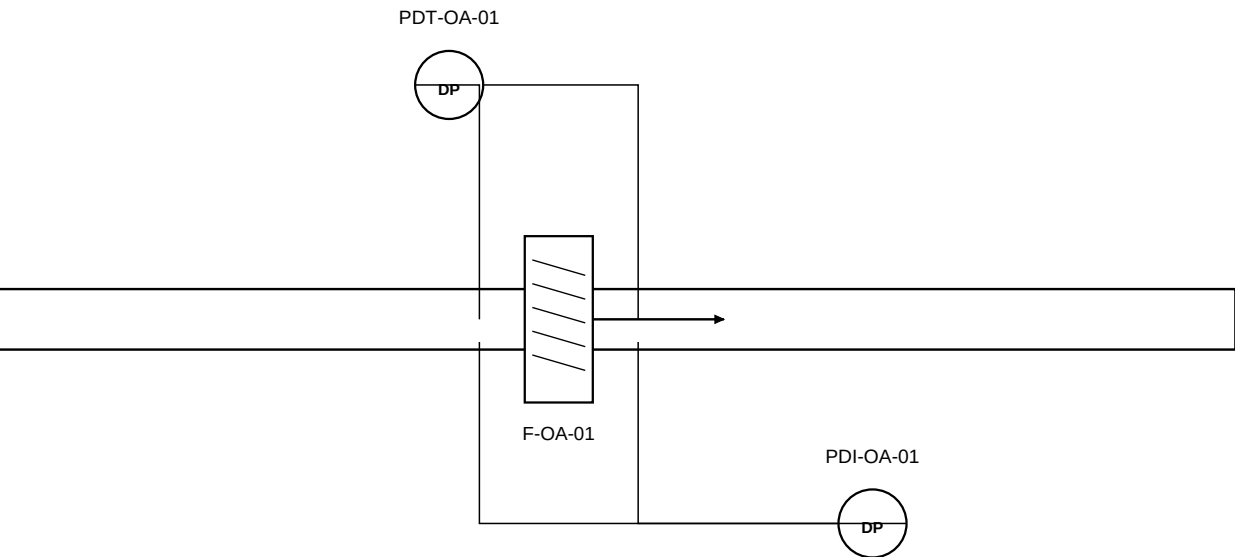
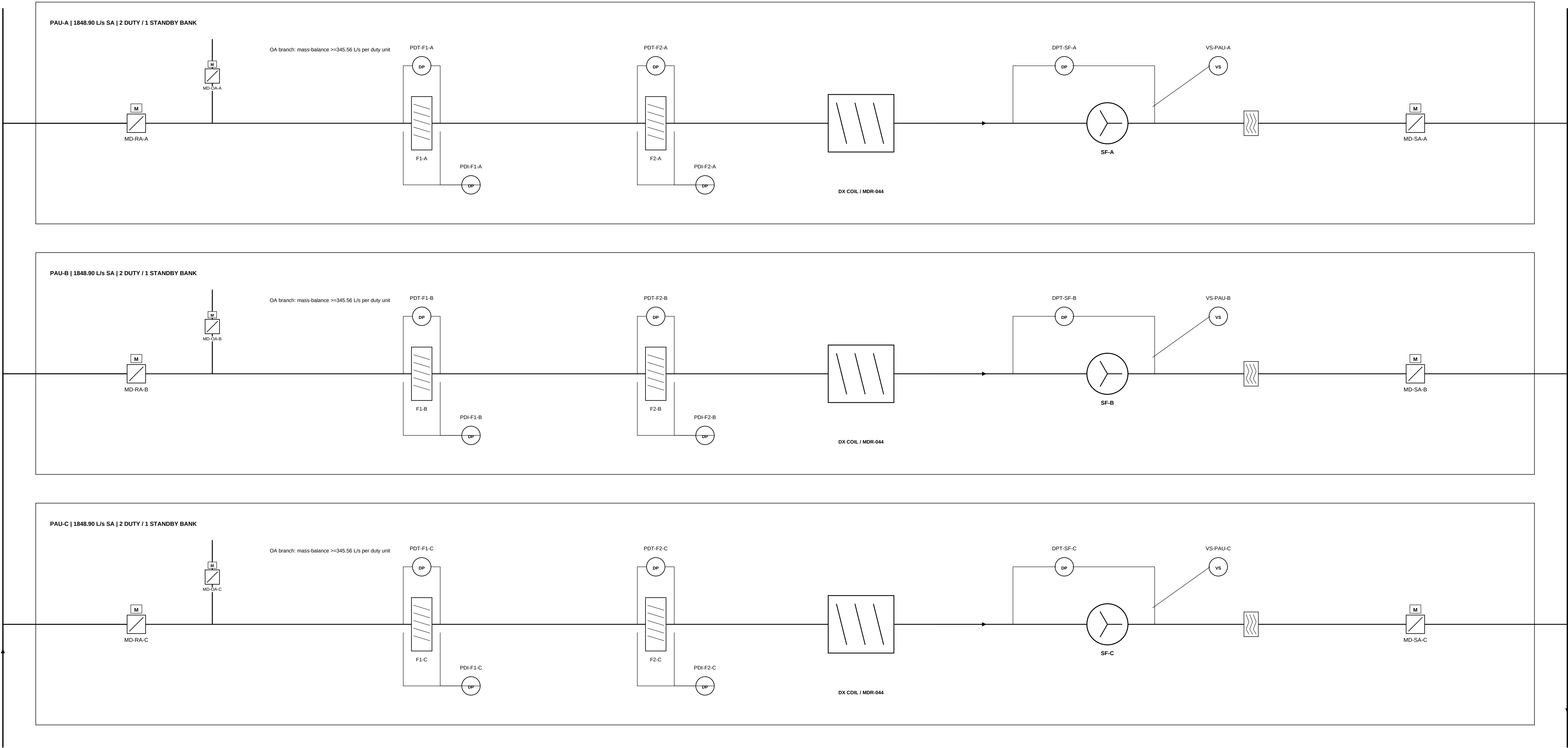
08 OCT 2026

SHEET 42 OF 50

MAIN DUCT AND RETURN - DX PAU BANK, COMMON HEADERS AND AIR-SIDE INSTRUMENTS

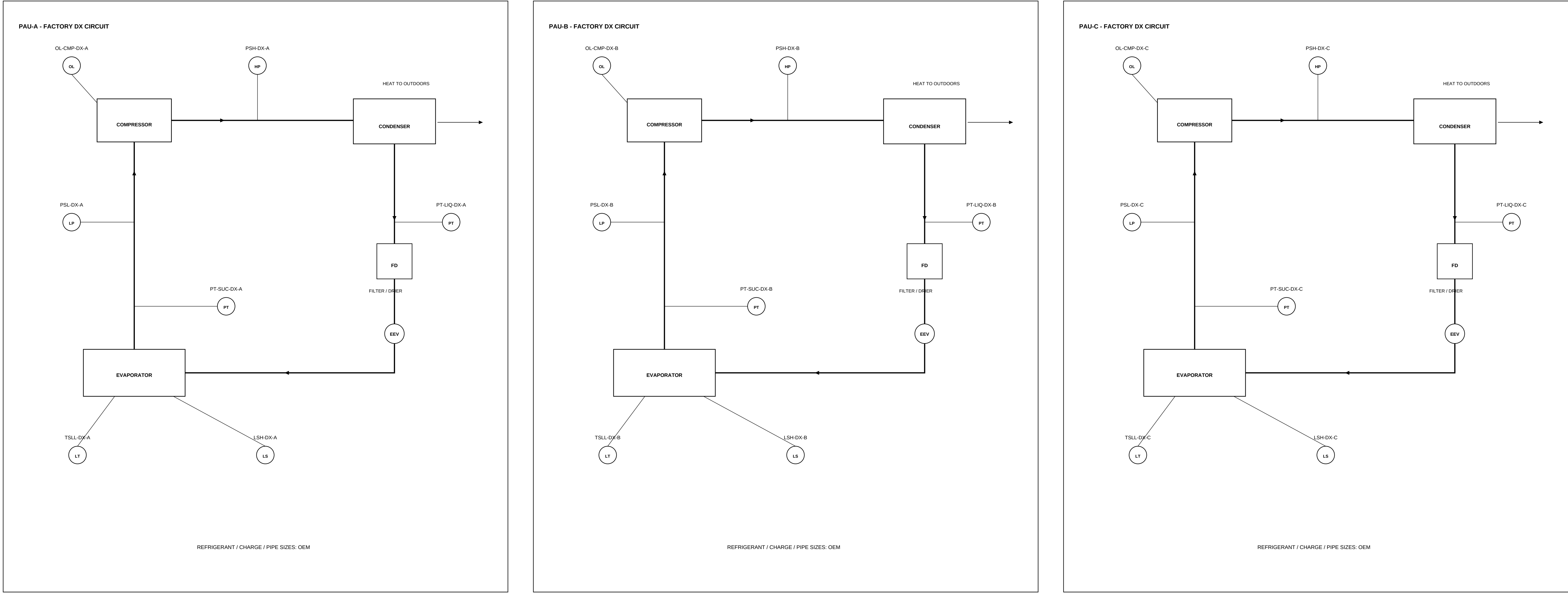
COMMON RA

COMMON SA

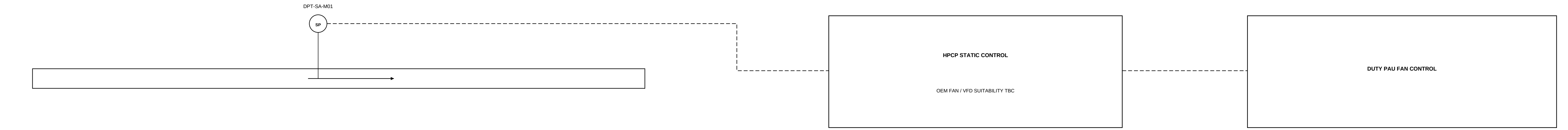


Standby PAU supply / return / OA paths close and prove isolated; duty paths open and prove before start. Common airflow stations require OEM straight runs. Unit nozzle routing, fan ESP, filters, cooling capacity and outdoor air remain unselected.

ADMINISTRATION HVAC / DX PAU OPTION COMMON SUPPLY MAIN + COMMON RETURN MAIN FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION	DRAWING TITLE DX PAU BANK, COMMON HEADERS AND AIR-SIDE INSTRUMENTS ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING	ADM-HVAC-MDR-043 SCALE: NTS AT A1	DRAWING REVISION 02 08 OCT 2026 SHEET 43 OF 50
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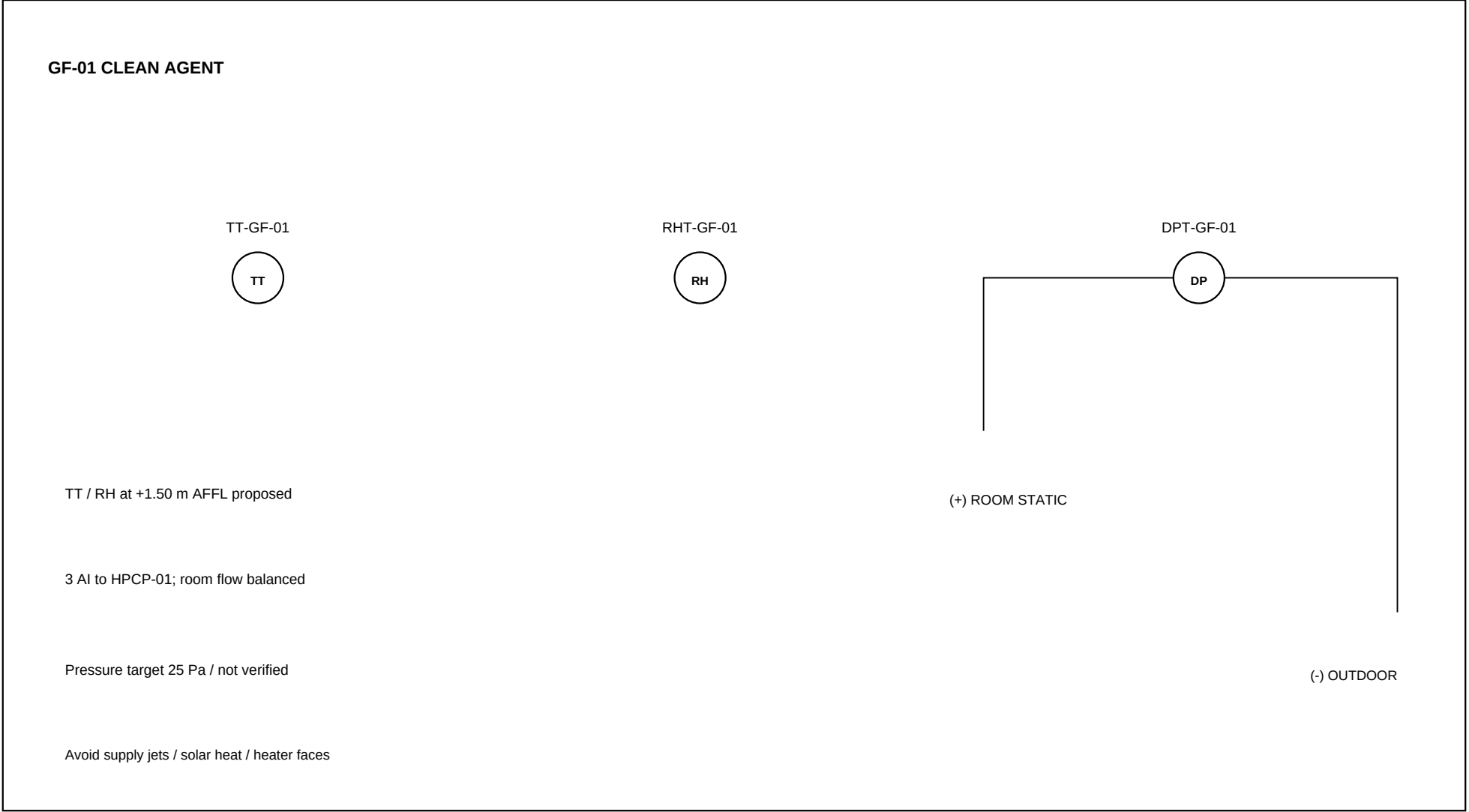
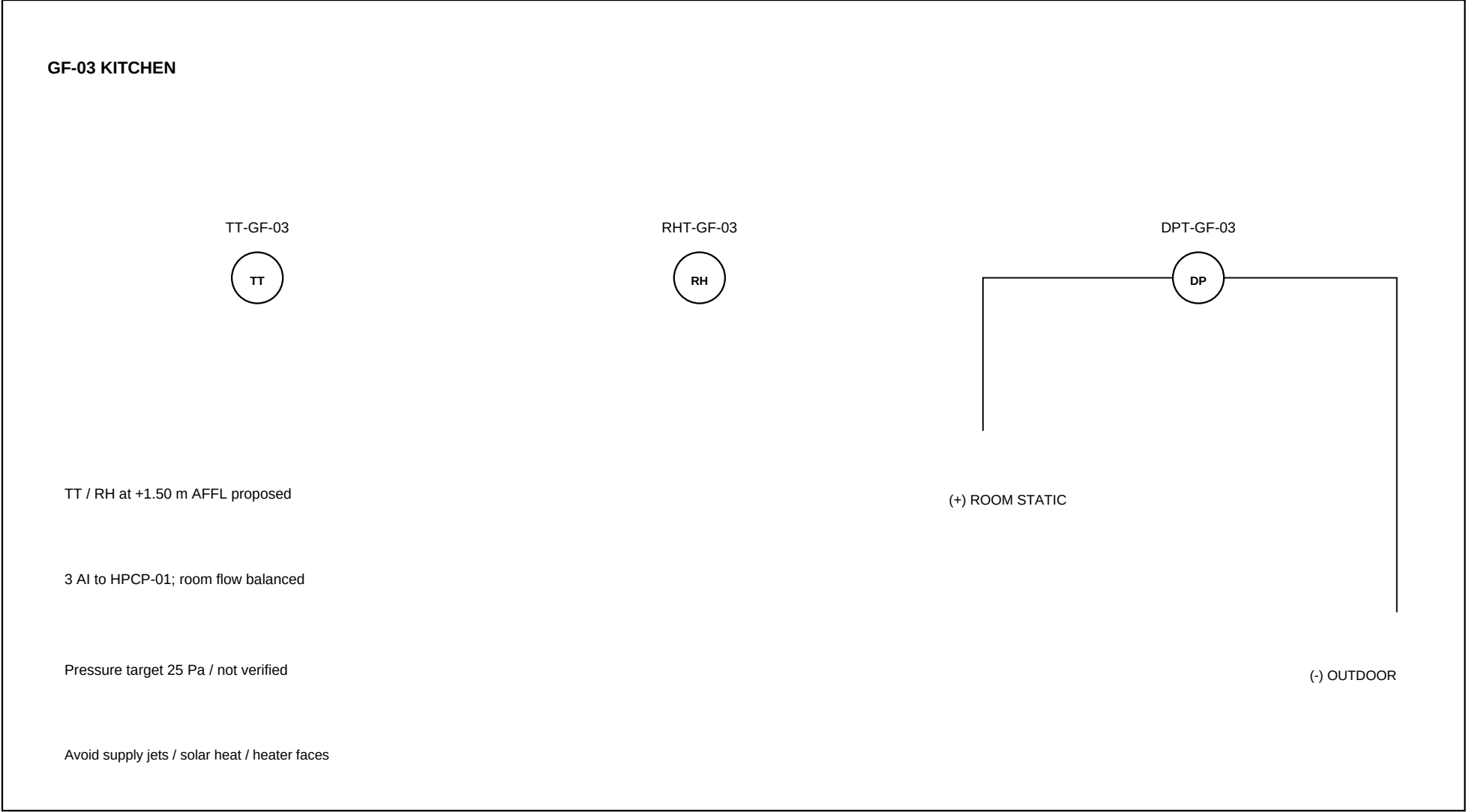
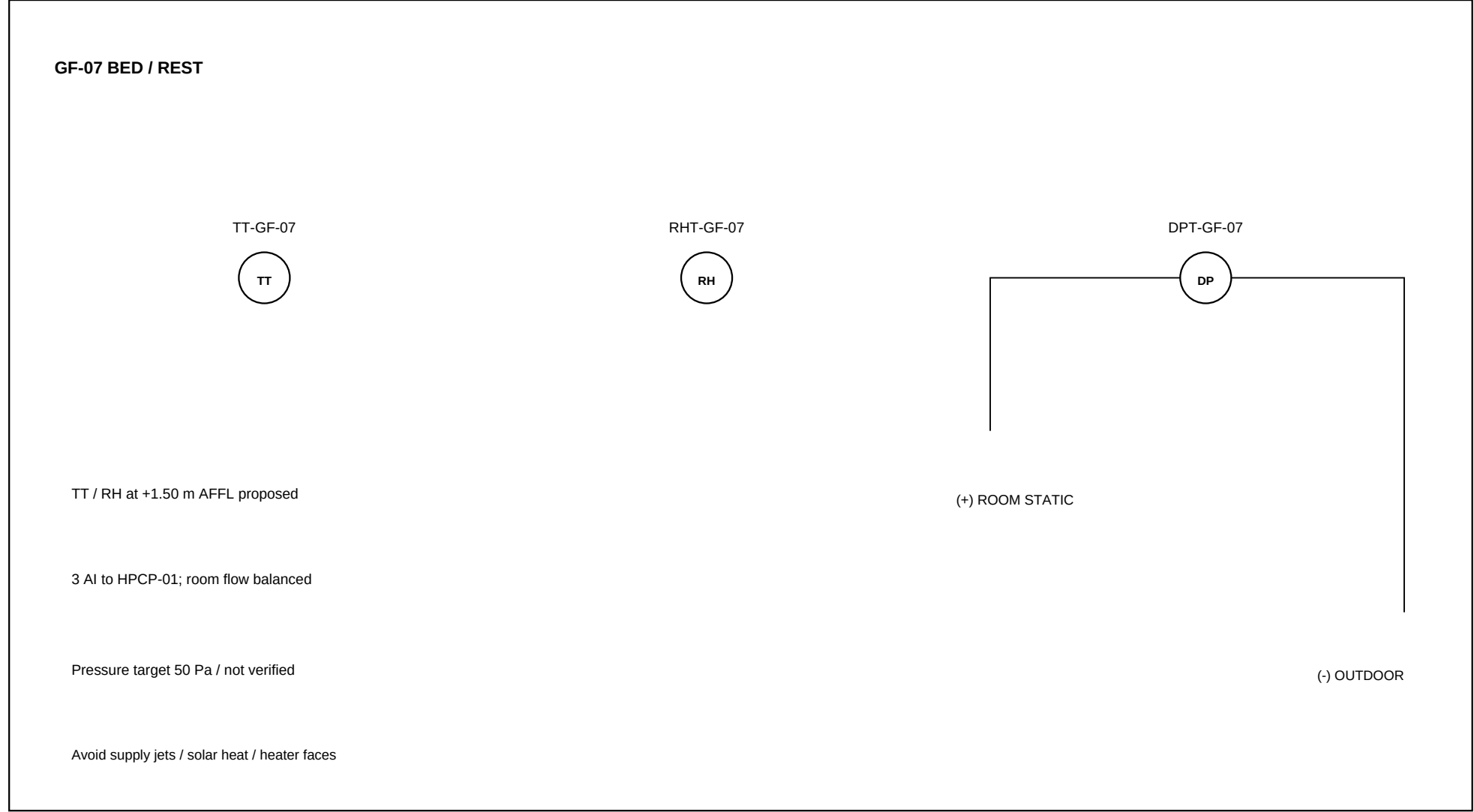
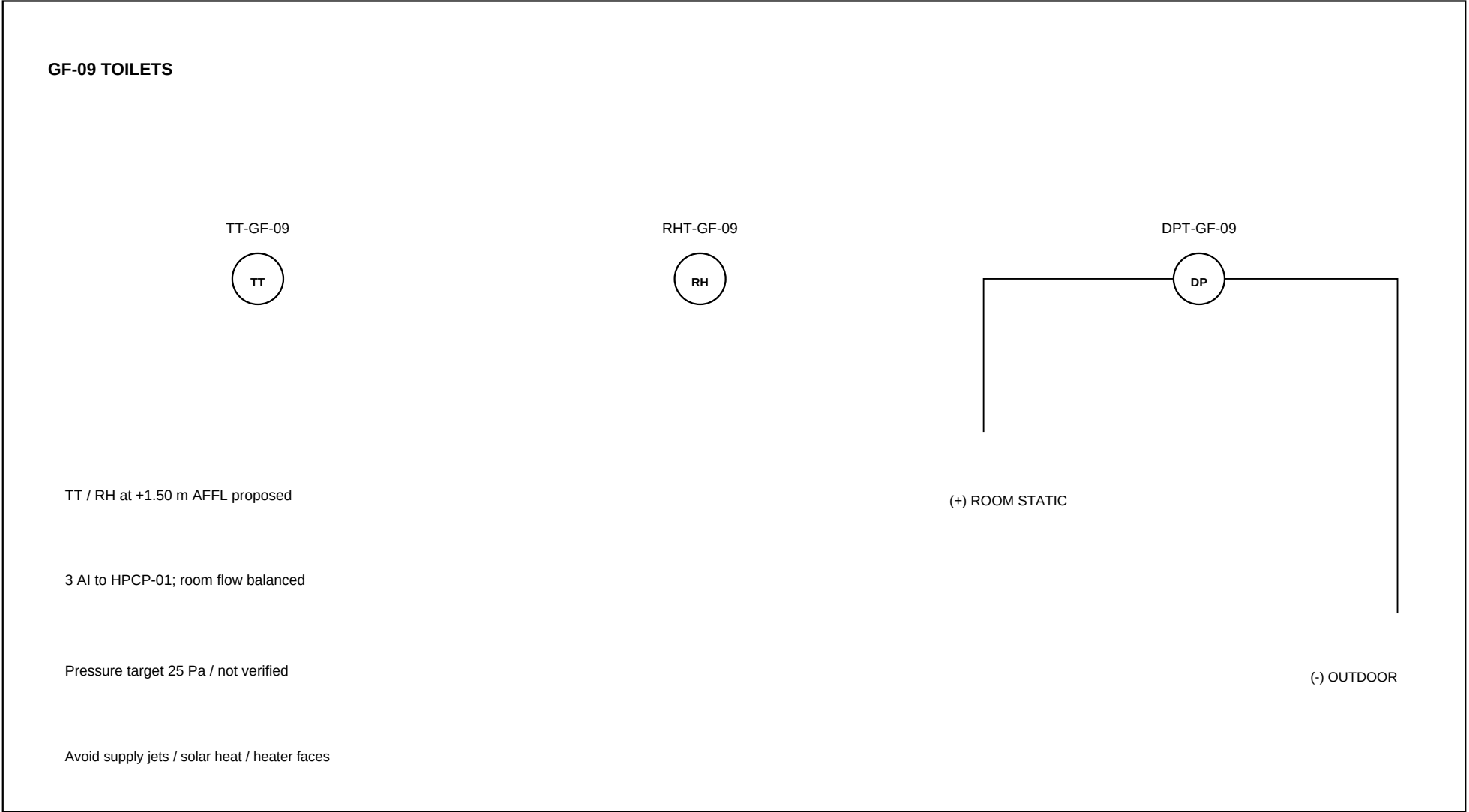
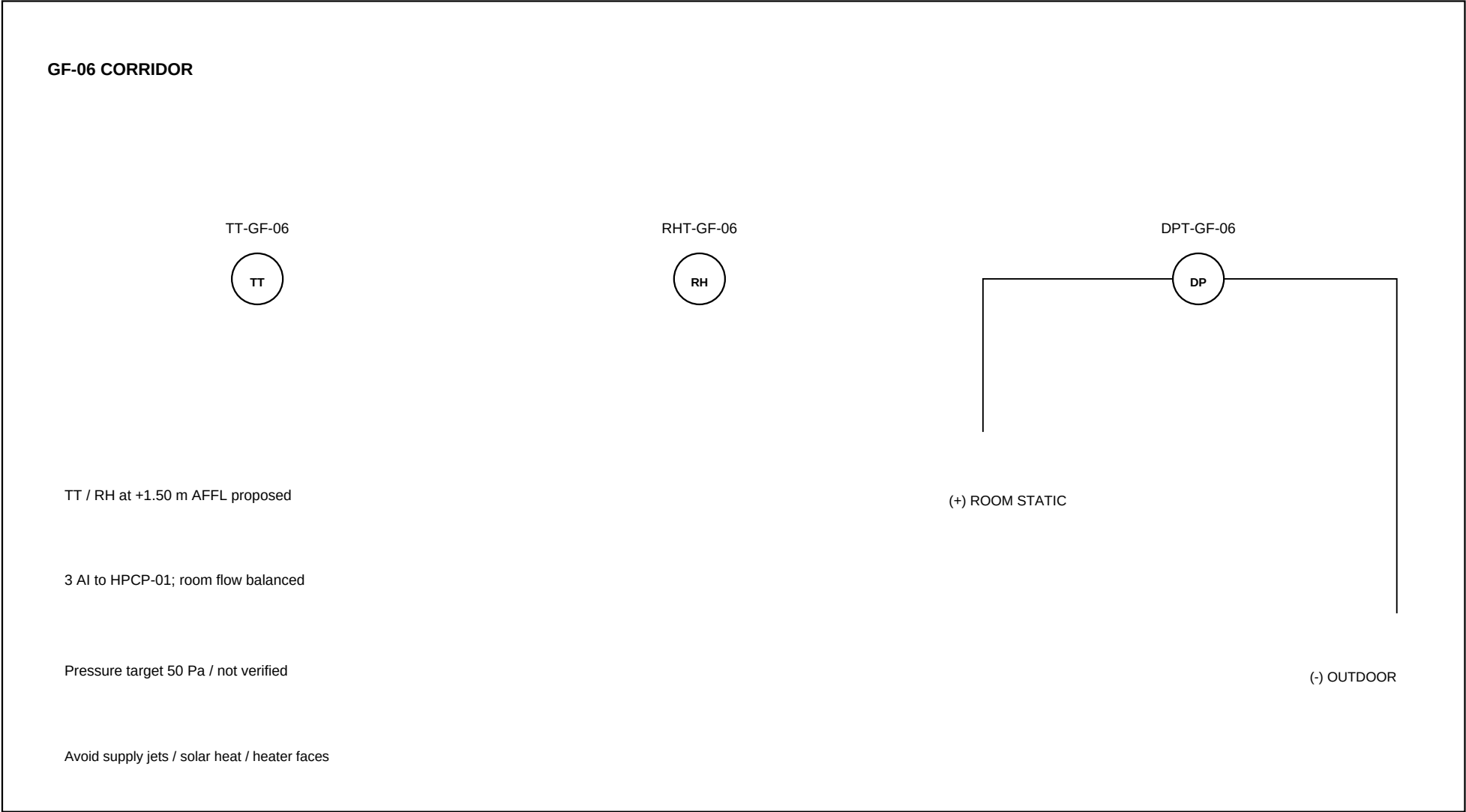
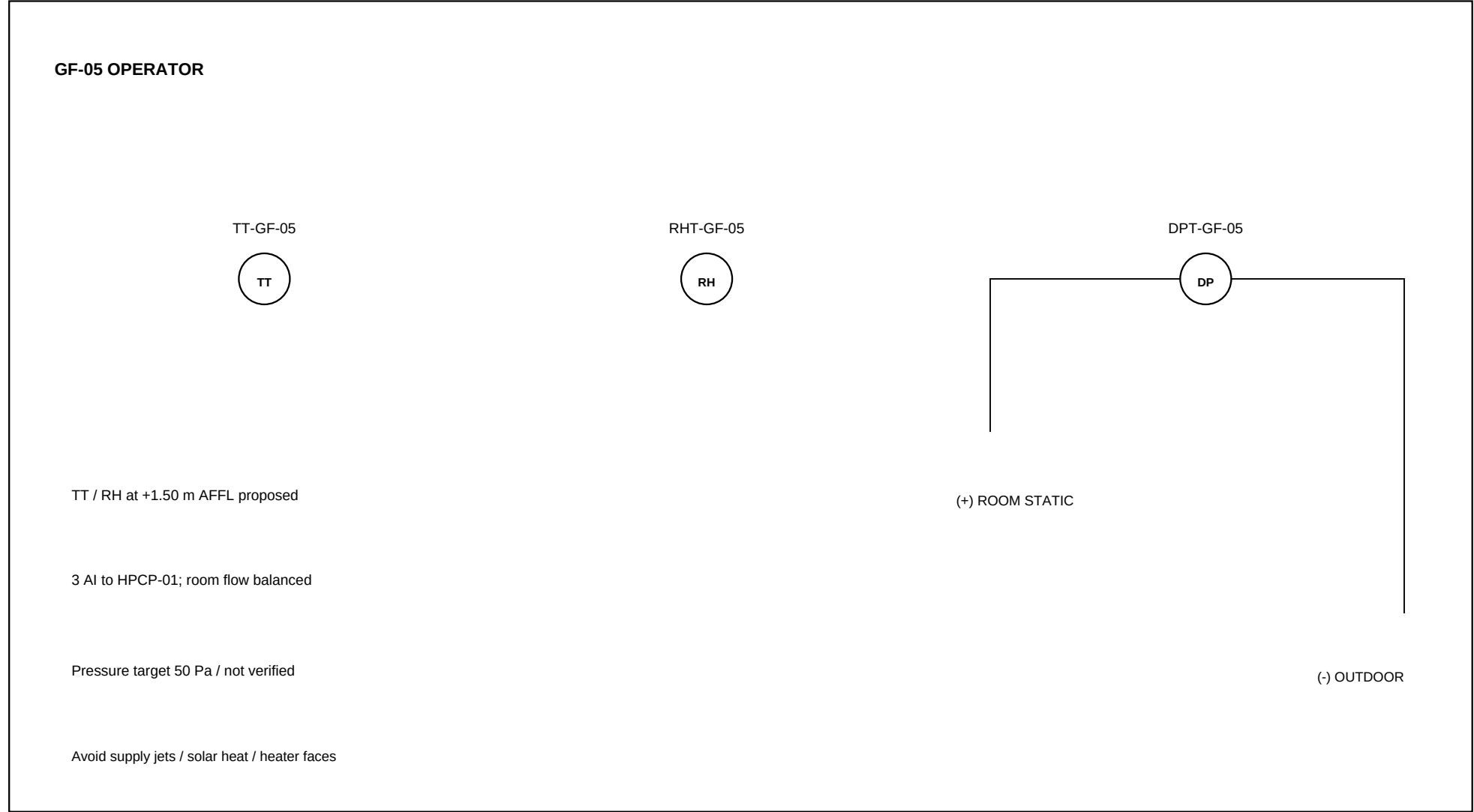
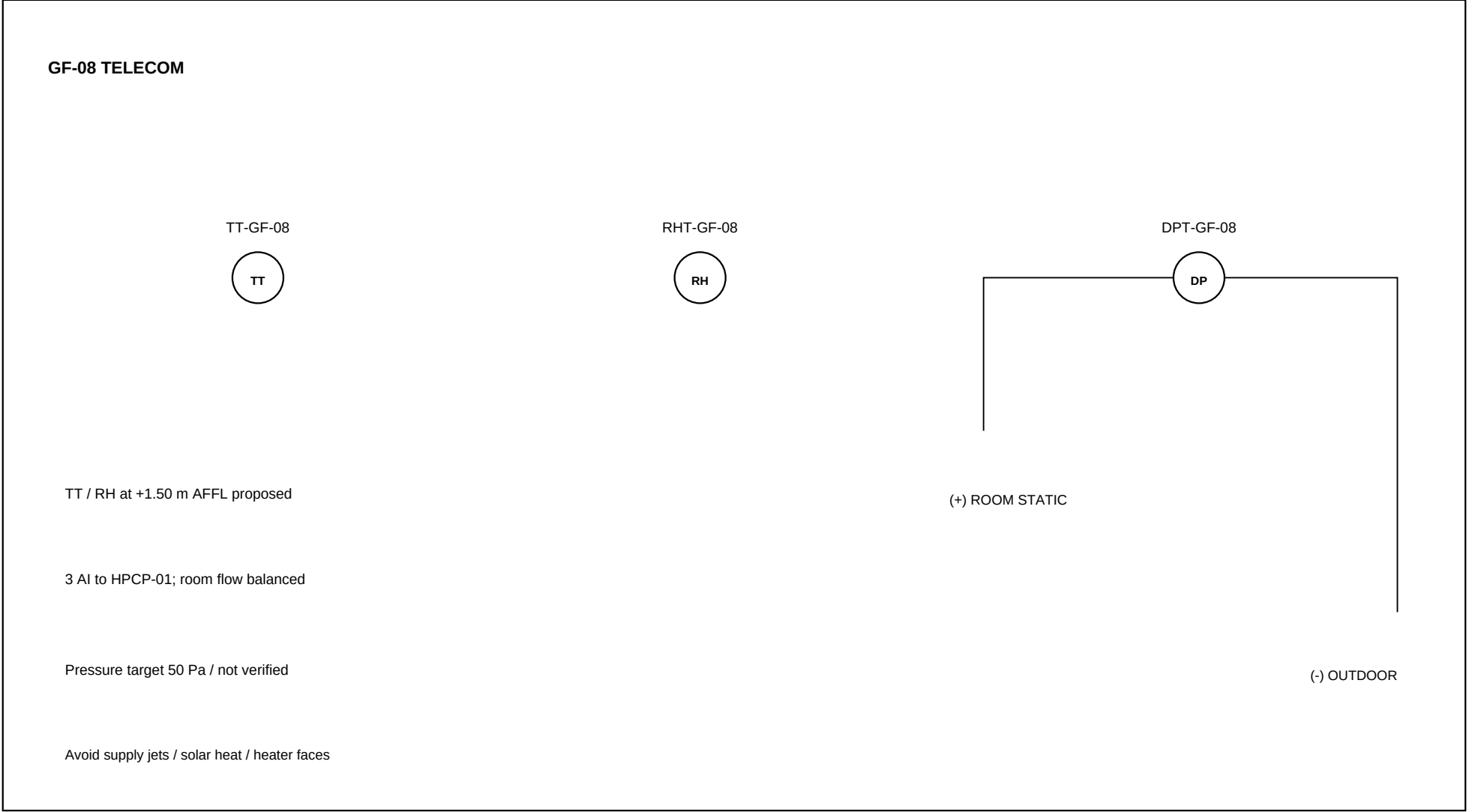
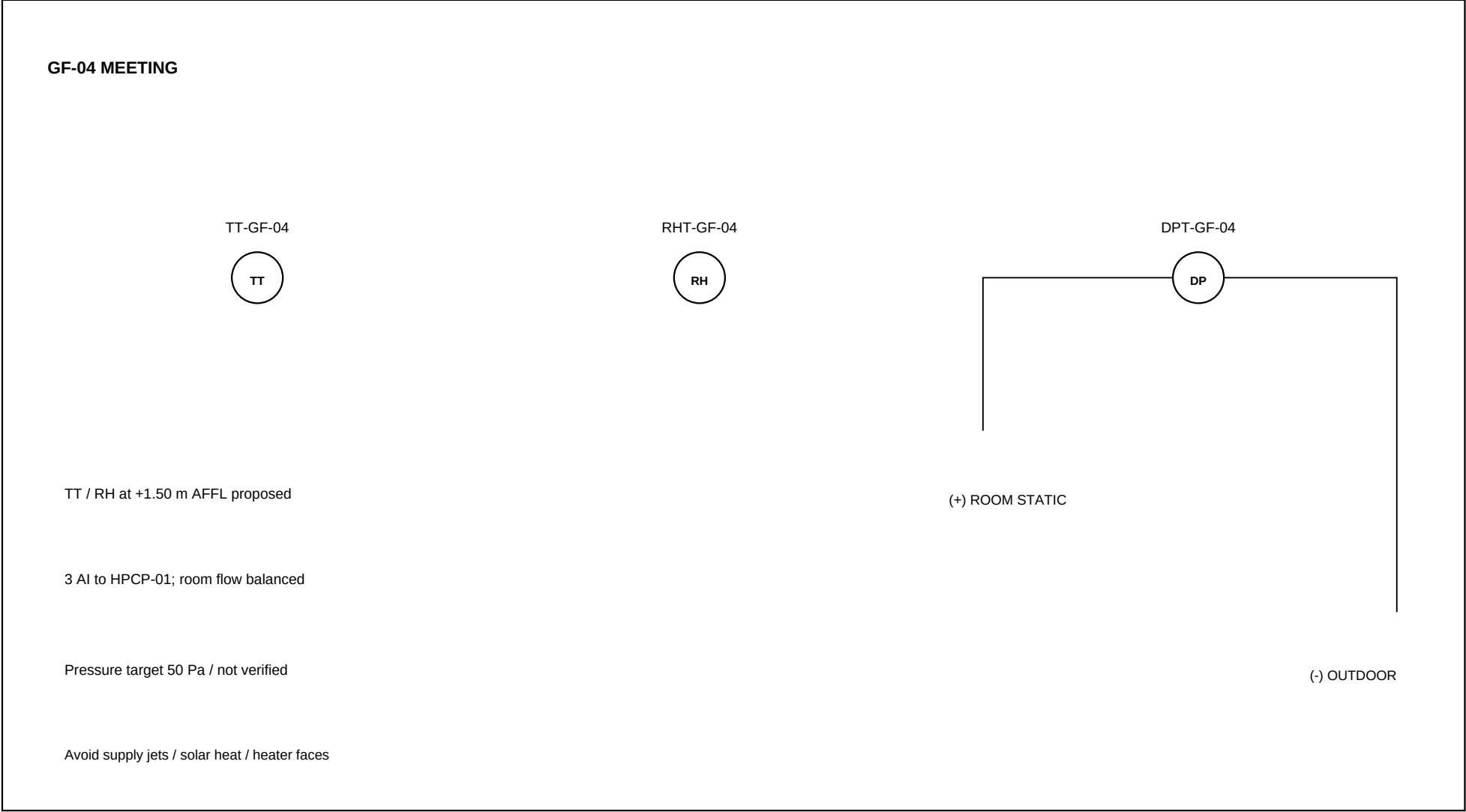
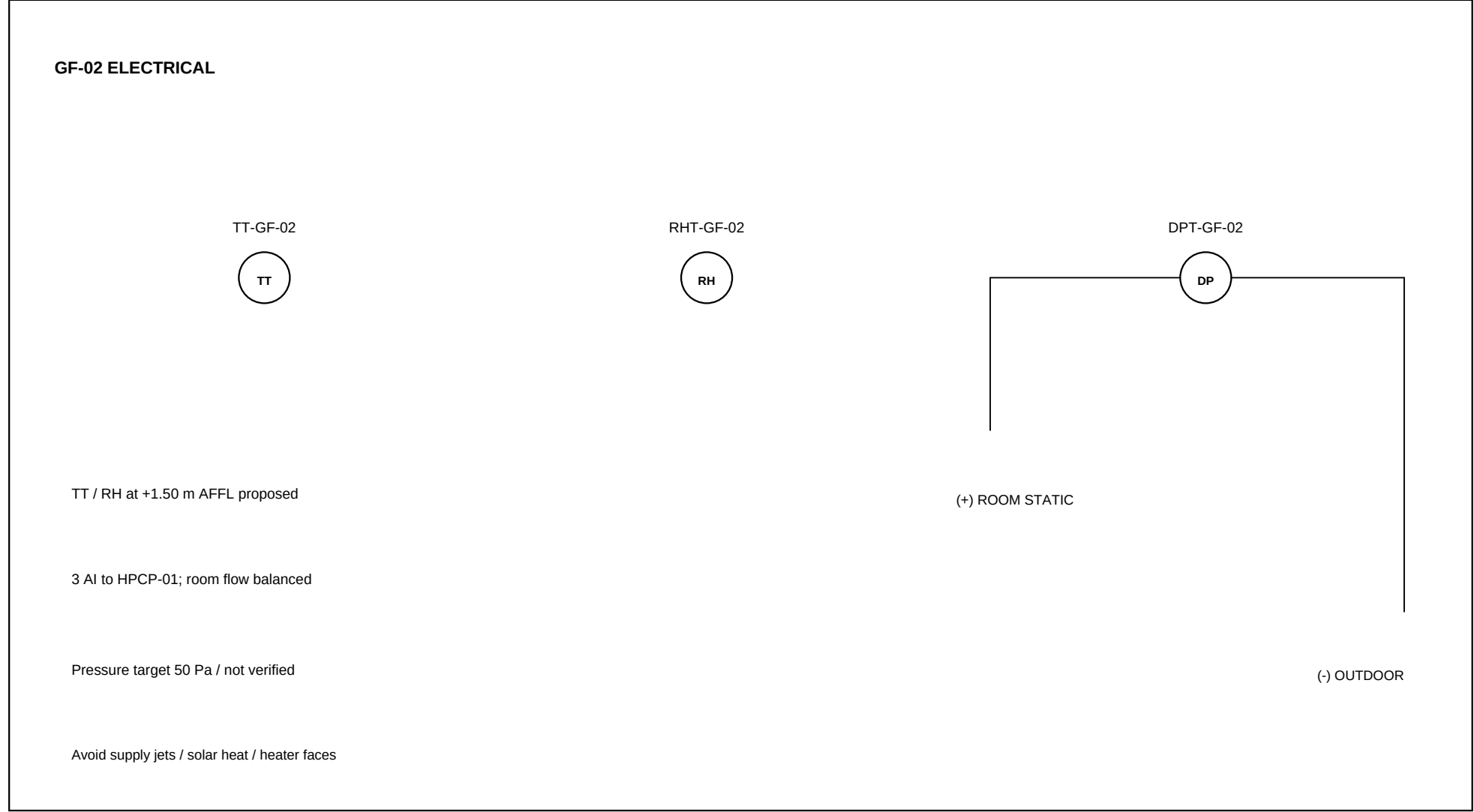
OPTIONAL COMMON-MAIN STATIC CONTROL / NEW PROPOSAL



22 new DX protection / monitoring candidates are separate from the 85 retained air-side instruments. Factory compressor / condenser / evaporator control governs. The shown circuit explains DX operation; receiver, accumulator, refrigerant, valve, oil return, relief and electrical arrangements follow the selected OEM circuit.

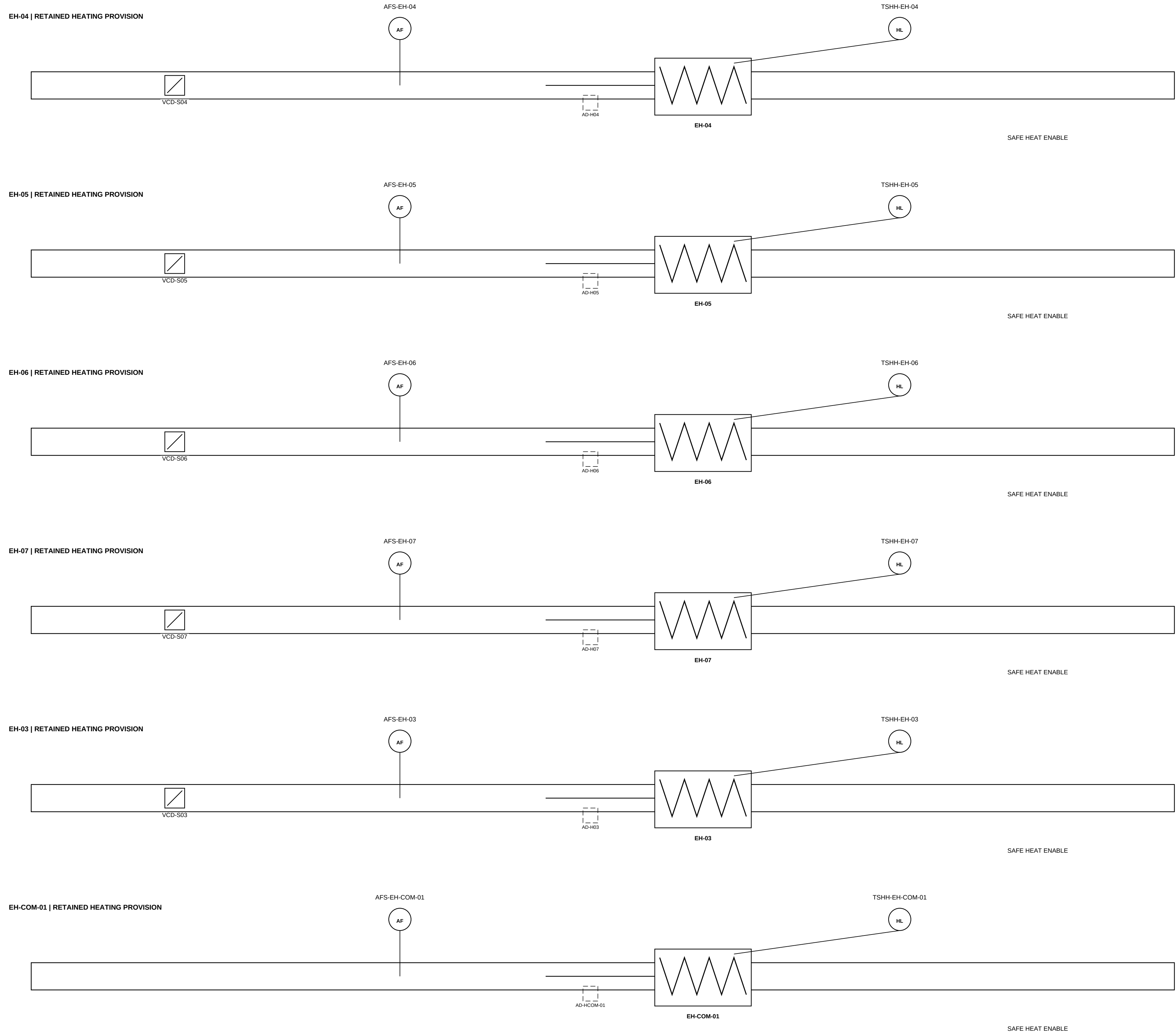
No field refrigerant network to the rooms is proposed. Cooling capacity, refrigerant pressure-temperature ranges and compressor quantity/staging are TBC. Do not modify factory safety wiring from this diagram.

ADMINISTRATION HVAC / DX PAU OPTION COMMON SUPPLY MAIN + COMMON RETURN MAIN FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION	DRAWING TITLE DX REFRIGERANT CYCLE AND PROPOSED OEM PROTECTION ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING	ADM-HVAC-MDR-044 SCALE: NTS AT A1	DRAWING REVISION 02 08 OCT 2026 SHEET 44 OF 50
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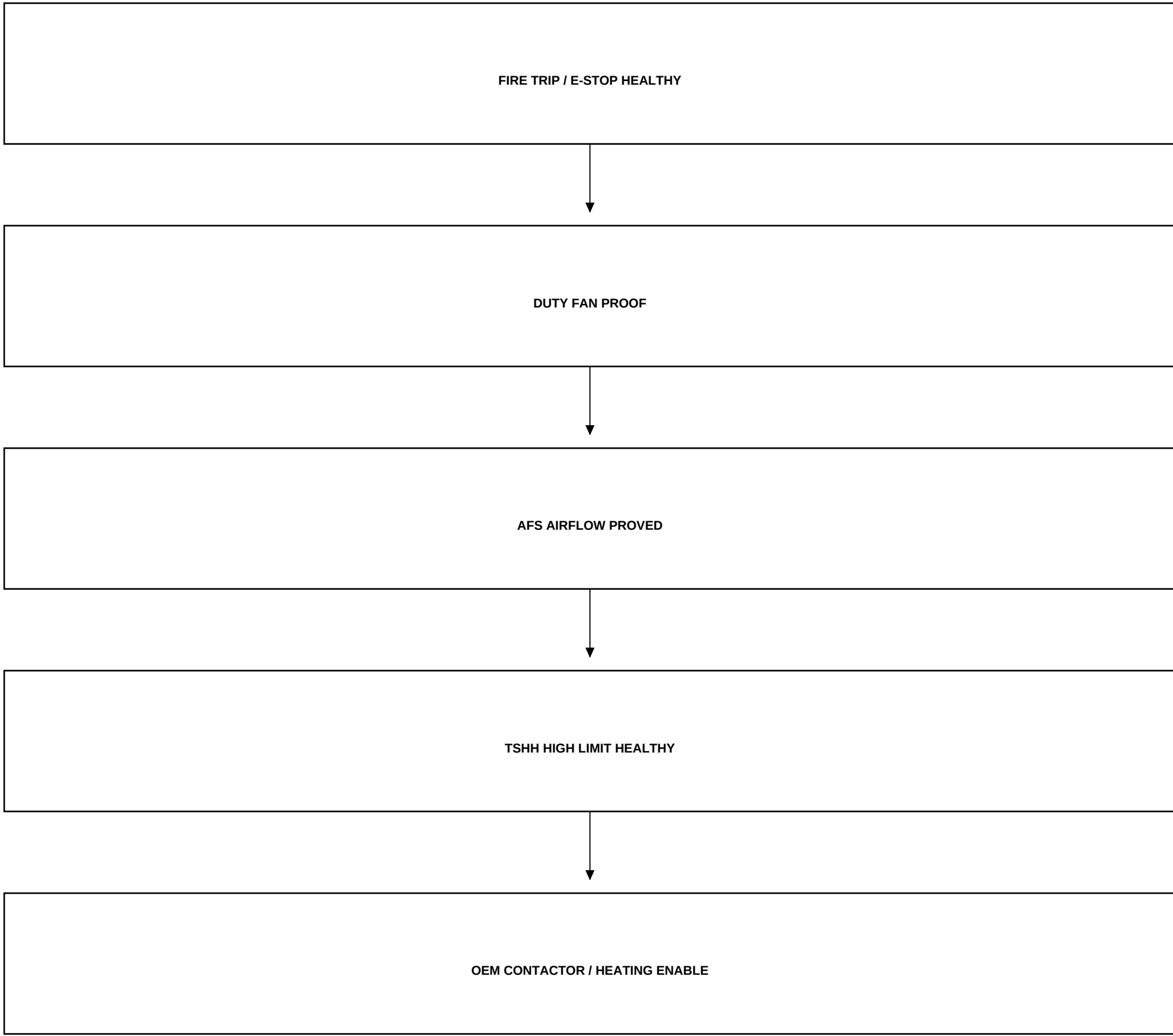
Room pressure uses a sheltered outdoor reference, equal datums and static pickups. RCD control is a proposed pressure trim; TT/RHT monitor the room and do not create independent DX zones. Final pressure, comfort control, ranges, setpoints and tubing are pending.

MAIN DUCT AND RETURN - HEATER PERMISSIVES, ACCESS AND HARDWIRED SAFETY



Six AFS and six independent TSHH provisions are retained. Heater dimensions, power, thermal clearances, airflow proof technology and contact reset are OEM selections. Dampers/access symbols are functional; MFD quantity/rating follows the fire strategy.

OEM SAFETY CHAIN / INTERFACE



Loss of airflow or high-limit trip inhibits heat through the independent OEM circuit. HPCP heating demand cannot bypass either safety. Contacts and final electrical terminals follow the selected heater manufacturer.

DX provides cooling. The retained room/common electric heaters and conditional HUM-01 serve distinct heating/humidity duties. Omitting them from a cooling-only operation requires confirmation of the project thermal/humidity requirements; no automatic deletion is made.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

HEATER PERMISSIVES, ACCESS AND HARDWIRED SAFETY

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-046

SCALE: NTS AT A1

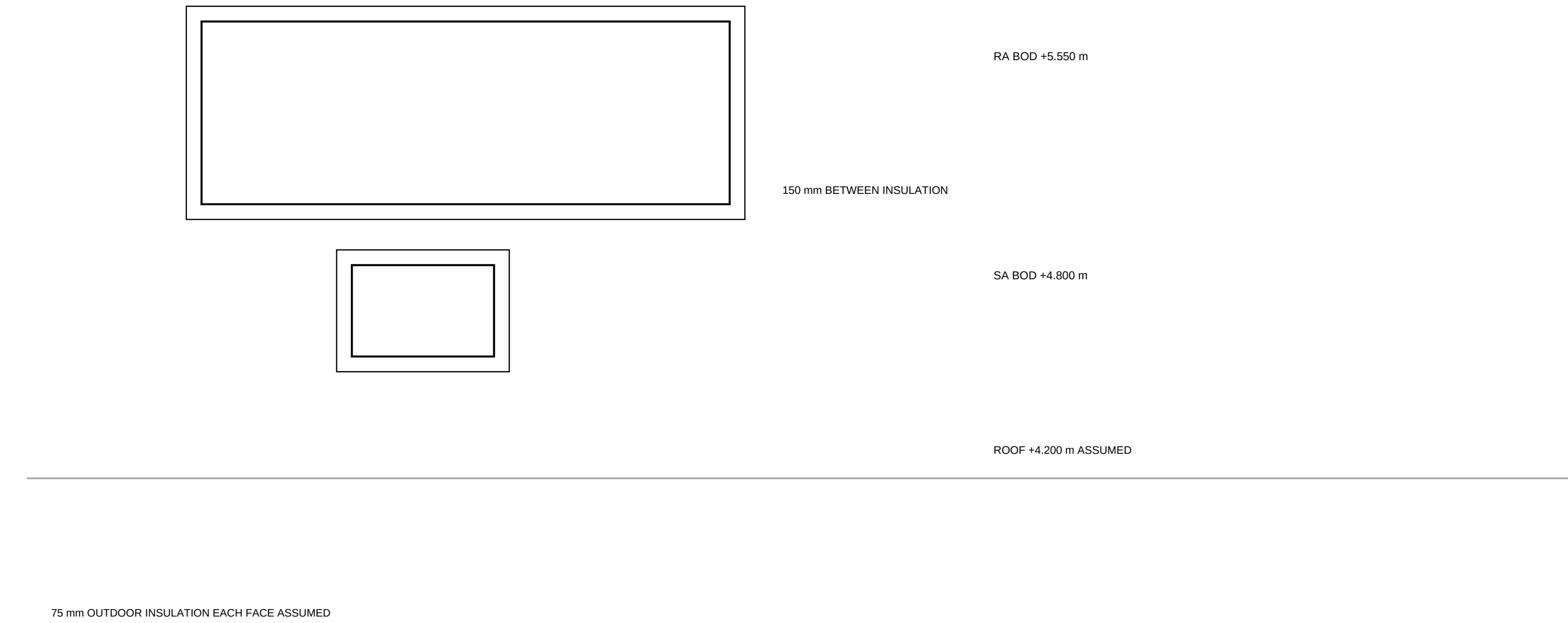
DRAWING REVISION 02

08 OCT 2026

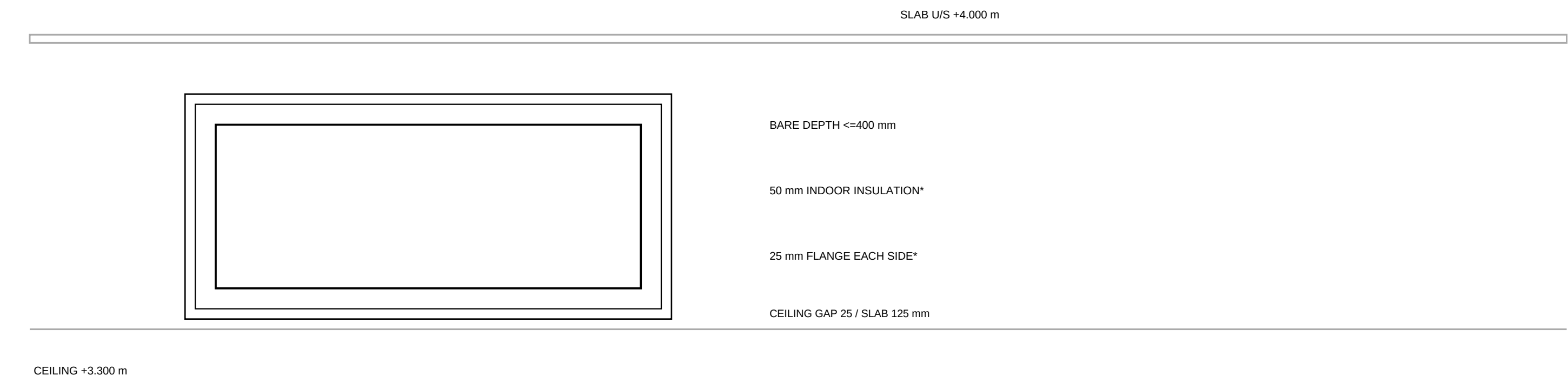
SHEET 46 OF 50

MAIN DUCT AND RETURN - ROOF CROSSING, ROOM DROPS AND CEILING ENVELOPES

A ROOF BRANCH CROSSING / RCX-01 | VERTICAL 1:20



B ROOM DUCT + FLANGES | SECTION 1:10



15 ROOM DROPS: 9 SUPPLY + 6 RETURN / PROPOSED LEVELS

Riser	Room	Clear W x H	Indoor BOD	Roof BOD	Rise m*
RS-SA02	02	500x350	+3.400	4.800	1.4
RS-RA02	02	650x400	+3.400	5.550	2.15
RS-SA04	04	350x400	+3.400	4.800	1.4
RS-RA04	04	500x350	+3.400	5.550	2.15
RS-SA08	08	600x400	+3.400	4.800	1.4
RS-RA08	08	850x400	+3.400	5.550	2.15
RS-SA05	05	600x350	+3.400	4.800	1.4
RS-RA05	05	700x400	+3.400	5.550	2.15
RS-SA06	06	450x300	+3.400	4.800	1.4
RS-RA06	06	500x400	+3.400	5.550	2.15
RS-SA09	09	200x150	+3.400	4.800	1.4
RS-SA07	07	300x250	+3.400	4.800	1.4
RS-RA07	07	350x250	+3.400	5.550	2.15
RS-SA03	03	300x200	+3.400	4.800	1.4
RS-SA01	01	300x200	+3.400	4.800	1.4

CONNECTION / INSTALLATION HOLDS

Roof SA main 600x1300 and RA 700x900 stay outdoors. Only local room ducts descend into the 700 mm ceiling void. Roof-header plan separation and registered crossing levels are shown on MDR-023.

Each RS tag matches its room branch. Two fitted elbows / roof transition, curb / weather seal / firestop and branch access require shop development. Roof +4.200 m assumes a 200 mm slab above the confirmed +4.000 m underside.

* Proposed envelope / rise only. Actual roof structure, beam downstands, insulation specification, actuators, heaters, supports and plenums require OEM/site coordination.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

ROOF CROSSING, ROOM DROPS AND CEILING ENVELOPES

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-047

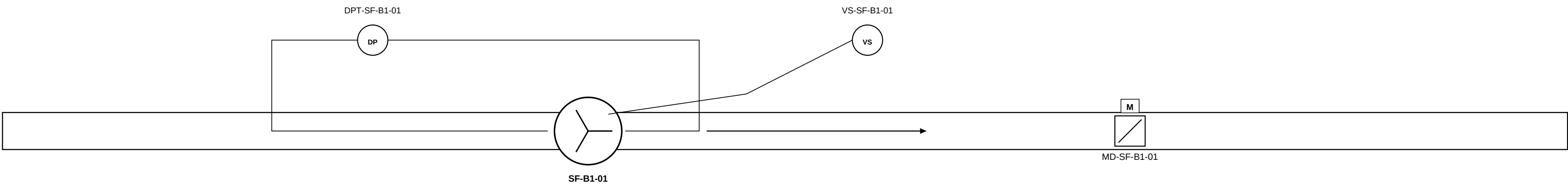
SCALE: NTS AT A1

DRAWING REVISION 02

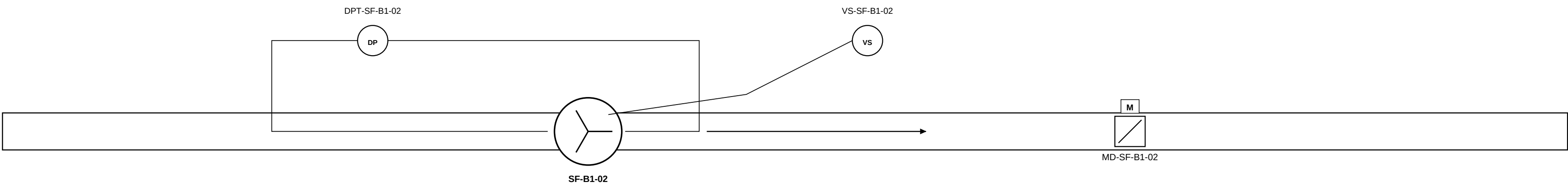
08 OCT 2026

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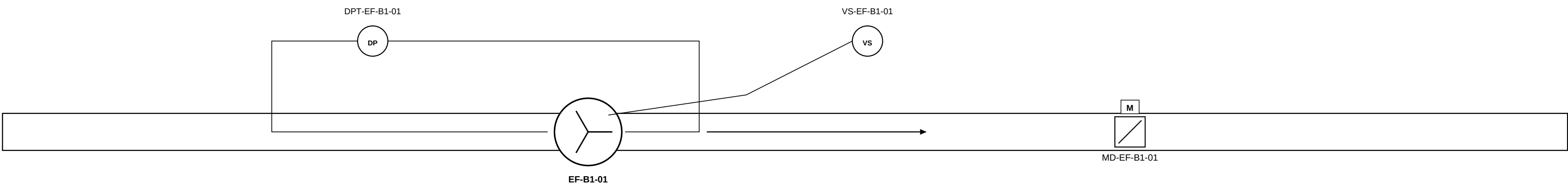
MAIN DUCT AND RETURN - BASEMENT, HPCP, ENERGY METERING AND CONTROL SEQUENCE



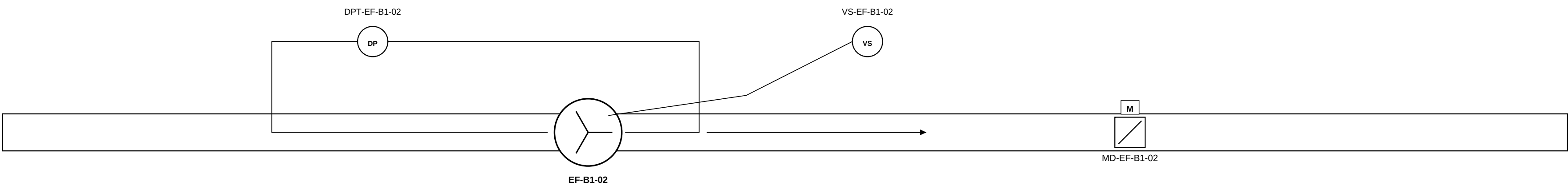
2 x 100% duty / standby, basement flow / heat / volume TBC



2 x 100% duty / standby, basement flow / heat / volume TBC



2 x 100% duty / standby, basement flow / heat / volume TBC



2 x 100% duty / standby, basement flow / heat / volume TBC

Retained I/O	AI	AO	DI	DO	COMM
275 baseline points	65	22	141	44	3

Basement remains an independent ventilation system; its flows/ESP/heat removal are not included in the 3697.80 U/s room supply. New DX proposals are separate from the unchanged 275-row I/O baseline; final channel allocation and protocols pending.

BASEMENT / POWER INSTRUMENTS



HPCP-01 / HMI / BMS SEQUENCE

Start: healthy fire/ESD and OEM safety; select the duty PAU pair; open/prove RA, OA and SA dampers before enabling selected fans/compressors.

Prove fan operation; verify common airflow; enable DX through the OEM controller. Heating requires separate hardwired AFS/TSHH healthy permissives.

Rotate duty pair; failed unit stops / isolates, standby opens/proves then starts. Avoid reverse airflow through an idle unit.

Confirmed fire stops all fans/heaters and latches ESD; manual reset after clearance. Unit compressor response and damper fire positions per approved OEM/fire design.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

BASEMENT, HPCP, ENERGY METERING AND CONTROL SEQUENCE

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-048

SCALE: NTS AT A1

DRAWING REVISION 02

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MAIN DUCT AND RETURN - COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

Retained tag	Parameter	Graphic ref / I/O
TT-GF-02	Dry bulb temperature	MDR-007 / MDR-045 / AI
RHT-GF-02	Relative humidity	MDR-007 / MDR-045 / AI
DPT-GF-02	Room pressure vs outdoor	MDR-007 / MDR-045 / AI
TT-GF-04	Dry bulb temperature	MDR-007 / MDR-045 / AI
RHT-GF-04	Relative humidity	MDR-007 / MDR-045 / AI
DPT-GF-04	Room pressure vs outdoor	MDR-007 / MDR-045 / AI
TSHH-EH-04	Heater high temperature limit	MDR-014 / MDR-046 / DI
TT-GF-08	Dry bulb temperature	MDR-007 / MDR-045 / AI
RHT-GF-08	Relative humidity	MDR-007 / MDR-045 / AI
DPT-GF-08	Room pressure vs outdoor	MDR-007 / MDR-045 / AI
TT-GF-05	Dry bulb temperature	MDR-007 / MDR-045 / AI
RHT-GF-05	Relative humidity	MDR-007 / MDR-045 / AI
DPT-GF-05	Room pressure vs outdoor	MDR-007 / MDR-045 / AI
TSHH-EH-05	Heater high temperature limit	MDR-014 / MDR-046 / DI
TT-GF-06	Dry bulb temperature	MDR-007 / MDR-045 / AI
RHT-GF-06	Relative humidity	MDR-007 / MDR-045 / AI
DPT-GF-06	Room pressure vs outdoor	MDR-007 / MDR-045 / AI
TSHH-EH-06	Heater high temperature limit	MDR-014 / MDR-046 / DI
TT-GF-09	Dry bulb temperature	MDR-007 / MDR-045 / AI
RHT-GF-09	Relative humidity	MDR-007 / MDR-045 / AI
DPT-GF-09	Room pressure vs outdoor	MDR-007 / MDR-045 / AI
TT-GF-07	Dry bulb temperature	MDR-007 / MDR-045 / AI
RHT-GF-07	Relative humidity	MDR-007 / MDR-045 / AI
DPT-GF-07	Room pressure vs outdoor	MDR-007 / MDR-045 / AI
TSHH-EH-07	Heater high temperature limit	MDR-014 / MDR-046 / DI
TT-GF-03	Dry bulb temperature	MDR-007 / MDR-045 / AI
RHT-GF-03	Relative humidity	MDR-007 / MDR-045 / AI
DPT-GF-03	Room pressure vs outdoor	MDR-007 / MDR-045 / AI
TSHH-EH-03	Heater high temperature limit	MDR-014 / MDR-046 / DI
TT-GF-01	Dry bulb temperature	MDR-007 / MDR-045 / AI
RHT-GF-01	Relative humidity	MDR-007 / MDR-045 / AI
DPT-GF-01	Room pressure vs outdoor	MDR-007 / MDR-045 / AI
TT-SA-01	Supply air temperature	MDR-008 / MDR-012 / MDR-036 / MDR-043 / AI
TT-RA-01	Return air temperature	MDR-008 / MDR-012 / MDR-036 / MDR-043 / AI
RHT-RA-01	Return relative humidity	MDR-008 / MDR-012 / MDR-036 / MDR-043 / AI
FIT-SA-01	Common supply airflow	MDR-008 / MDR-012 / MDR-036 / MDR-043 / AI
FIT-RA-01	Common return airflow	MDR-008 / MDR-012 / MDR-036 / MDR-043 / AI
PDT-OA-01	Outdoor intake filter differential pressure	MDR-012 / MDR-043 / AI
PDI-OA-01	Local intake filter differential pressure gauge	MDR-012 / MDR-043 / LOCAL
TSHH-EH-COM-01	Common heater high-limit	MDR-014 / MDR-046 / DI
PDT-F1-A	Filter differential pressure	MDR-012 / MDR-043 / AI
PDI-F1-A	Local filter differential pressure gauge	MDR-012 / MDR-043 / LOCAL
PDT-F2-A	Filter differential pressure	MDR-012 / MDR-043 / AI

Retained tag	Parameter	Graphic ref / I/O
PDI-F2-A	Local filter differential pressure gauge	MDR-012 / MDR-043 / LOCAL
DPT-SF-A	Evaporator fan differential pressure	MDR-012 / MDR-043 / AI
VS-PAU-A	RMS vibration velocity	MDR-012 / MDR-043 / AI
PDT-F1-B	Filter differential pressure	MDR-012 / MDR-043 / AI
PDI-F1-B	Local filter differential pressure gauge	MDR-012 / MDR-043 / LOCAL
PDT-F2-B	Filter differential pressure	MDR-012 / MDR-043 / AI
PDI-F2-B	Local filter differential pressure gauge	MDR-012 / MDR-043 / LOCAL
DPT-SF-B	Evaporator fan differential pressure	MDR-012 / MDR-043 / AI
VS-PAU-B	RMS vibration velocity	MDR-012 / MDR-043 / AI
PDT-F1-C	Filter differential pressure	MDR-012 / MDR-043 / AI
PDI-F1-C	Local filter differential pressure gauge	MDR-012 / MDR-043 / LOCAL
PDT-F2-C	Filter differential pressure	MDR-012 / MDR-043 / AI
PDI-F2-C	Local filter differential pressure gauge	MDR-012 / MDR-043 / LOCAL
DPT-SF-C	Evaporator fan differential pressure	MDR-012 / MDR-043 / AI
VS-PAU-C	RMS vibration velocity	MDR-012 / MDR-043 / AI
DPT-EF-03-01	Extract fan pressure proof	MDR-013 / MDR-042 / AI
VS-EF-03-01	RMS vibration velocity	MDR-013 / MDR-042 / AI
DPT-EF-03-02	Extract fan pressure proof	MDR-013 / MDR-042 / AI
VS-EF-03-02	RMS vibration velocity	MDR-013 / MDR-042 / AI
DPT-EF-09-01	Extract fan pressure proof	MDR-013 / MDR-042 / AI
VS-EF-09-01	RMS vibration velocity	MDR-013 / MDR-042 / AI
DPT-EF-09-02	Extract fan pressure proof	MDR-013 / MDR-042 / AI
VS-EF-09-02	RMS vibration velocity	MDR-013 / MDR-042 / AI
DPT-EF-01-01	Extract fan pressure proof	MDR-013 / MDR-042 / AI
VS-EF-01-01	RMS vibration velocity	MDR-013 / MDR-042 / AI
DPT-SF-B1-01	Fan pressure proof	MDR-013 / MDR-048 / AI
VS-SF-B1-01	RMS vibration velocity	MDR-013 / MDR-048 / AI
DPT-SF-B1-02	Fan pressure proof	MDR-013 / MDR-048 / AI
VS-SF-B1-02	RMS vibration velocity	MDR-013 / MDR-048 / AI
DPT-EF-B1-01	Fan pressure proof	MDR-013 / MDR-048 / AI
VS-EF-B1-01	RMS vibration velocity	MDR-013 / MDR-048 / AI
DPT-EF-B1-02	Fan pressure proof	MDR-013 / MDR-048 / AI
VS-EF-B1-02	RMS vibration velocity	MDR-013 / MDR-048 / AI
TT-B1-01	Cable basement air temperature	MDR-015 / MDR-048 / AI
DPT-B1-01	Basement pressure vs outdoor	MDR-015 / MDR-048 / AI
EM-HPCP-01	Energy metering: V, I, kW, kWh, PF, Hz, THD for VFD	MDR-015 / MDR-048 / COMM
AFS-EH-04	Airflow proving switch for heater safety permissive	MDR-014 / MDR-046 / DI
AFS-EH-05	Airflow proving switch for heater safety permissive	MDR-014 / MDR-046 / DI
AFS-EH-06	Airflow proving switch for heater safety permissive	MDR-014 / MDR-046 / DI
AFS-EH-07	Airflow proving switch for heater safety permissive	MDR-014 / MDR-046 / DI
AFS-EH-03	Airflow proving switch for heater safety permissive	MDR-014 / MDR-046 / DI
AFS-EH-COM-01	Airflow proving switch for heater safety permissive	MDR-014 / MDR-046 / DI

85 retained instruments are graphically referenced on MDR-042 through MDR-048. 22 new DX candidates are on MDR-044 and in Additional_DX_proposals. Original instrument requirements and I/O are unchanged; final OEM selection / safety settings remain open.

MAIN DUCT AND RETURN - COMMON HEADER FLOW ACCOUNTING AND ENGINEERING BASIS

Main	Takeoff GF	Before L/s	Branch L/s	After L/s	Clear W x H	m/s	Straight Pa/m
SA-M01	02	3697.80	664.80	3033.00	600x1300	4.741	0.260
SA-M02	04	3033.00	449.20	2583.80	600x1300	3.888	0.180
SA-M03	08	2583.80	884.60	1699.20	600x1300	3.313	0.134
SA-M04	05	1699.20	777.70	921.50	600x1300	2.178	0.062
SA-M05	06	921.50	418.20	503.30	600x1300	1.181	0.021
SA-M06	09	503.30	46.20	457.10	600x1300	0.645	0.007
SA-M07	07	457.10	168.20	288.90	600x1300	0.586	0.006
SA-M08	03	288.90	147.60	141.30	600x1300	0.370	0.003
SA-M09	01	141.30	141.30	0.00	600x1300	0.181	0.001
RA-M01	02	3006.68	605.46	2401.22	700x900	4.773	0.277
RA-M02	04	2401.22	389.86	2011.35	700x900	3.811	0.183
RA-M03	08	2011.35	825.26	1186.09	700x900	3.193	0.132
RA-M04	05	1186.09	718.36	467.73	700x900	1.883	0.050
RA-M05	06	467.73	358.86	108.86	700x900	0.742	0.009
RA-M06	07	108.86	108.86	0.00	700x900	0.173	0.001

CALCULATION / FABRICATION BASIS

Retained roof spines: SA600x1300, RA700x900. This sequential room flow accounting explains the method; actual grouped roof route / branch dimensions and duties are authoritative in Roof_routes_Rev12 and MDR-023. Colebrook/Darcy-Weisbach, rho1.2, mu1.81e-5, galvanized roughness0.09 mm. These are preliminary design choices, not selected acoustic performance.

At each room SA takeoff, remaining flow decreases. On RA the sequence is listed away from the PAU; actual airflow collects toward the PAU. The scaled roof uses two grouped SA branch columns, so these sequential steps are not a fabrication main schedule. No toilet/kitchen/CAG return enters this header.

Develop each reducer with L >= max(300, max((delta W)/(delta H))^2 tan15deg)). Flared three-leg boots and open throats; selected turning vanes/radii and device face-to-face lengths govern shop drawings.

THERMAL / PRESSURE / SELECTION HOLD

SA 3697.80 minus RA 3006.68 requires makeup >= 691.12 L/s. OA reference420.70 is short by 270.42 L/s. This is a mass-balance lower bound, not an approved outdoor-air duty.

No complete fan ESP is calculated for this method: surveyed common-main/riser lengths, nozzle losses, branch layout, fitted K, terminals, coils/dirty filters, intake and exhaust discharge are unselected. The Rev12 allowance budget applies only to its measured proposed room-drop routes, excludes PAU bank/nozzle/OA details and does not select a fan for this method package.

DX coil total/sensible/latent capacity and refrigerant must be selected from revised room/OA loads and design outdoor conditions. One common supply condition alone does not prove independent room comfort; verify loads, reheat/humidity requirements and zoning.

Review index: 001 whole method; 002 all terminals; 003 extracts; 004 PAU/air instruments; 005 DX circuit; 006 room instruments; 007 heater safety; 008 sections/risers; 009 controls/basement; 010 all 85 instruments; 011 functional flow steps; 012 scaled roof plan; 013 indoor coordination; 014 room instrument locations.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

COMMON HEADER FLOW ACCOUNTING AND ENGINEERING BASIS

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-050

SCALE: NTS AT A1

DRAWING REVISION 02

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